

FDA IRVINE

RM 2309 EXHAUST DESIGN

PROJECT IDENTIFICATION

REGION: CO
DIVISION: GSA OFFICE OF DESIGN & CONSTRUCTION / SPECIAL PROGRAMS DIVISION
BUILDING NAME: SOUTHWEST PACIFIC LABORATORY
ADDRESS: 19701 FAIRCHILD RD IRVINE, CA 92612
CONTRACTING OFFICER: KALISH POOLE
PROJECT MANAGER: DEMIAN GO P.E.

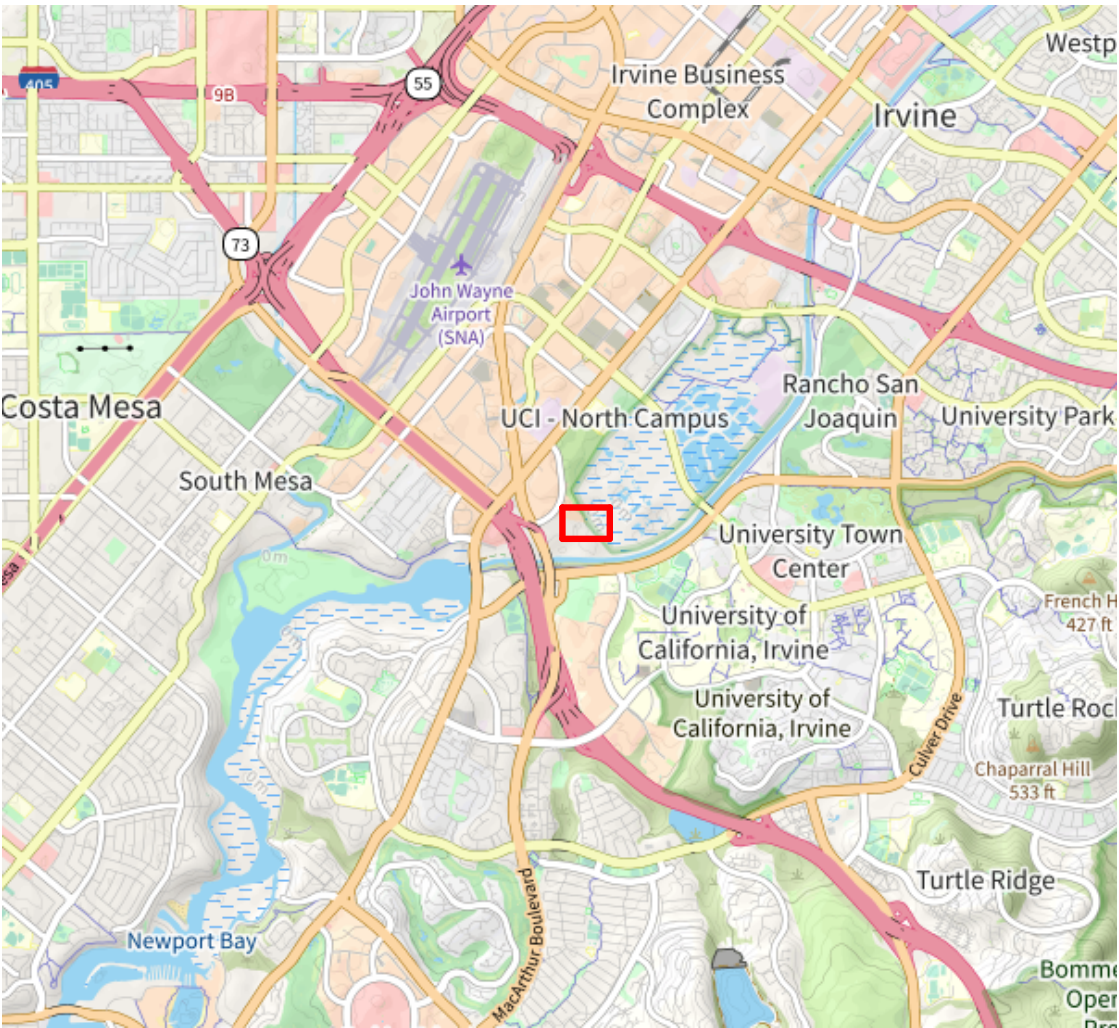
PROJECT SUMMARY: THIS PROJECT IS FOR MECHANICAL EQUIPMENT, DUCTWORK AND CONTROLS TO REDUCE CONCENTRATION OF VAPORIZED HYDROGEN PEROXIDE (VHP) GENERATED BY LABORATORY ISOLATOR EQUIPMENT IN ROOM 2309. VHP AND ROOM AIR WILL BE CONVEYED FROM THE EQUIPMENT EXHAUST PORTS TO ROOF TOP STACKS VIA DEDICATED EXHAUST SYSTEM. PROJECT INCLUDES LIMITED ARCHITECTURAL AND ELECTRICAL MODIFICATIONS TO EXISTING SYSTEMS TO SUPPORT MECHANICAL EXHAUST SYSTEM PROPOSED. PROPOSED EXHAUST SYSTEM WILL ALSO REQUIRE INTEGRATION AND MODIFICATIONS TO THE EXISTING BUILDING AUTOMATION SYSTEM (BAS) AND HVAC SYSTEM FOR THE ROOM, AS WELL AS THE INTEGRATION OF THE EXISTING ROOM 2309 LABORATORY CONTROLS WITH THE EXISTING BAS.

BID SET
AUGUST 15, 2025

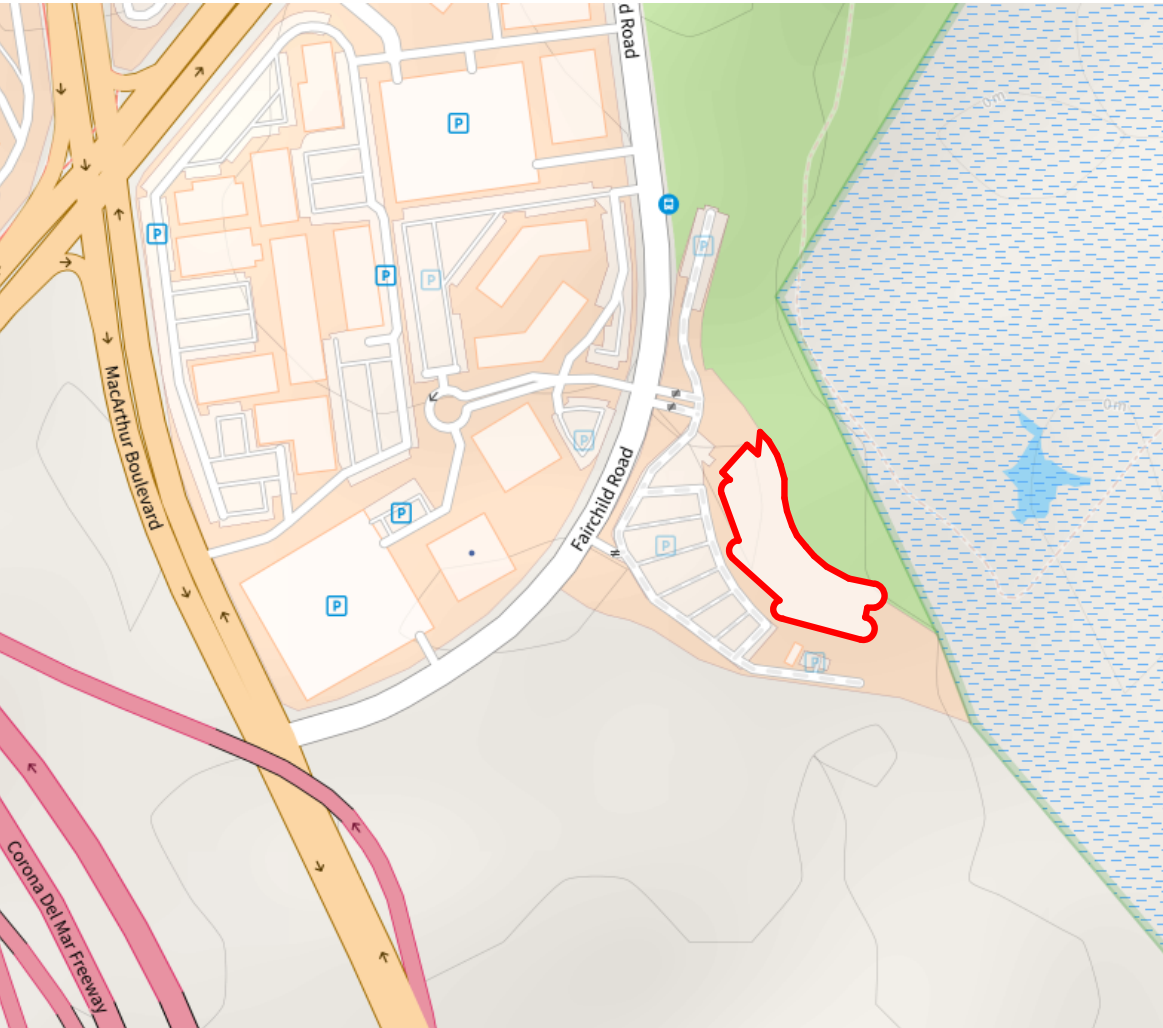
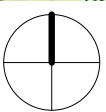


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POINT LOMA MARINA 4980 NORTH HARBOR DRIVE, SUITE 100 SAN DIEGO, CA 92106

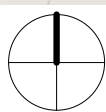
SHEET INDEX	
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VICINITY MAP



LOCATION MAP



1. INTRODUCTION

THIS BASIS OF DESIGN DOCUMENT ESTABLISHES THE KEY STRUCTURAL AND MECHANICAL CONSIDERATIONS FOR THE PROPOSED MODIFICATIONS TO THE EXISTING BUILDING. THE INFORMATION HEREIN IS BASED ON FINDINGS FROM THE SITE VISIT CONDUCTED BY MILLER HULL AND INTERFACE ENGINEERS ON DECEMBER 11, 2024.

2.1 MAIN ROOF STRUCTURE

THE EXISTING MAIN ROOF CONSISTS OF A 7-INCH CONCRETE SLAB SUPPORTED BY CONCRETE BEAMS. NO BEAMS ARE PRESENT IN THE IMMEDIATE VICINITY OF THE PROPOSED SLAB PENETRATIONS.

REINFORCING IN THE SLAB IS SPACED AT APPROXIMATELY 10 INCHES (NORTH-SOUTH) AND 16 INCHES (EAST-WEST).

A SINGLE 12-INCH BY 12-INCH PENETRATION IS FEASIBLE WITHOUT SLAB STRENGTHENING, PROVIDED SCANNING IS PERFORMED TO ADJUST CORE LOCATIONS TO AVOID EXISTING REINFORCEMENT.

A SINGLE 12-INCH BY 12-INCH PENETRATION IS POSSIBLE WITH MINIMAL LOCALIZED REINFORCEMENT ADDED TO THE UNDERSIDE OF THE SLAB.

A NEW PREFABRICATED CURB WILL BE ANCHORED TO THE EXISTING SLAB TO. THIS CURB WILL SERVE WATERPROOFING AND STRUCTURAL SUPPORT PURPOSES FOR THE NEW UNITS AND DUCTWORK, IF REQUIRED.

THE NEW CURB AND NEW UNITS (EACH ESTIMATED AT 300 POUNDS MAX) ARE EXPECTED TO BE ACCOMMODATED BY THE EXISTING SLAB WITHOUT REINFORCEMENT, PENDING STRUCTURAL ANALYSIS UPON FINAL LAYOUT CONFIRMATION.

- FRAMING.
- ROOF DECK PENETRATIONS ARE ALLOWED WITH MINOR LOCALIZED REINFORCEMENT TO STIFFEN THE DECKING.
- NO STEEL BEAMS ARE DIRECTLY IN LINE WITH THE PROPOSED PENETRATIONS, BUT EXISTING STEEL BRIDGING (RUNNING NORTH-SOUTH AT THIRD POINTS BETWEEN THE MAIN PENTHOUSE ROOF GIRDERS) MUST BE AVOIDED.
- THE ROOF DECKING IS EXPECTED TO SUPPORT CABLING FOR BRACING THE UPPER PORTION OF THE EXHAUST STACK. ASSUMING THE STACK EXTENDS LESS THAN 6 FEET ABOVE THE ROOF DECK.
- BRACING TO THE EXTERIOR CLADDING SYSTEM SHOULD BE AVOIDED.

- THIS WALL CAN LIKELY BE USED FOR LATERAL BRACING OF THE EXHAUST DUCTS AND FOR SUPPORTING SMALL EQUIPMENT (UP TO 50 POUNDS).
- HEAVIER EQUIPMENT MAY REQUIRE ADDITIONAL STRUCTURAL SUPPORT, LIKELY CONSISTING OF A UNISTRUT SYSTEM.

3.1 ROOF PENETRATIONS

- QUANTITY: ONE (1) ROOF PENETRATION.
- SIZE: 14-15 INCHES IN DIAMETER.

- **CORROSION RESISTANCE:** 316 STAINLESS STEEL IS THE TYPICAL MECHANICAL MATERIAL USED TO RESIST HYDROGEN PEROXIDE EXPOSURE.

- CURBS WILL BE PROVIDED AROUND PENETRATIONS TO SUPPORT SMALL ROOF FANS (MAXIMUM 300 LBS EACH).
- THE EXHAUST DUCT WILL EXTEND AT LEAST 6 INCHES ABOVE THE EXISTING MECHANICAL SCREEN. THE VERTICAL DUCT WILL REQUIRE SUPPORT OFF THE BACK WALL USING UNISTRUT.

- SCOPE TBD: WORK RELATED TO BAS INTEGRATION MAY BE REQUIRED IN OTHER BUILDING AREAS SUBJECT TO FURTHER REVIEW.

SPECIFICATIONS AND DEMOLITION PLANS MUST INCLUDE PROVISIONS FOR SAFE ENCAPSULATION DURING ROOF SLAB CUTTING TO PROTECT THE OCCUPIED LABORATORY SPACE AND MISSION-CRITICAL EQUIPMENT.



- MECHANICAL DUCTS AND FITTINGS TO BE ROUTED THROUGH THE EXISTING CEILING USING EXISTING PANELS OR SHEET METAL PENETRATION PLATES.
- EXISTING INTERIOR LIGHTING TO BE MAINTAINED.



NOTE: ABBREVIATIONS NOTED IN THE DRAWINGS THAT ARE FOLLOWED BY A MODIFIER SUCH AS "-1", "-A" ARE FURTHER DEFINED IN THE SPECIFICATIONS SECTION ASSOCIATED WITH THE MATERIAL OR SYSTEM ASSEMBLY AS NOTED. REFERENCE FINISH LEGEND A070

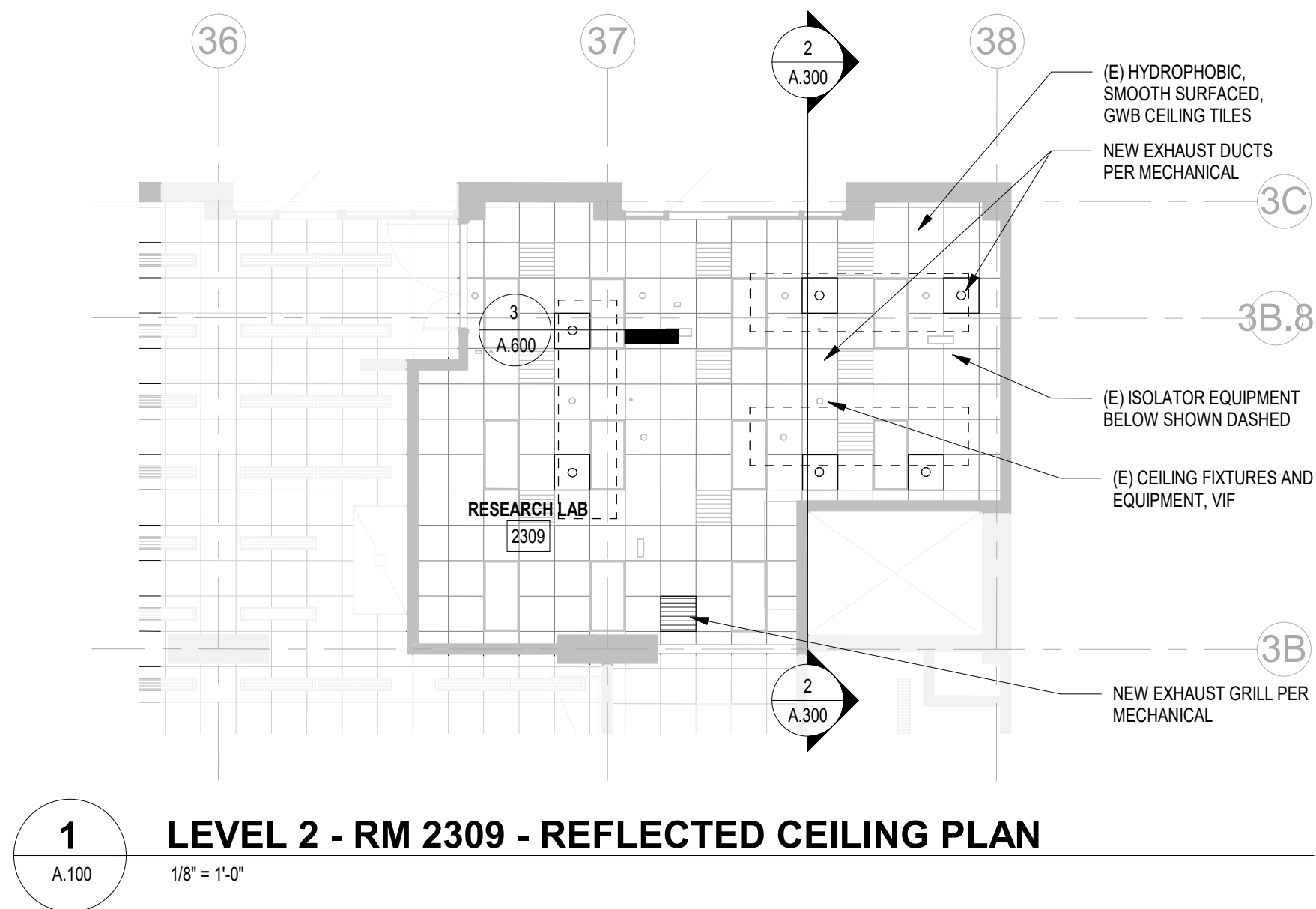
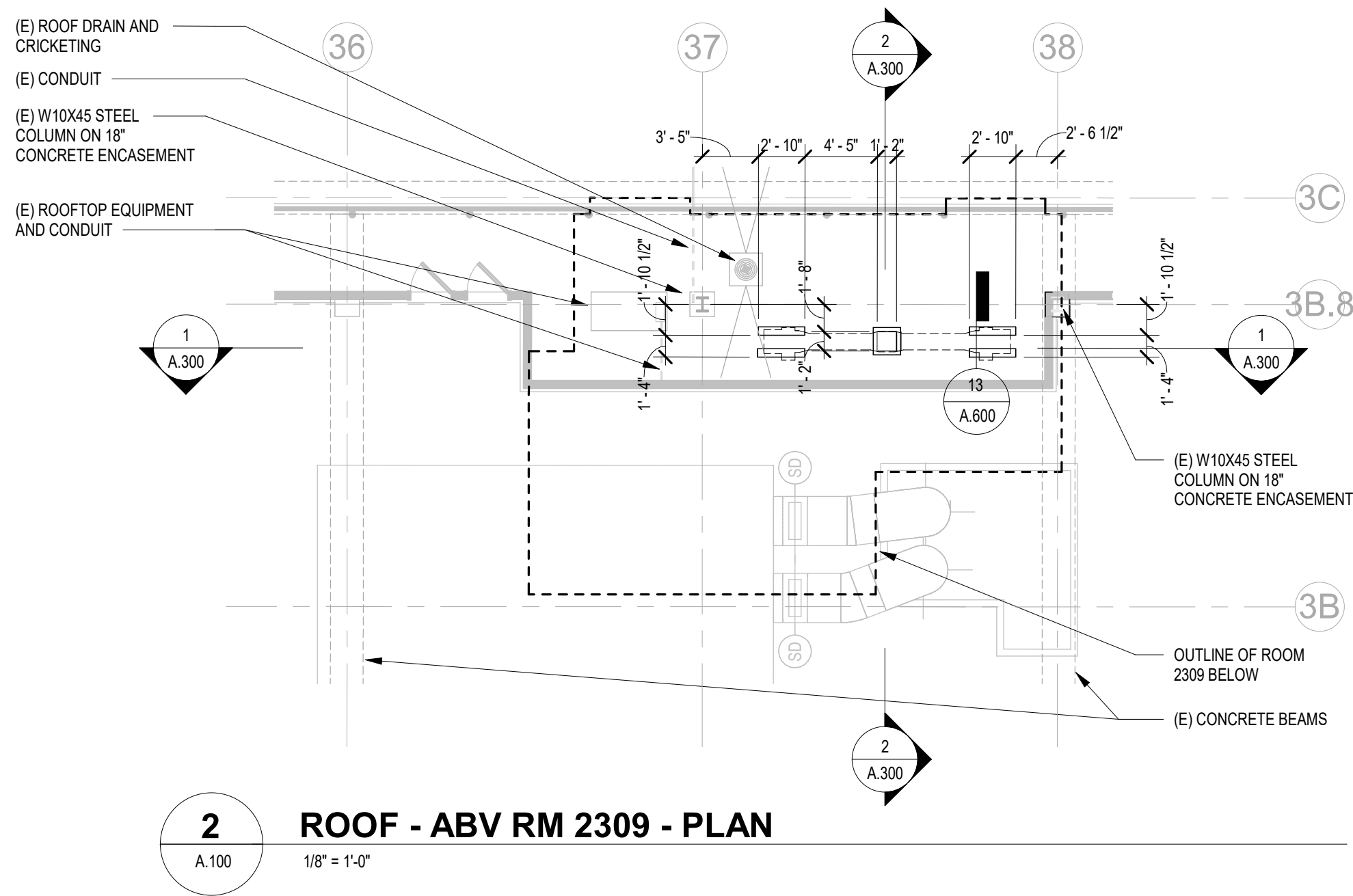
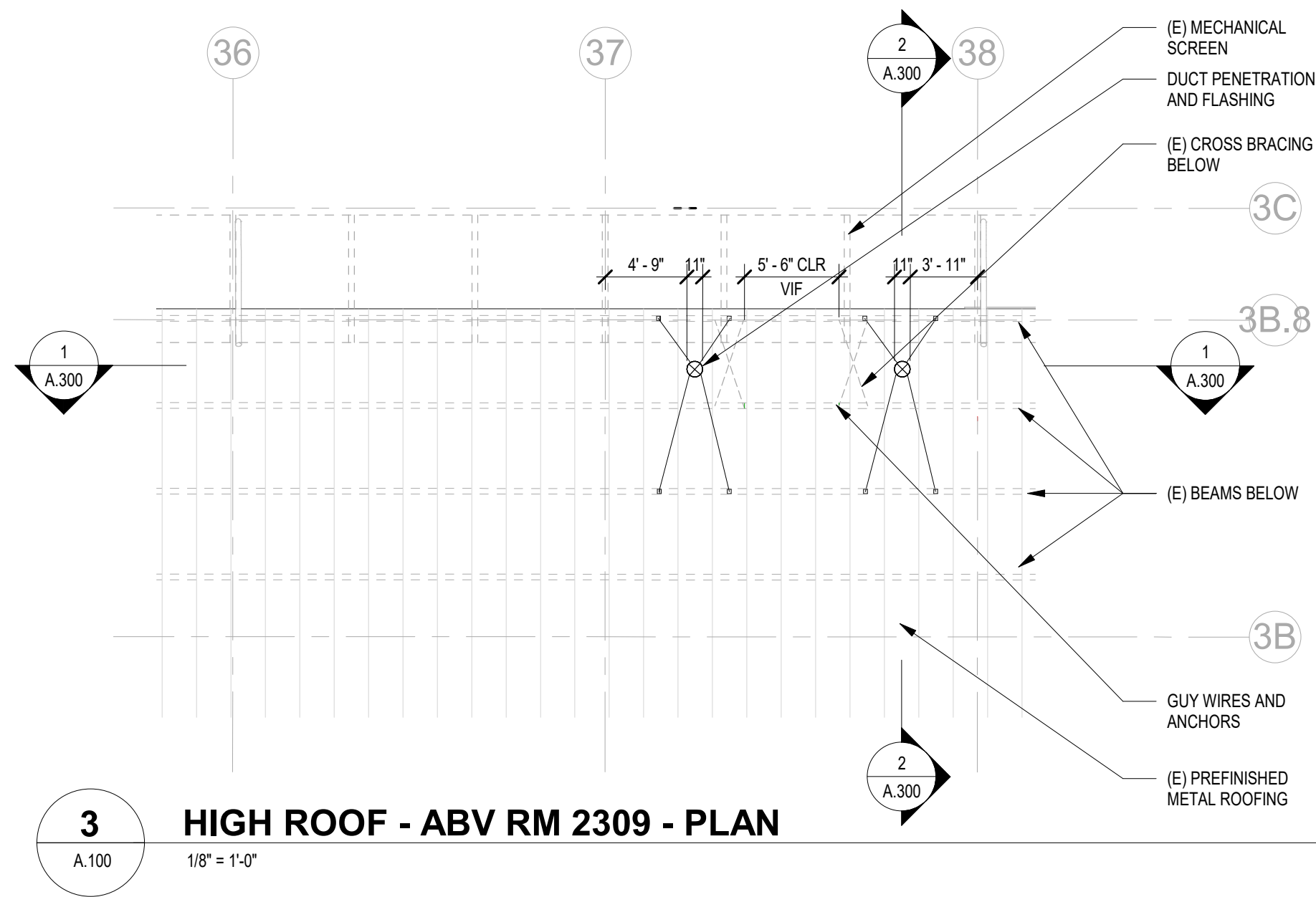
- National Institutes of Health Design Requirements Manual 1 (Rev 2.1 08/02/24)
- National Institutes of Health Design Requirements Manual 2 (Rev 1.5 12/12/16)



**LEGENDS,
NOTES &
ABBREVIATIONS**

A.000

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GENERAL NOTES

1. PROTECT IN PLACE ALL EXISTING TO REMAIN EQUIPMENT, LIGHT FIXTURES, AND DEVICES.
2. CONTRACTOR TO VIF LOCATIONS OF ISOLATORS AND COORDINATE IN FIELD ALL CEILING PENETRATIONS TO MAINTAIN REQUIRED CLEARANCES BETWEEN ISOLATOR EXHAUST CONNECTIONS AND PENETRATIONS.

REFLECTED CEILING PLAN LEGEND

	(E) SUSPENDED T-GRID & CEILING
	(E) RECESSED LIGHT FIXTURE
	(E) MECHANICAL GRILL
	NEW MECHANICAL GRILL
	(E) CEILING FIXTURES AND EQUIPMENT

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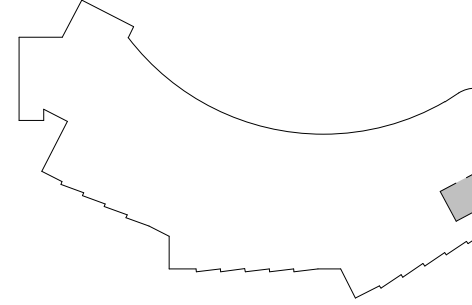
GSA

FDA U.S. FOOD & DRUG
ADMINISTRATION

STAMP



KEY PLAN



FDA IRVINE

RM 2309
EXHAUST
DESIGN

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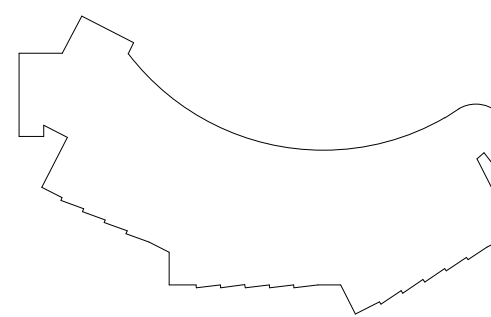
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SHEET

FLOOR PLANS
A.100



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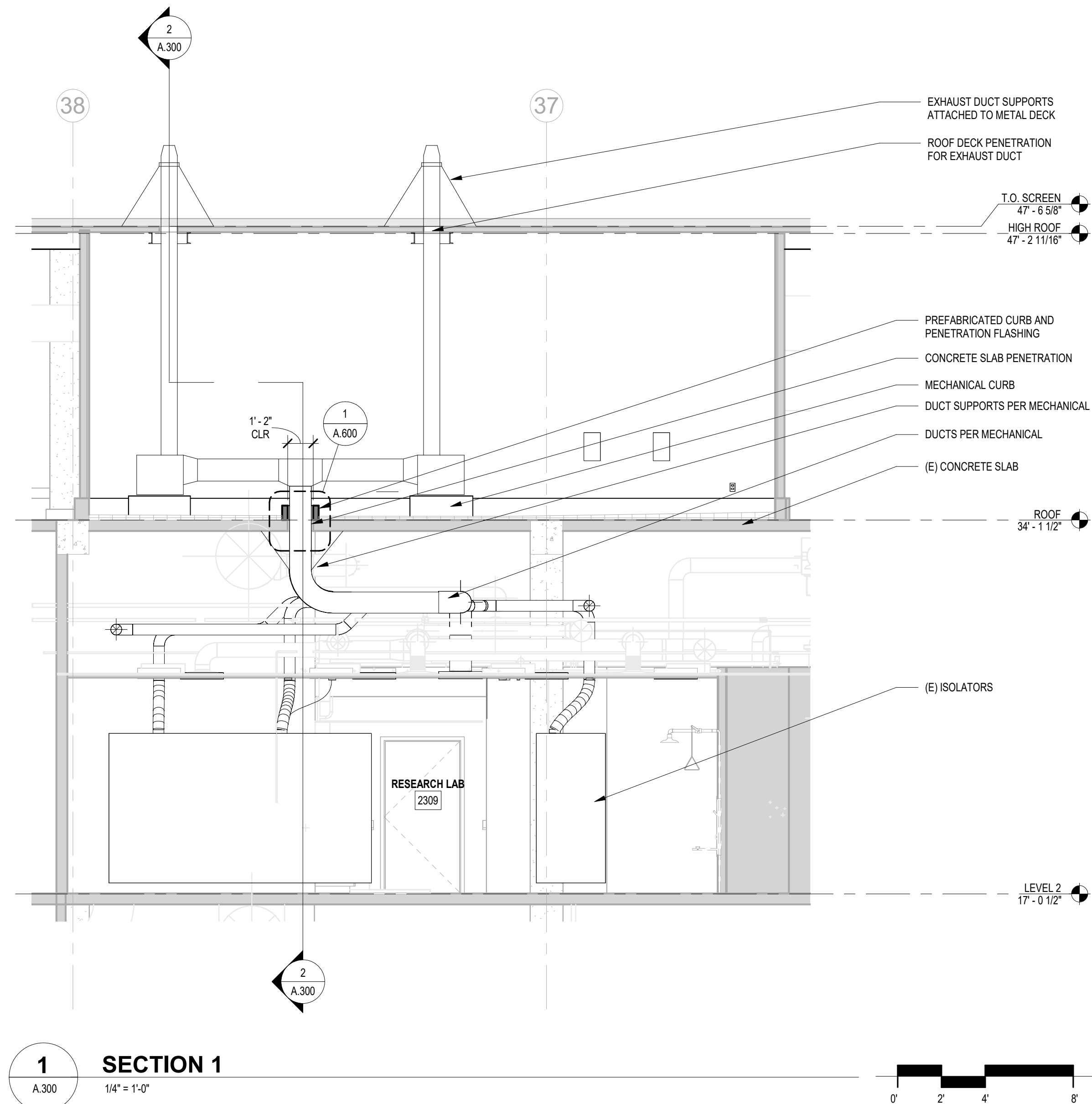
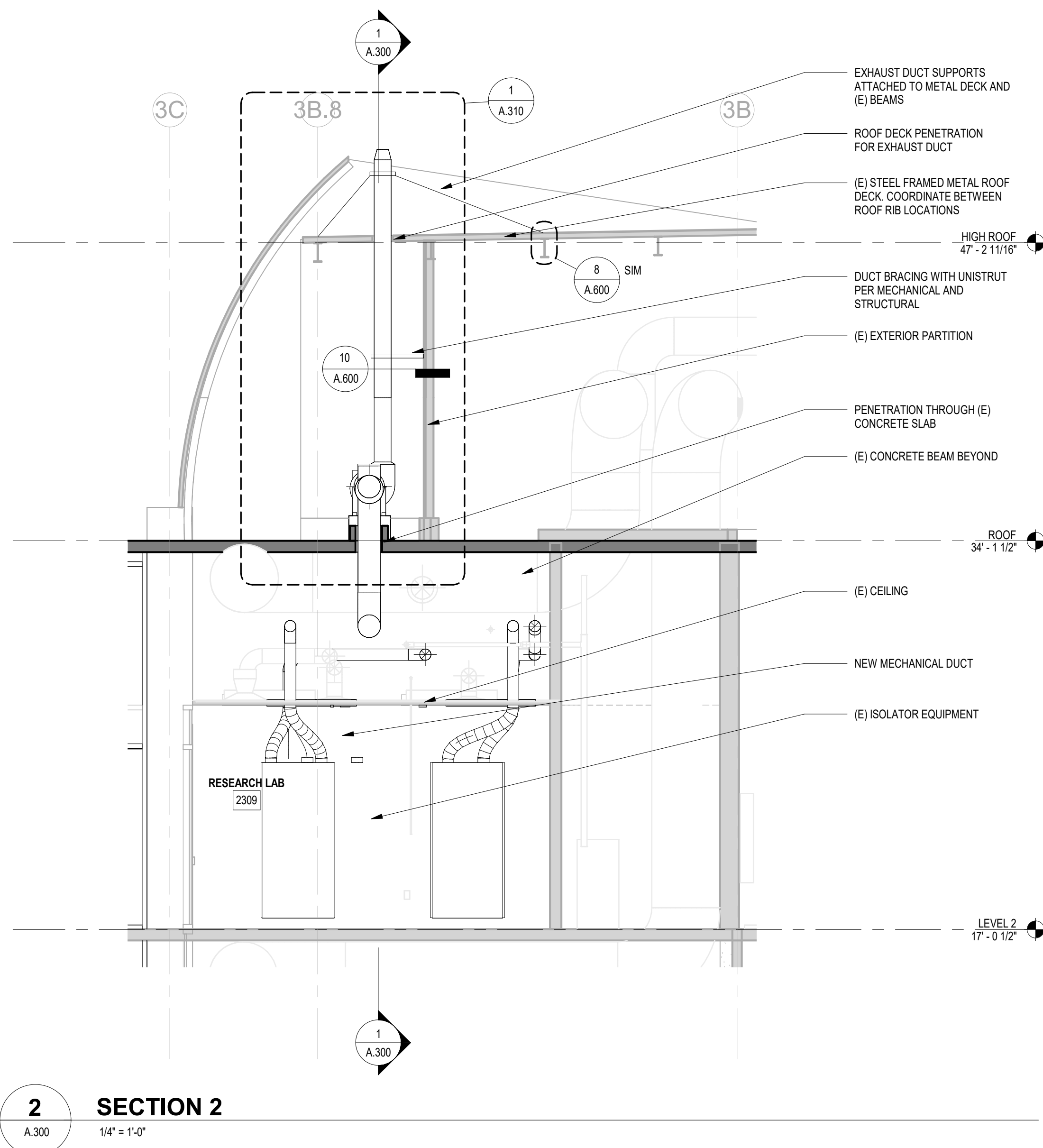
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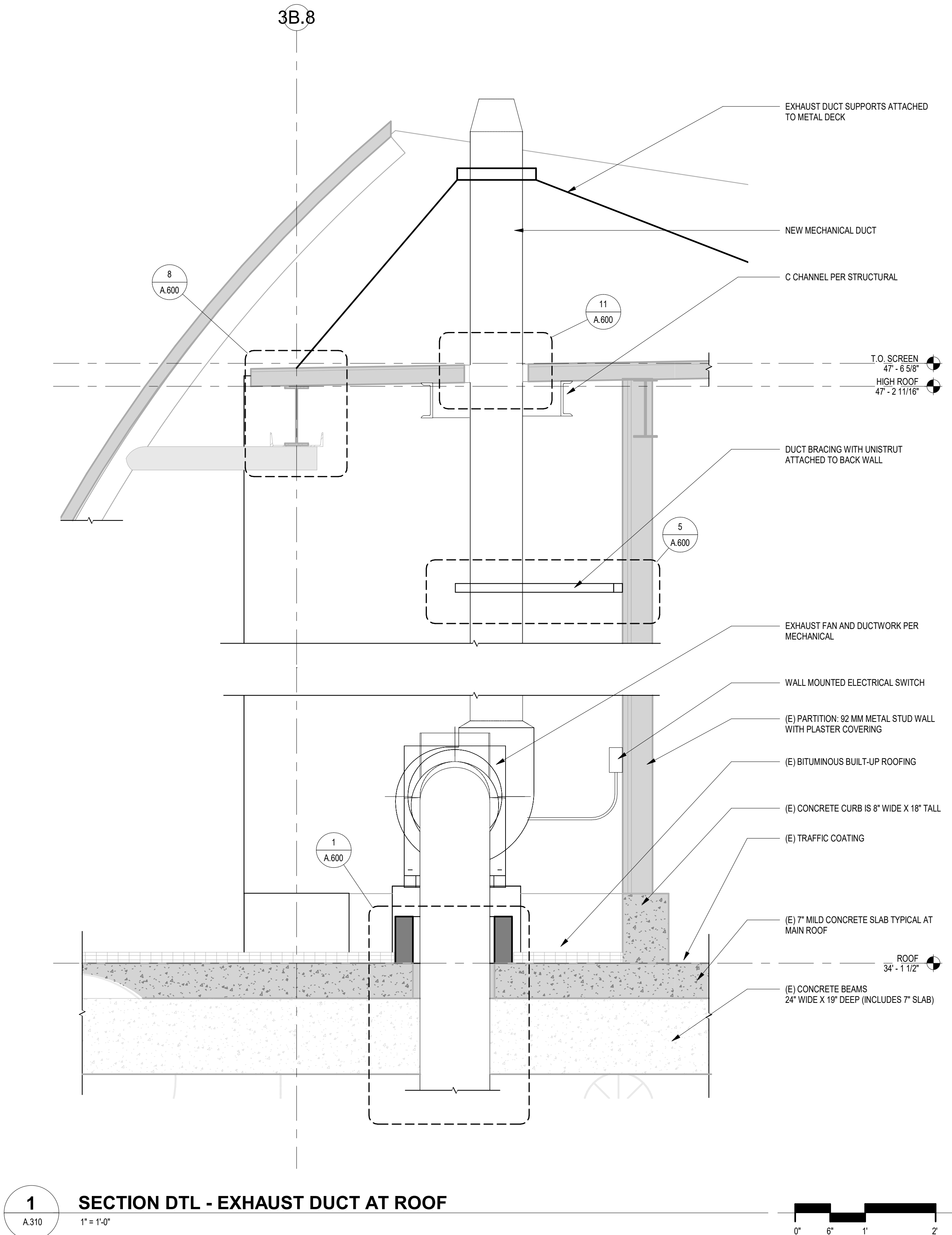
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BUILDING SECTIONS A.300

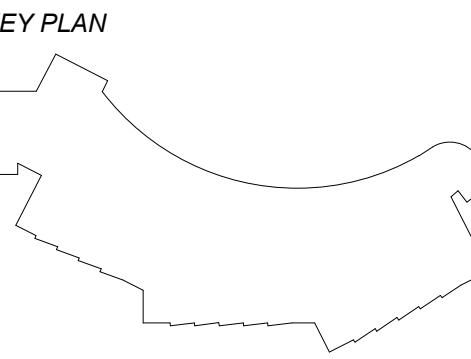


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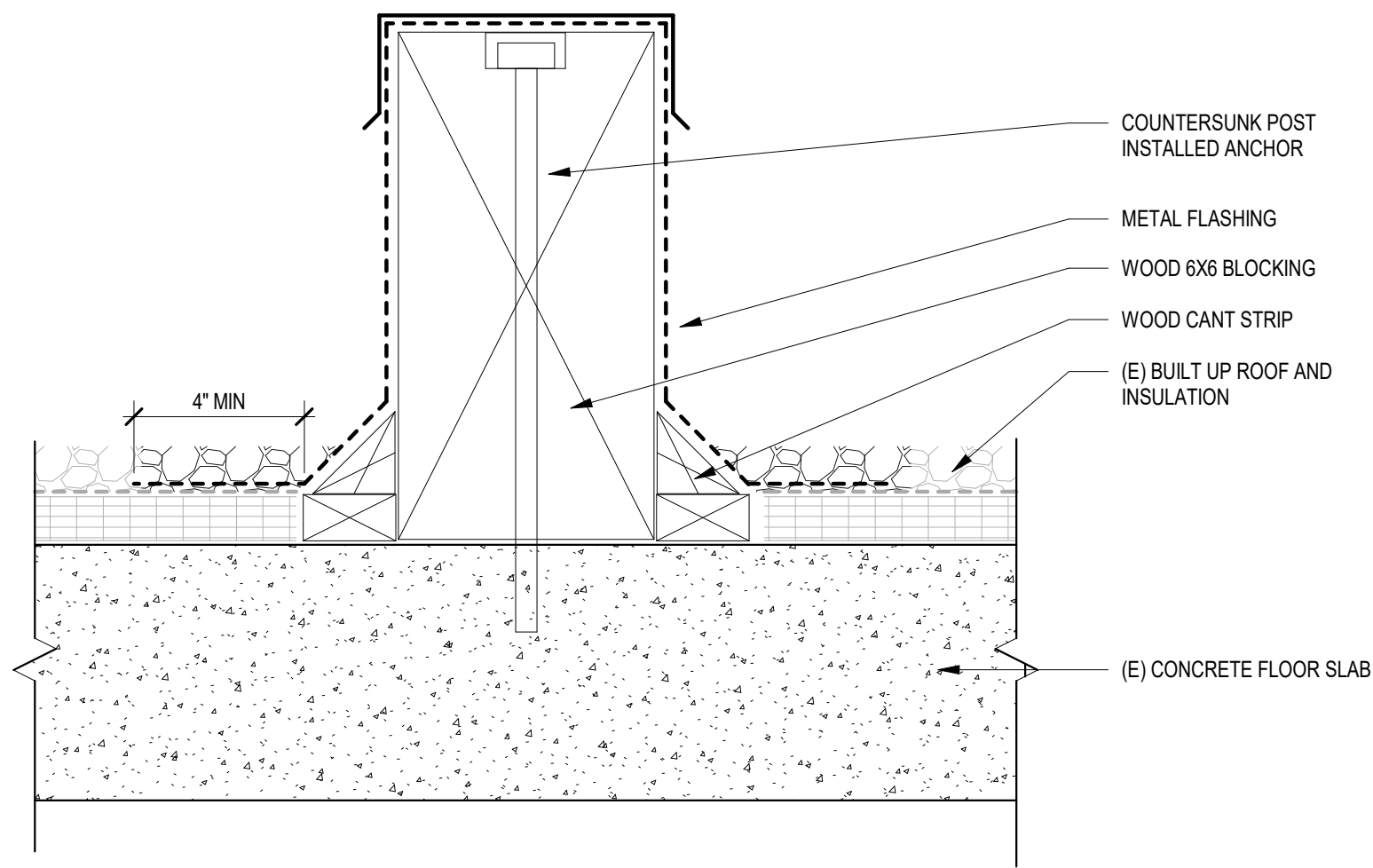
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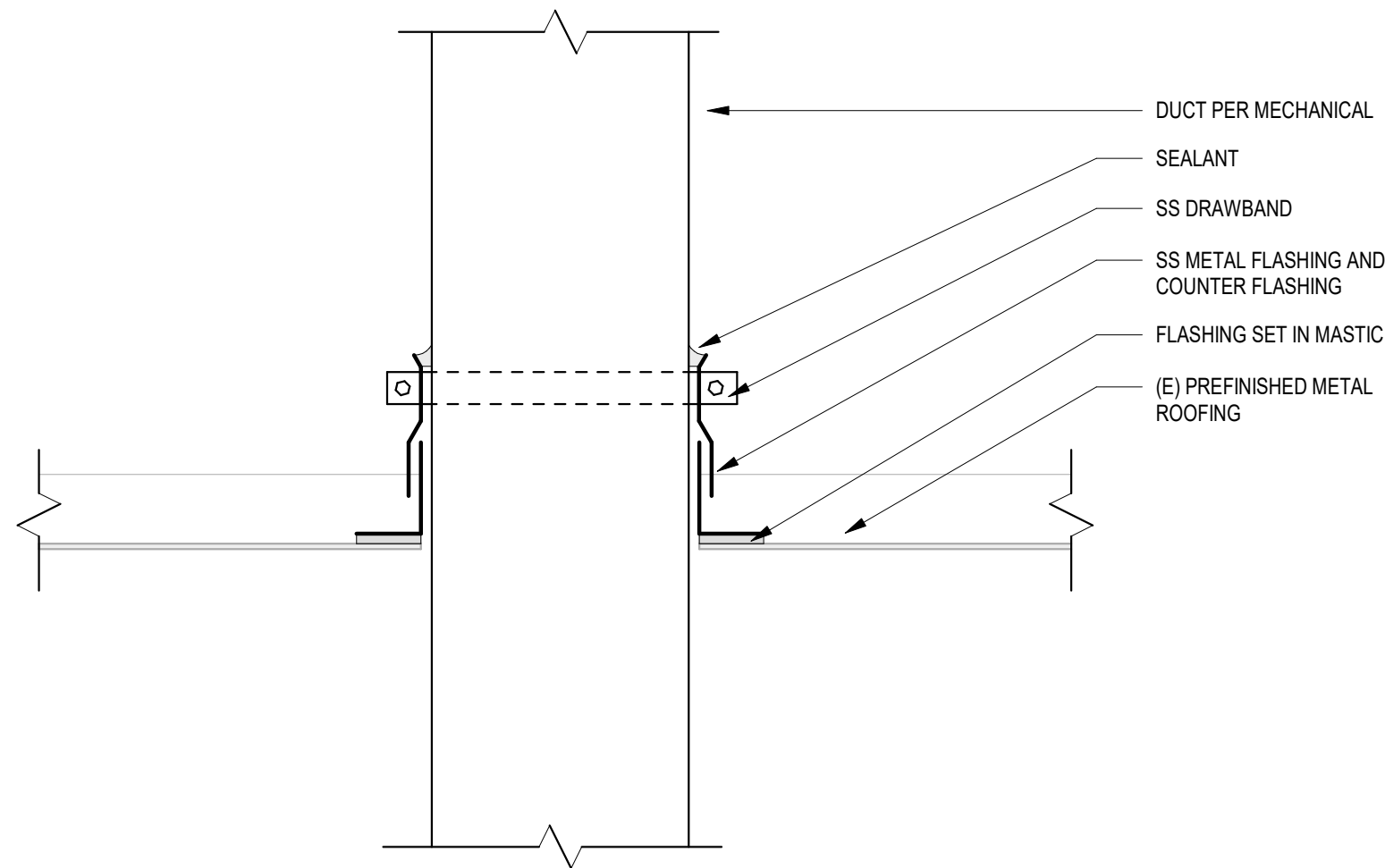
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Issue Date: 01/06/25

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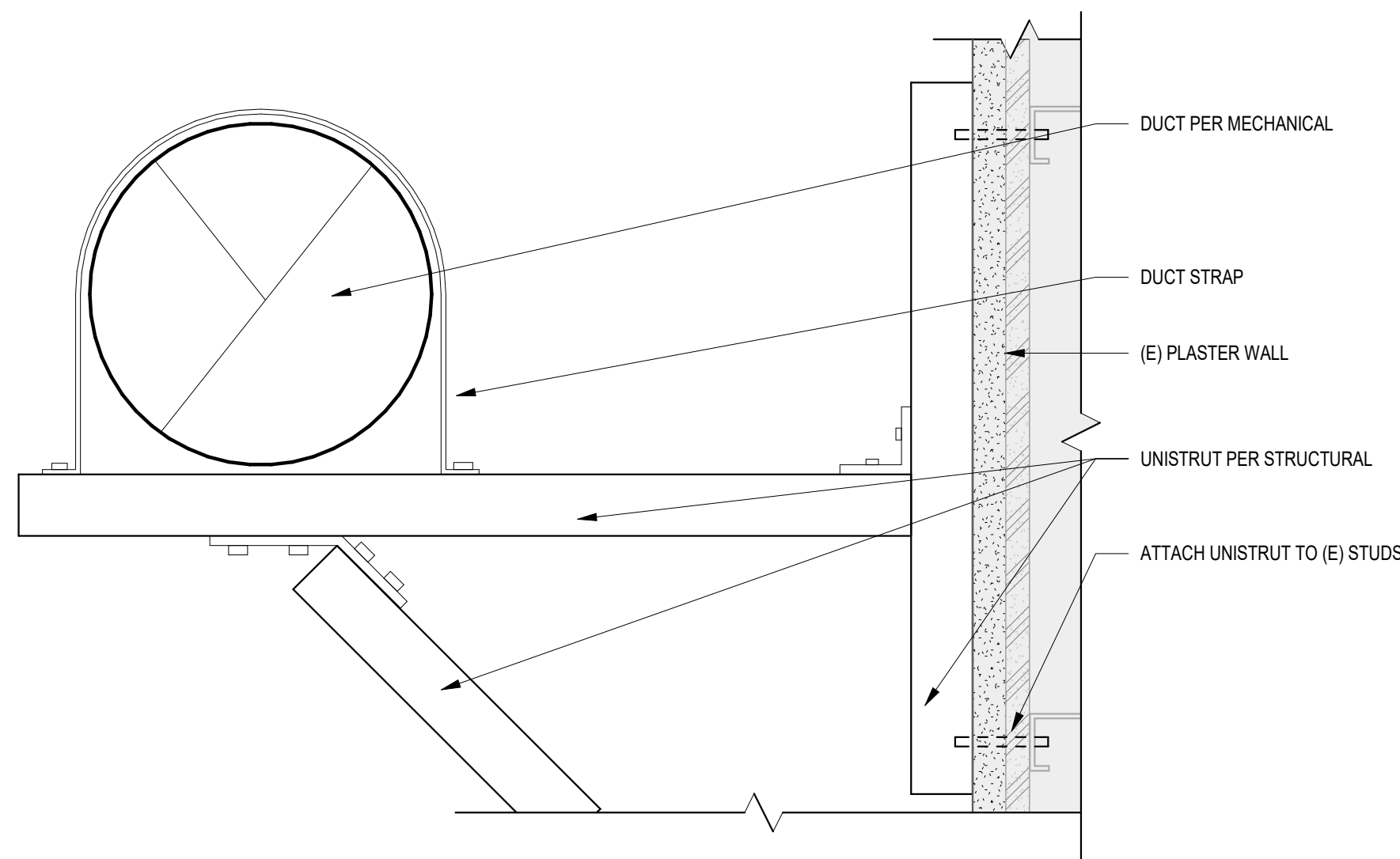
WALL SECTIONS A.310



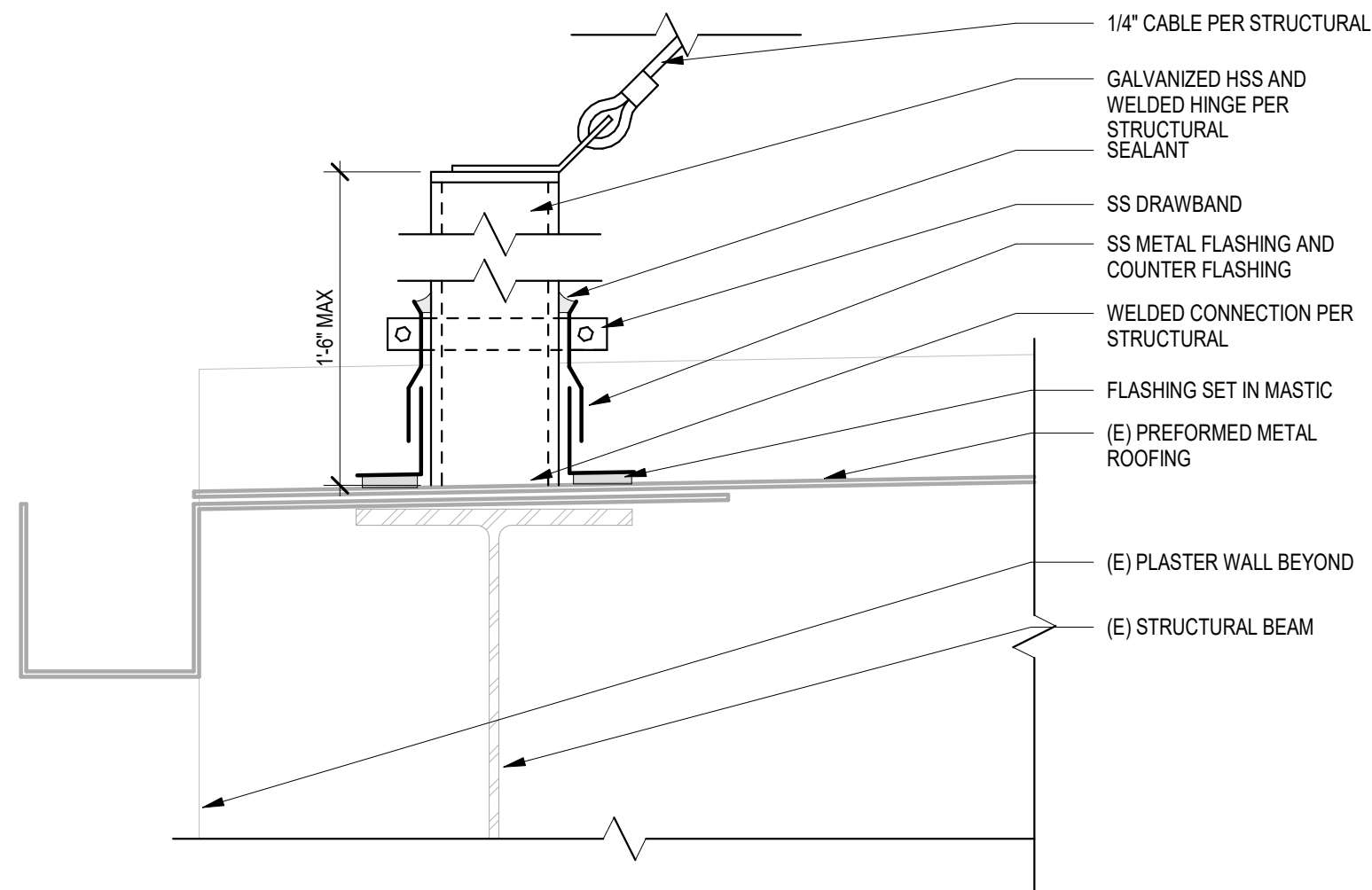
13 FAN CURB
A.600 3" = 1'-0"



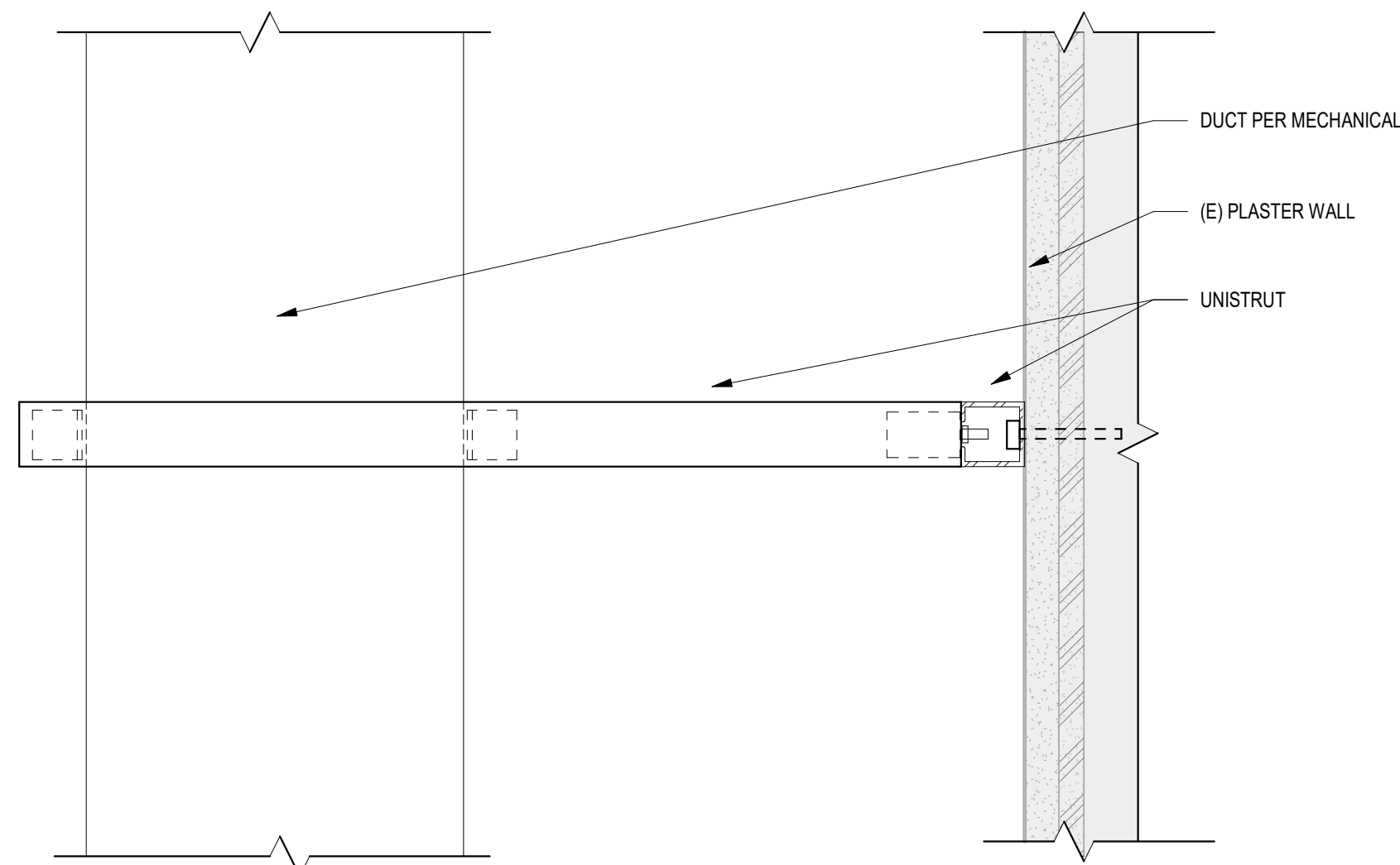
11 DUCT @ PREFORMED METAL
A.600 3" = 1'-0"



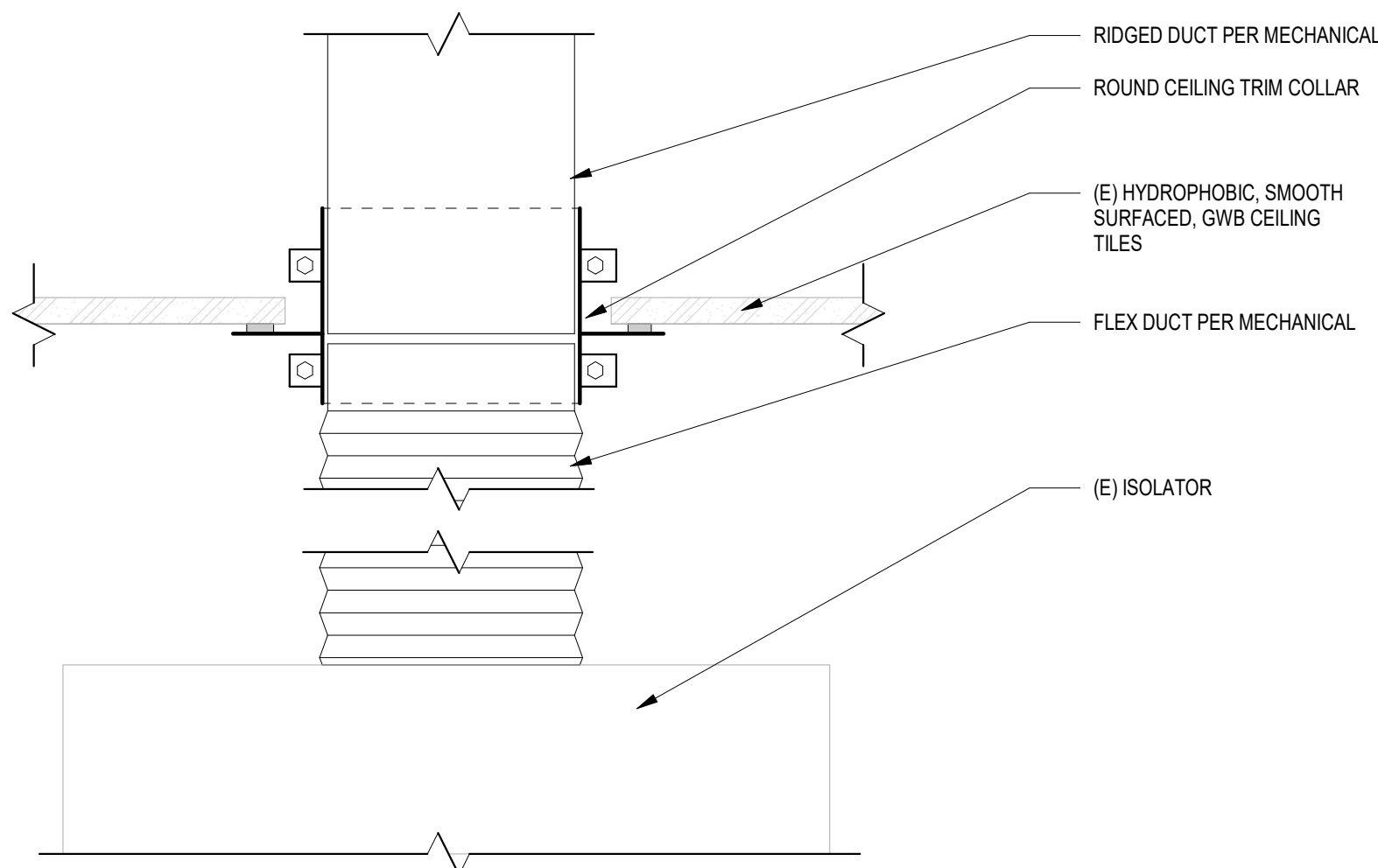
10 DUCT BRACING - PLAN
A.600 3" = 1'-0"



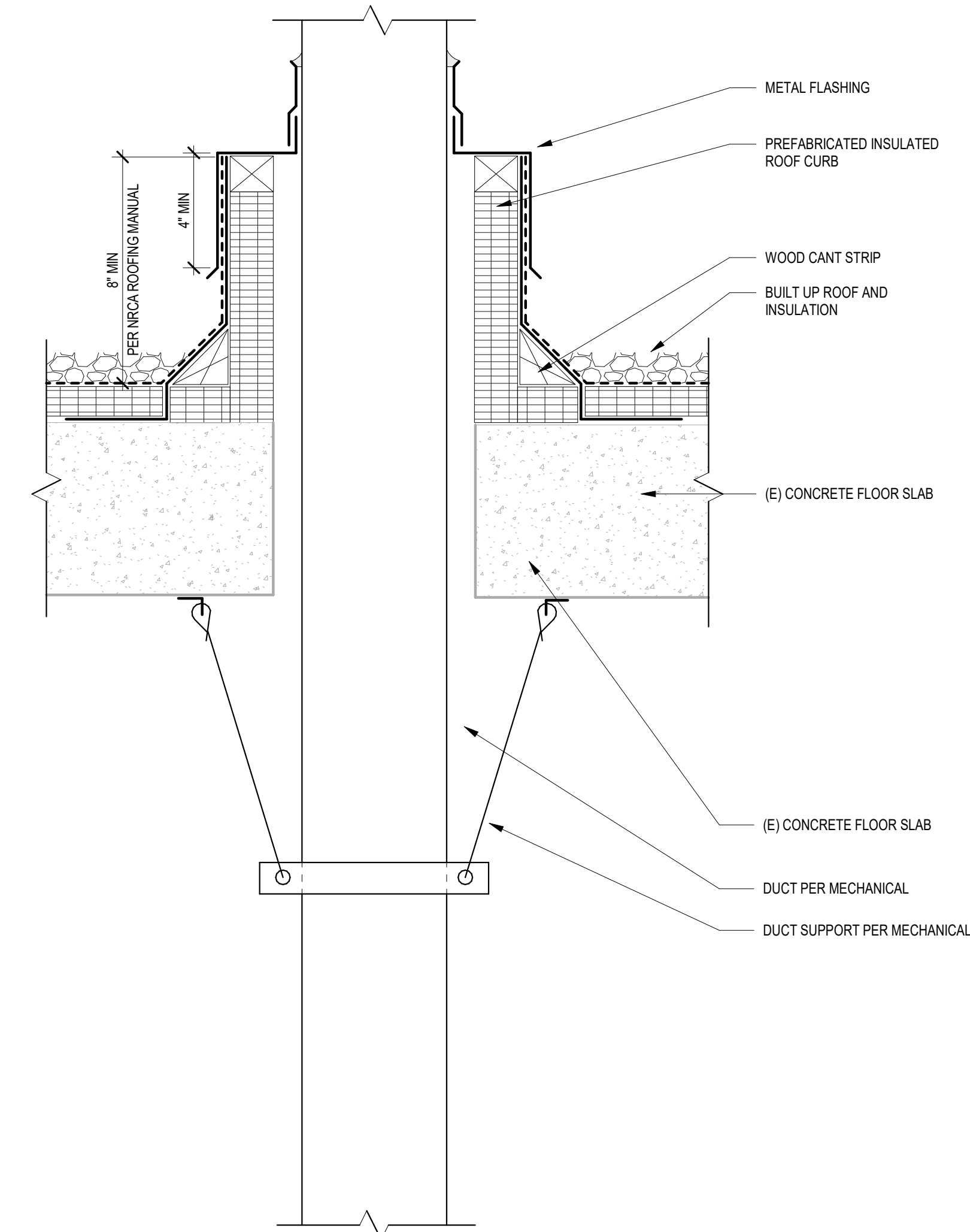
8 GUY WIRE CONNECTION
A.600 3" = 1'-0"



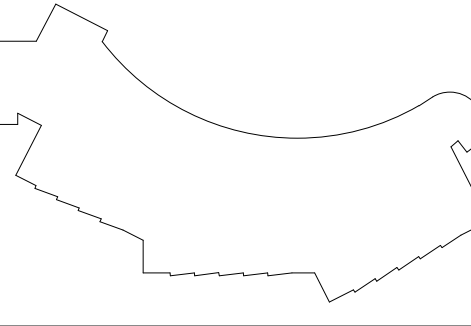
5 DUCT BRACING - SECTION
A.600 3" = 1'-0"



3 DUCT @ CEILING
A.600 3" = 1'-0"



1 CONCRETE SLAB PENETRATION
A.600 3" = 1'-0"



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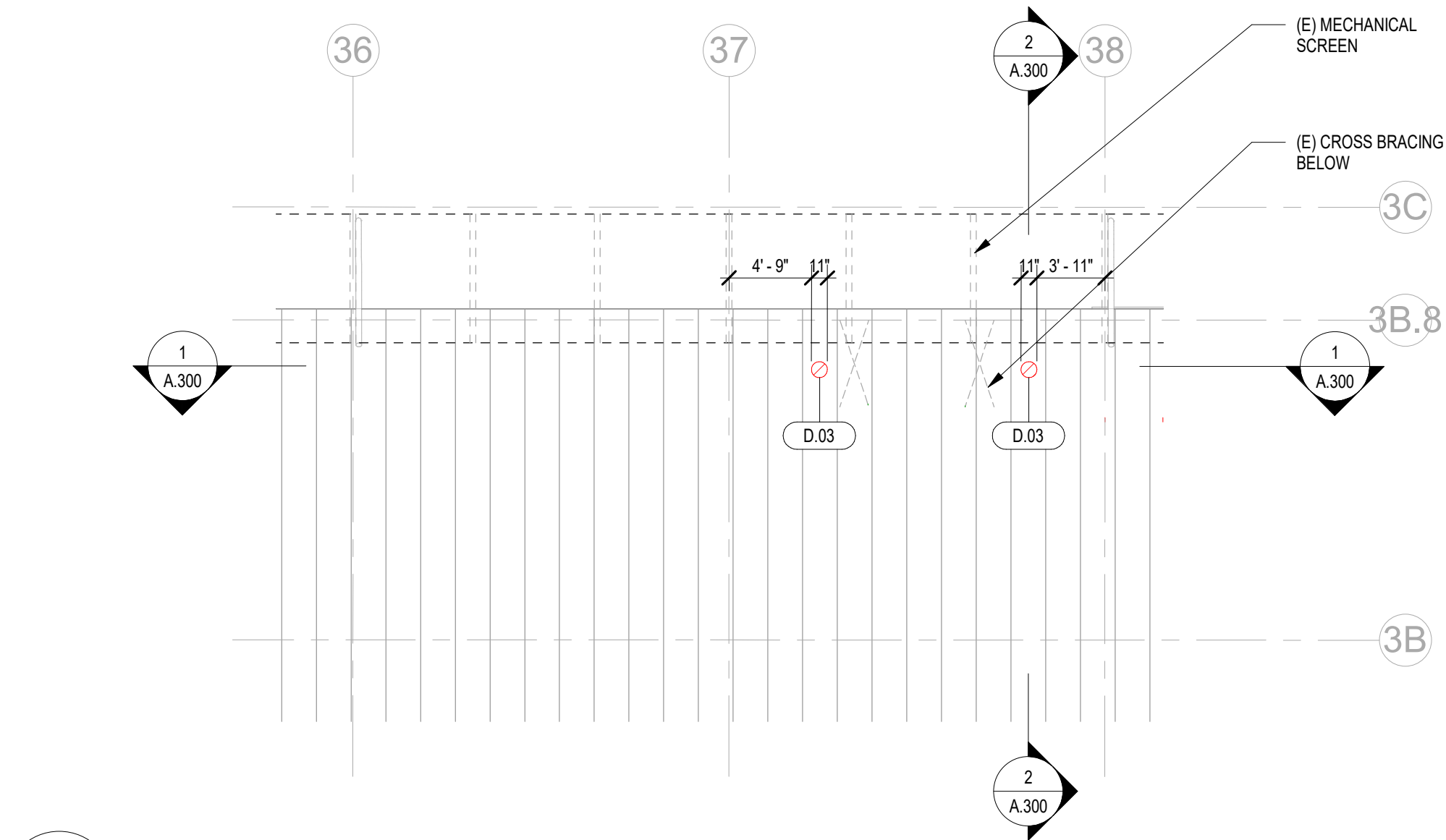
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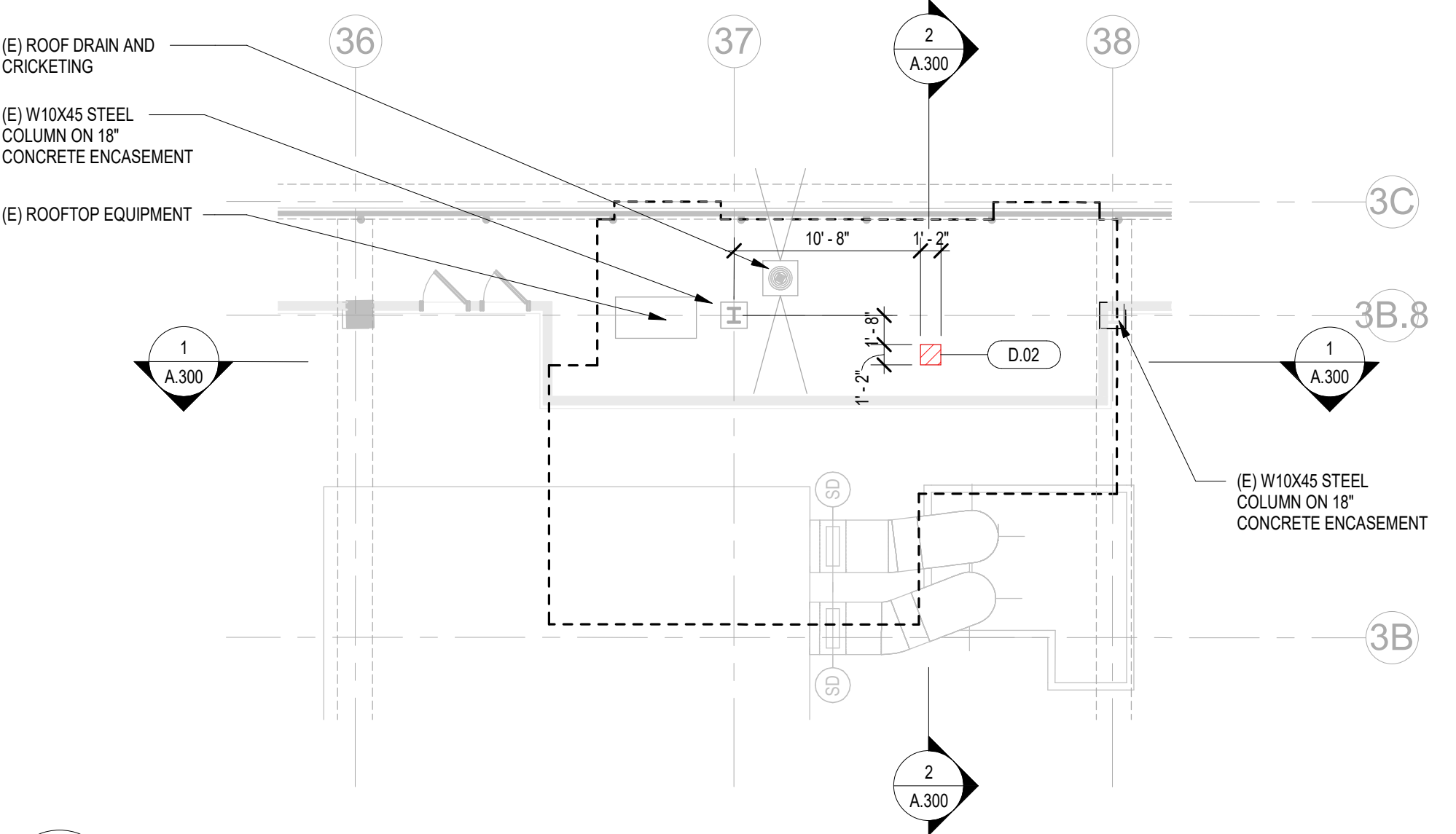
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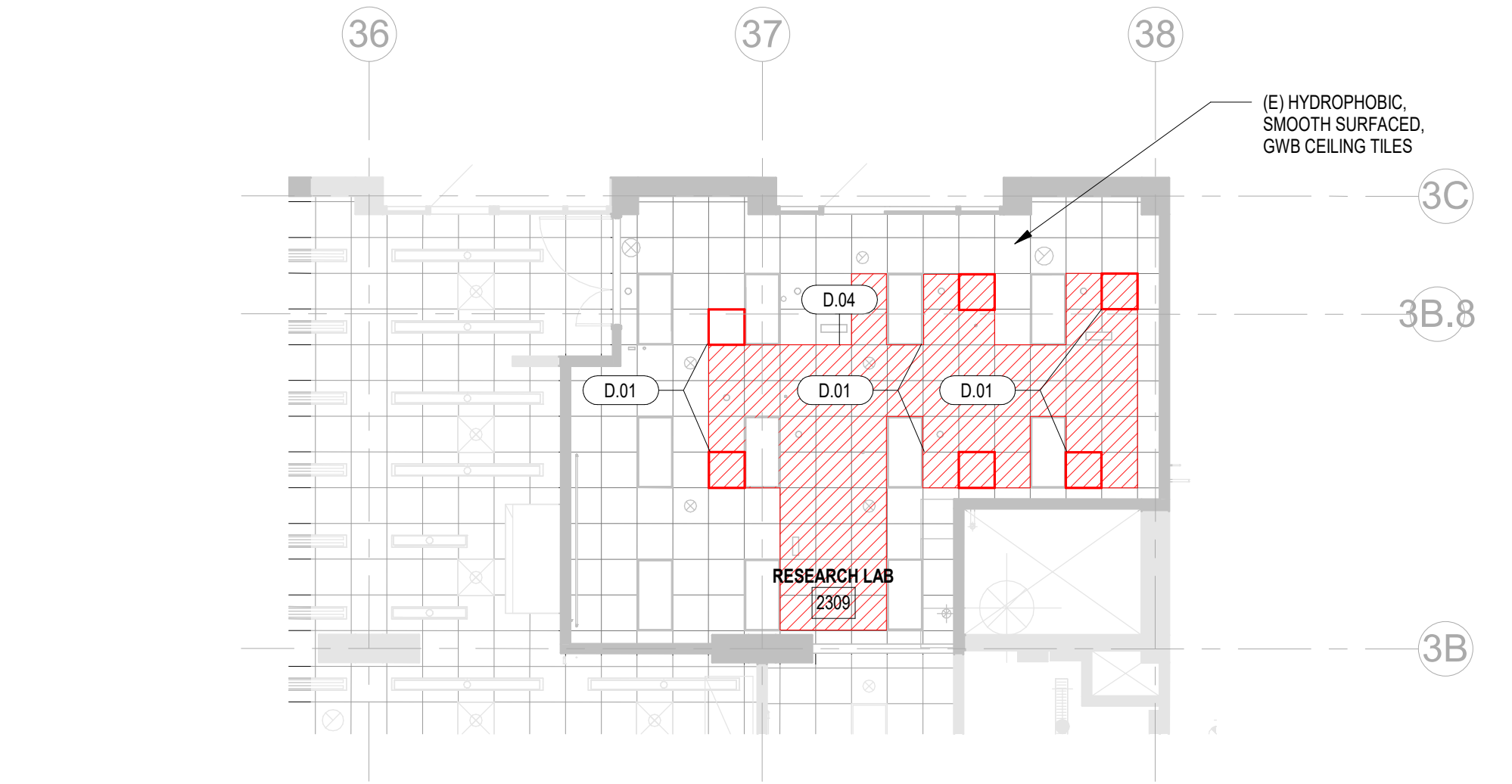
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1 HIGH ROOF - ABV RM 2309 - DEMO PLAN
AD.100 1/8" = 1'-0"



2 ROOF - ABV RM 2309 - DEMO PLAN
AD.100 1/8" = 1'-0"



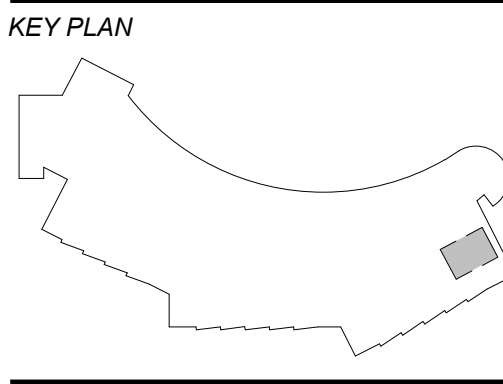
3 LEVEL 2 - RM 2309 - DEMO REFLECTED CEILING PLAN
AD.100 1/8" = 1'-0"

DEMOLITION KEYNOTES	
D.01	MODIFY CEILING TILE FOR NEW DUCTWORK PENETRATION
D.02	CREATE NEW OPENING IN EXISTING CONCRETE SLAB
D.03	CREATE NEW OPENING IN EXISTING METAL ROOF
D.04	REMOVE AND REPLACE CEILING TILES AND GRID AS REQUIRED TO PERFORM WORK SHOWN ON MECHANICAL DRAWINGS. REPLACE ANY DAMAGED TILES WITH NEW TO MATCH EXISTING IN SIZE, TYPE, AND COLOR.



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DEMOLITION PLANS AD.100

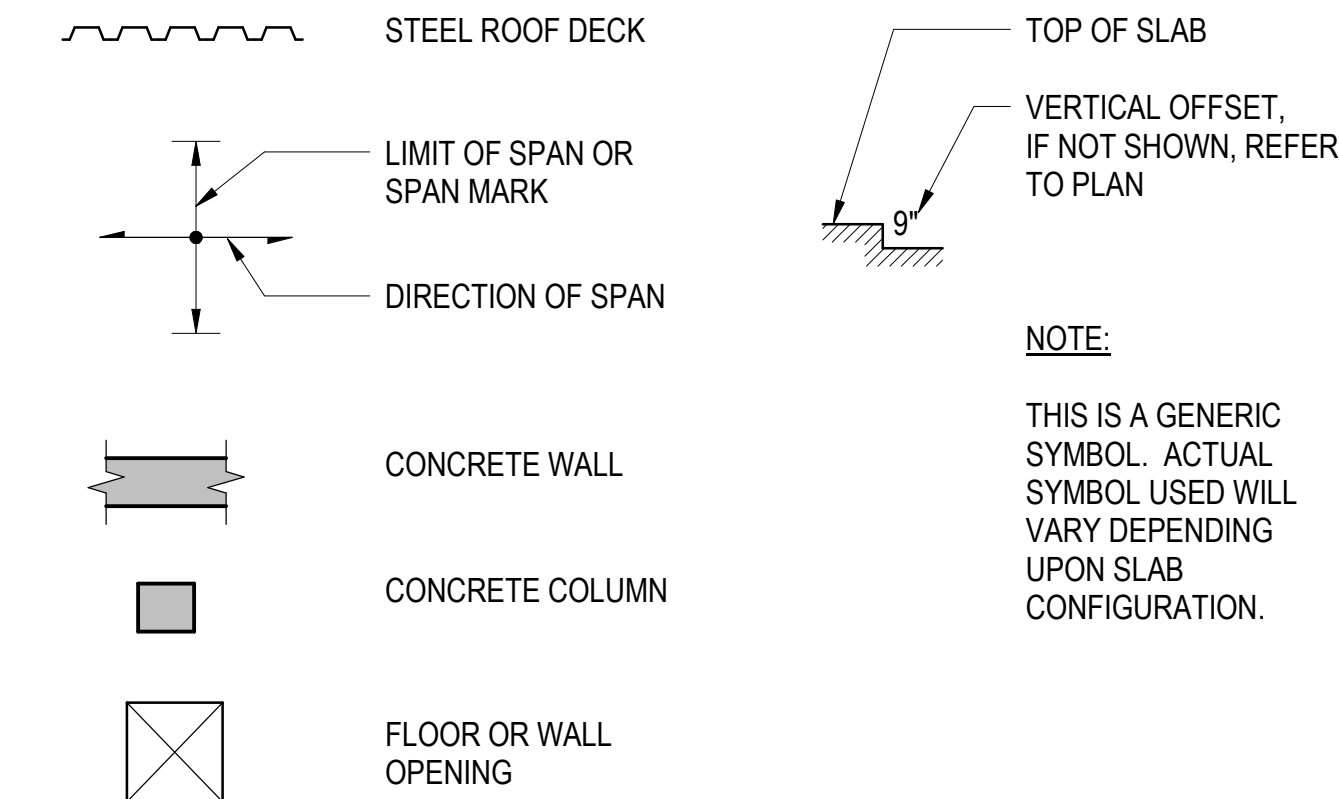


9 ABBREVIATIONS

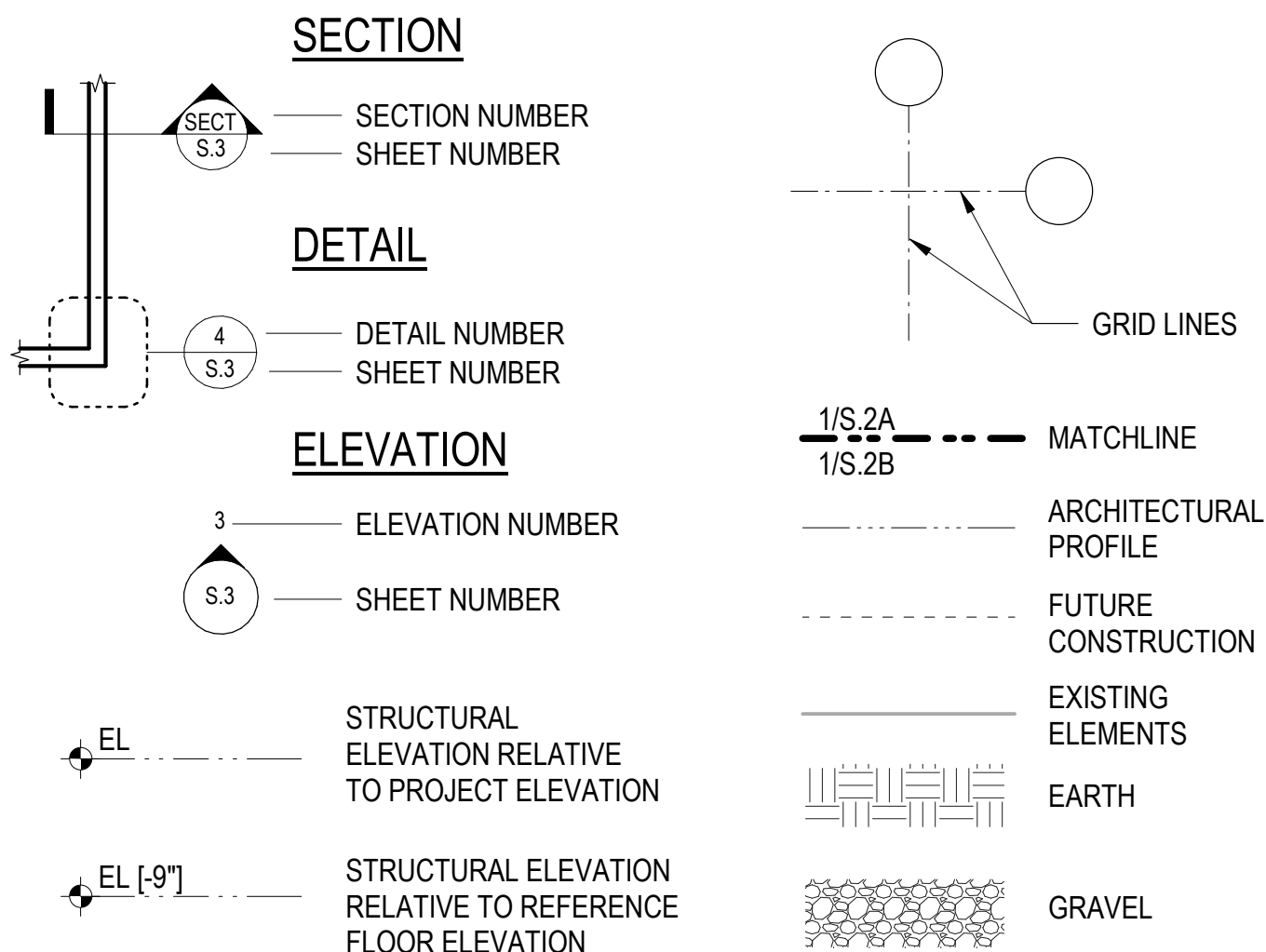
& ° ° DEG ø DIA #	AND AT DEGREE DIAMETER NUMBER, POUND	F FD FDN FF FG FIN FL FLG FOS FP FRMG FS FT FTG FV	DEGREES FAHRENHEIT FLOOR DRAIN FOUNDATION FAR FACE FRICTION GRIP BOLT FINISH FLOOR; FLOOR LINE FLANGE FACE OF STUD FIREPROOF; FULL PENETRATION FRAMING FULL SIZE; FAR SIDE FOOT; FEET; FINGER TIGHT FOOTING FIELD VERIFY
AB ACI ADDL ADJ AESS	ANCHOR BOLT AMERICAN CONCRETE INSTITUTE ADDITIONAL ADJACENT ARCHITECTURAL EXPOSED STRUCTURAL STEEL	GA GALV GB GFRG GL GR GRND	GAGE, GAUGE GALVANIZED GRADE BEAM GLASS FIBER REINFORCED CONCRETE GLUED LAMINATED (BEAM) GRADE GROUND
AGGR AISC	AGGREGATE AMERICAN INSTITUTE OF STEEL CONSTRUCTION	H HEF HGR HIF HOF HORIZ	HORIZONTAL HORIZONTAL EACH FACE HANGER HORIZONTAL INSIDE FACE HORIZONTAL OUTSIDE FACE HORIZONTAL
ALT ALUM ANSI	ALTERNATE ALUMINUM AMERICAN NATIONAL STANDARDS INSTITUTE	HP HS HSS HT	HP SHAPES; HIGH POINT HIGH STRENGTH HOLLOW STRUCTURAL SECTION HEIGHT
APA APPD APPROX AR ARCH ASSY ASTM AWS	AMERICAN PLYWOOD ASSOCIATION APPROVED APPROXIMATE ANCHOR RODS ARCHITECTURAL; ARCHITECT ASSEMBLY AMERICAN SOCIETY FOR TESTING AND MATERIALS AMERICAN WELDING SOCIETY	ICC ID IN INCL INFO INSUL INT	INTERNATIONAL CODE COUNCIL INSIDE DIAMETER INCH INCLUDE INFORMATION INSULATION INTERIOR
BAL BD BF BLDG BLK BM BMU BOS BOT BRB BRCG BRG BRKT BSMT BTWN BU	BALANCE BOARD BRACED FRAME BUILDING BLOCK; BLOCKING BEAM BRICK MASONRY UNIT BOTTOM OF STEEL; BOSOM (WELD) BOTTOM BUCKLING RESTRAINED BRACE BRACING BEARING BRACKET BASEMENT BETWEEN BUILT-UP	JST JT K KO KSI L LAB LB, # LF LIN LL LLBB LLH LLV LOC LONGIT LP LSL LSW LTWT LVL LWC	JOIST JOINT KIP (1,000 POUNDS) KNOCK-OUT KIPS PER SQUARE INCH ANGLE LABORATORY POUND LINEAL FOOT LINEAL; LINEAR LIVE LOAD LONG LEGS BACK-TO-BACK LONG LEG HORIZONTAL LONG LEG VERTICAL LOCATION; LOCATE LONGITUDINAL LOW POINT LONG SLOTTED (HOLES) LIGHT GAGE SHEAR WALL LIGHTWEIGHT LEVEL LIGHTWEIGHT CONCRETE
c C CANT CC CG CIP CJ CJP CL CLR CLT CMU COL COMP CONC CONFIG CONN CONST CONT CONTR COORD CORR CTR CTSK CU	CAMBER STANDARD CHANNEL CANTILEVER CENTER TO CENTER CENTER OF GRAVITY CAST-IN-PLACE CONSTRUCTION JOINT COMPLETE JOINT PENETRATION WELD CENTERLINE CLEARANCE; CLEAR CROSS LAMINATED TIMBER CONCRETE MASONRY UNIT COLUMN COMPRESSION CONCRETE CONFIGURATION CONNECTION; CONNECT CONSTRUCTION CONTINUE; CONTINUOUS CONTRACTOR COORDINATE; COORDINATION CORRUGATED CENTER COUNTERSINK; COUNTERSUNK CUBIC	MAS MATL MAX MB MC MECH MEMB MEP MEZZ MF MFB MFC MFR MFRG MIN MISC ML MO MS N N-S NF NIC NLT NS NTS NWC	MASONRY MATERIAL MAXIMUM MACHINE BOLT MISCELLANEOUS CHANNEL MECHANICAL MEMBRANE MECHANICAL/ ELECTRICAL / PLUMBING MEZZANINE MOMENT FRAME MOMENT FRAME BEAM MOMENT FRAME COLUMN MANUFACTURE; MANUFACTURER MANUFACTURING MINIMUM; MINUTE MISCELLANEOUS MATCH LINE MASONRY OPENING MECHANICAL SPLICE
d db DBA DBL DC DEG, ° DEMO DEPT DET DIA, ø DIAG DIAPHR DICA DIM DISC DL DLT DN DO DWG DWL	PENNY (NAIL) NOMINAL BAR DIAMETER (INCHES) DEFORMED BAR ANCHOR DOUBLE DEMAND CRITICAL WELD DEGREE DEMOLISH; DEMOLITION DEPARTMENT DETAIL DIAMETER DIAGONAL DIAPHRAGM DRILLED-IN CONCRETE ANCHOR DIMENSION DISCONTINUED; DISCONTINUOUS DEAD LOAD DOWEL LAMINATED TIMBER DOWN DO OVER; DITTO DRAWING DOWEL	OC OD OPNG OPP OPT OVS OWJ	ON CENTER OUTSIDE DIAMETER OPENING OPPOSITE (HAND) OPTION; OPTIONAL OVERSIZED (HOLES) OPEN WEB JOIST
(E) E E-W EA EF EJ EL ELEC ELEV EMBED ENGR EQ EQUIP ES ETC EW EXIST EXP EXT EXTD	EXISTING EAST EAST-WEST EACH EACH FACE EXPANSION JOINT ELEVATION ELECTRICAL ELEVATOR EMBEDDED ENGINEER EQUAL; EARTHQUAKE EQUIPMENT EACH SIDE ET CETERA EACH WAY EXISTING EXPANSION EXTERIOR EXTEND; EXTENDED		

P PC PCF PCP PEN PERP PH PJP PL PLC PLF PLYWD PREFAB PS PSF PSI PT PV PVC	PIPE PRECAST POUNDS PER CUBIC FOOT PRECAST CONCRETE PANEL PENETRATION PERPENDICULAR PENTHOUSE PARTIAL JOINT PENETRATION WELD PLATE PLACE POUNDS PER LINEAL FOOT PLYWOOD PREFABRICATED PRESTRESSED POUNDS PER SQUARE FOOT POUNDS PER SQUARE INCH POST-TENSIONED PHOTOVOLTAICS POLYVINYL CHLORIDE	R RB RC RCMD REF REINF REQD REQT	RADIUS RISER BAR REINFORCED CONCRETE RECOMMEND REFERENCE REINFORCE; REINFORCING; REINFORCEMENT REQUIRED REQUIREMENT
S1S S2S S4S S SB SC SCC	SURFACED ONE SIDE SURFACED TWO SIDES SURFACED FOUR SIDES AMERICAN STANDARD SHAPE; SOUTH SPACER BAR; SUPPORT BAR SLIP CRITICAL STRUCTURAL CONSULTANT TO THE CONTRACTOR	SCHED SDQ SECT SEOR SFRS SHT SHTG SIM SLBB SOG SOSD SP SPC SPCG SPEC SQ SSL STD STIFF STIRR STL STR STRUC SUPT SW SYM	SCHEDULE, SCHEDULED SPECIAL DUCTILE QUALITY SECTION STRUCTURAL ENGINEER OF RECORD SEISMIC FORCE RESISTING SYSTEM SHEET SHEATHING SIMILAR SHORT LEGS BACK-TO- BACK SLAB ON GRADE SLAB ON STEEL DECK SPIRAL SPACE SPACING SPECIFICATION SQUARE SHORT SLOTTED (HOLES) STANDARD STIFFENER STIRRUP STEEL STRAIGHT STRUCTURAL SUPPORT SHEAR WALL SYMMETRICAL
T&B T&G T&R TBD TEMP THK THRU TOC TOF TOS TOW TRANS TYP	TOP AND BOTTOM TONGUE AND GROOVE TREAD AND RISER TO BE DETERMINED TEMPERATURE; TEMPORARY THICK; THICKNESS THROUGH TOP OF CURB; TOP OF CONCRETE TOP OF FOOTING TOP OF STEEL TOP OF WALL TRANSVERSE TYPICAL	UB UBC UC UL UNO UT	UNIVERSAL BEAM UNIFORM BUILDING CODE UNIVERSAL COLUMN UNDERWRITERS' LABORATORY, INC. UNLESS NOTED OTHERWISE ULTRASONIC TEST
V, VERT VEF VG VIF VOF	VERTICAL VERTICAL EACH FACE VERTICAL GRAIN VERTICAL INSIDE FACE VERTICAL OUTSIDE FACE		
W W/ W/O WD WF WH WL WP WPJ WT WV WWR	WIDE FLANGE; WIDE; WEST WITH WITHOUT WOOD WIDE FLANGE WEEP HOLE WORK LINE WORK POINT, WEB PERPENDICULAR WEAKENED PLANE JOINT WEIGHT; STRUCTURAL TEE CUT FROM W SHAPE WEB VERTICAL WELDED WIRE REINFORCING		
YD	YARD		

11 CONCRETE SYMBOLS



12 MISCELLANEOUS SYMBOLS



4 DRAWING LIST

SECTION	PLAN VIEW	FRONT ELEVATION	BACK ELEVATION
WIDE FLANGE			
CHANNEL			
ANGLE			
DOUBLE ANGLE			
SQUARE OR RECTANGULAR HOLLOW SECTION			
CIRCULAR HOLLOW SECTION, PIPE			

COLUMN IN SECTION	H
CONCRETE ENCASED STEEL COLUMN SECTION	H
COLUMN STARTS HERE	H
COLUMN ENDS HERE	H
MOMENT CONNECTION	H
MOMENT FRAME CONNECTION	H
SHEAR CONNECTION	H
SPLICE	H
DIAGONAL BRACING ABOVE (DASHED LINE)	H
KICKER BRACING (x INDICATES HIGH POINT WHERE SPECIFIED)	x
HANGER	H
HANGER BELOW FLOOR	H
ELEVATOR RAIL SUPPORT POST SEE "TYPICAL ELEVATOR RAIL SUPPORT POST" SCHEDULE & DETAILS	RX

8 STEEL SYMBOLS

SECTION

SECTION NUMBER

SHEET NUMBER

DETAIL

DETAIL NUMBER

SHEET NUMBER

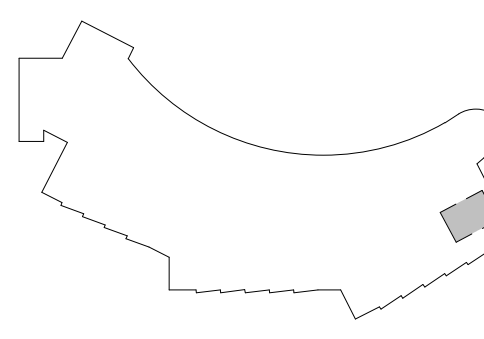
ELEVATION

ELEVATION NUMBER

SHEET NUMBER

STRUCTURAL ELEVATION RELATIVE TO PROJECT ELEVATION

STRUCTURAL ELEVATION RELATIVE TO REFERENCE FLOOR ELEVATION



FDA IRVINE RM 2309 EXHAUST DESIGN

19701 Fairchild Rd, Irvine, CA 92612

BID SET

AUGUST 15, 2025

No.	Description	Date
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Drawn: CK
Checked: GJR
M/H Proj No.: B2291.01

Issue Date: 06/20/25

GENERAL

ALL TYPICAL DETAILS AND NOTES SHOWN ON DRAWINGS SHALL APPLY UNLESS NOTED OTHERWISE. TYPICAL DETAILS MAY NOT NECESSARILY BE INDICATED ON THE PLANS BUT SHALL STILL APPLY AS SHOWN OR DESCRIBED IN THE DETAILS. WHERE TYPICAL DETAILS ARE NOTED ON THE DRAWINGS, THE SPECIFIED TYPICAL DETAIL SHALL BE USED. WHERE NO DETAIL IS NOTED, IT SHALL BE THE CONTRACTOR'S RESPONSIBILITY TO CHOOSE THE APPROPRIATE TYPICAL DETAIL FROM THOSE PROVIDED. THE CONTRACTOR SHALL SUBMIT ALL PROPOSED ALTERNATE TYPICAL DETAILS TO THOSE PROVIDED WITH RELATED CALCULATIONS TO THE ENGINEER FOR APPROVAL PRIOR TO SHOP DRAWING PRODUCTION AND FIELD USE.

BUILDING CODE

ALL CONSTRUCTION SHALL BE IN ACCORDANCE WITH THE BUILDING CODE. THE PUBLICATIONS LISTED BELOW ARE THE GOVERNING CODES AND STANDARDS AND ARE REFERENCED BY THEIR BASIC DESIGNATION. IN THE CASE OF CONFLICTING REQUIREMENTS, THE BUILDING CODE SHALL GOVERN.

APPLICABLE CODES AND STANDARDS

BUILDING CODE	CALIFORNIA BUILDING CODE (CBC), 2022 EDITION
ACI 318	AMERICAN CONCRETE INSTITUTE, "BUILDING CODE REQUIREMENTS FOR STRUCTURAL CONCRETE," 2019 EDITION
RCSC	RESEARCH COUNCIL ON STRUCTURAL CONNECTIONS, "SPECIFICATION FOR STRUCTURAL JOINTS USING HIGH-STRENGTH BOLTS," 2014 EDITION
AISC 360	AMERICAN INSTITUTE OF STEEL CONSTRUCTION, "SPECIFICATION FOR STRUCTURAL STEEL BUILDINGS," 2016 EDITION
ASCE 7	AMERICAN SOCIETY OF CIVIL ENGINEERS, "MINIMUM DESIGN LOADS AND ASSOCIATED CRITERIA FOR BUILDINGS AND OTHER STRUCTURES," 2016 EDITION, INCLUDING SUPPLEMENTS NO. 1, 2, AND 3
ASTM	AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM INTERNATIONAL)
AWS A2.4	AMERICAN WELDING SOCIETY, "STANDARD SYMBOLS FOR WELDING, BRAZING, AND NONDESTRUCTIVE EVALUATION," 2020 EDITION
AWS D1.1	AMERICAN WELDING SOCIETY, "STRUCTURAL WELDING CODE - STEEL," 2015 EDITION
AWS D1.3	AMERICAN WELDING SOCIETY, "STRUCTURAL WELDING CODE - SHEET STEEL," 2008 EDITION
AWS D1.4	AMERICAN WELDING SOCIETY, "STRUCTURAL WELDING CODE - REINFORCING STEEL INCLUDING METAL INSERTS AND CONNECTIONS IN REINFORCED CONCRETE CONSTRUCTION," 2018 EDITION
ICC	INTERNATIONAL CODE COUNCIL, INTERNATIONAL CODE COUNCIL - EVALUATION SERVICES (ICC-ES)
IAPMO	INTERNATIONAL ASSOCIATION OF PLUMBING AND MECHANICAL OFFICIALS – UNIFORM EVALUATION SERVICE (IAPMO-UES)

ANCHORAGE TO HARDENED CONCRETE

ANCHORAGE TO HARDENED CONCRETE SHALL INCLUDE MECHANICAL AND ADHESIVE ANCHORS OF SIZE, NUMBER, AND SPACING AS SHOWN ON THE DRAWINGS. HOLES SHALL BE DRILLED AND CLEANED AND ANCHORS SHALL BE INSTALLED IN STRICT ACCORDANCE WITH THE MANUFACTURER'S PUBLISHED INSTRUCTIONS AND AN APPROVED ICC–ES (OR IAPMO-UES EQUIVALENT) REPORT. INSPECTION AND TESTING SHALL BE PROVIDED IN ACCORDANCE WITH THE GENERAL NOTES AND THE APPROVED ICC–ES (OR IAPMO-UES EQUIVALENT) REPORT.

WHERE THE ANCHOR TYPE IS SPECIFIED ON THE DRAWINGS, SUBSTITUTION FOR A DIFFERENT TYPE OF ANCHORAGE (INCLUDING SUBSTITUTING FOR CAST-IN-PLACE ANCHORAGE) SHALL NOT BE PERMITTED WITHOUT PRIOR CONSENT OF THE ENGINEER.

ACCEPTABLE ANCHORS SHALL HAVE A CURRENT ICC–ES (OR IAPMO-UES EQUIVALENT) REPORT INDICATING THAT THE ANCHOR IS PERMITTED FOR RESISTING SEISMIC LOADS IN CRACKED CONCRETE. UNLESS NOTED OTHERWISE, ANCHORS SHALL BE ASTM A36 THREADED ROD OR ASTM A615, GRADE 60 REINFORCING STEEL DOWELS.

WHEN EMBEDMENT IS NOTED ON THE DRAWINGS, THE ANCHOR EFFECTIVE EMBEDMENT DEPTH SHALL EQUAL OR EXCEED THE NOTED EMBEDMENT DEPTH. WHERE NO EMBEDMENT IS NOTED ON THE DRAWINGS, THE MINIMUM EFFECTIVE ANCHOR EMBEDMENT DEPTH SHALL BE 6.5 ANCHOR DIAMETERS, MINIMUM DISTANCE TO THE NEAREST CONCRETE EDGE SHALL BE 12 ANCHOR DIAMETERS, AND MINIMUM ANCHOR SPACING SHALL BE 8 ANCHOR DIAMETERS.

HOLES SHALL BE DRILLED WITH ROTARY IMPACT HAMMER OR EQUIVALENT METHOD TO PRODUCE A HOLE WITH A ROUGH INSIDE SURFACE. CORE DRILLING HOLES IS NOT PERMITTED. NO STEEL REINFORCEMENT SHALL BE CUT TO INSTALL ANCHORS. THE ADHESIVE SHALL BE MIXED, APPLIED, AND CURED IN STRICT ACCORDANCE WITH THE MANUFACTURER'S PUBLISHED INSTALLATION INSTRUCTIONS IN THE ICC–ES (OR IAPMO-UES EQUIVALENT) REPORT. ALL PLACEMENT AND CURING SHALL BE CONDUCTED WITH CONCRETE AND AIR TEMPERATURES ABOVE 50 DEGREES FAHRENHEIT. ADHESIVE SHALL BE APPLIED ONLY TO CLEAN, DRY CONCRETE. POSITIVE PROTECTION SHALL BE PROVIDED SO THAT ANCHORS ARE NOT DISTURBED DURING THE CURING PERIOD. STAINLESS STEEL ANCHORS SHALL BE USED AT ALL EXTERIOR LOCATIONS AND WHERE SPECIFICALLY INDICATED ON THE DRAWINGS. DEFECTIVE OR ABANDONED HOLES SHALL BE FILLED WITH NON-SHRINK GROUT OR AN INJECTABLE ADHESIVE MATCHING THE ADJACENT CONCRETE COMPRESSIVE STRENGTH. NOTIFY THE STRUCTURAL ENGINEER OF DEFECTIVE OR ABANDONED HOLES IN WALLS AND COLUMNS. THESE ELEMENTS MAY REQUIRE NON-SHRINK GROUT WITH A COMPRESSIVE MODULUS OF ELASTICITY MATCHING THAT OF THE ADJACENT CONCRETE.

STRUCTURAL STEEL

ALL STEEL SHALL CONFORM TO THE FOLLOWING:

W-SHAPES	ASTM A992, Fy=50 KSI ASTM A913, Fy=50 KSI
W-SHAPES CALLED OUT ON PLANS AS (Fy=65 KSI)	ASTM A913, Fy=65 KSI
ALL ANGLES AND CHANNELS UNLESS NOTED OTHERWISE	ASTM A36, Fy=36 KSI
SQUARE OR RECTANGULAR STRUCTURAL TUBE (HSS)	ASTM A500, GRADE C, Fy=50 KSI
ROUND STRUCTURAL TUBE (HSS)	ASTM A500, GRADE C, Fy=46 KSI
STEEL PIPE DIAMETER LESS THAN OR EQUAL TO 12 INCHES	ASTM A53, TYPE E OR S, GRADE B, Fy=35 KSI
MATERIAL CALLED OUT ON PLANS AS (Fy=36 KSI)	ASTM A36, Fy=36 KSI
MATERIAL CALLED OUT ON PLANS AS (Fy=65 KSI)	ASTM A572, Fy=65 KSI
ALL OTHER STEEL UNLESS NOTED OTHERWISE	ASTM A572, Fy=50 KSI ASTM A588, Fy=50 KSI

GENERAL NOTES FOR STEEL CONNECTIONS SHALL APPLY TO ALL STEEL CONNECTIONS UNLESS NOTED OTHERWISE.

ALL WORK SHALL BE IN ACCORDANCE WITH THE AISC SPECIFICATION. SHOP DRAWINGS SHALL BE SUBMITTED AND REVIEWED BY THE ARCHITECT/ENGINEER BEFORE COMMENCING FABRICATION. ALL STEEL ANCHORS AND TIES AND OTHER MEMBERS EMBEDDED IN CONCRETE OR MASONRY SHALL BE LEFT UNPAINTED. DIMENSIONAL TOLERANCE FOR BUILT-UP MEMBERS SHALL BE PER AWS D1.1.

STEEL BEAMS ARE EQUALLY SPACED BETWEEN DIMENSION POINTS AT THE MAXIMUM DECK SPAN LOCATION, UNLESS NOTED OTHERWISE. STEEL BEAMS ARE ORIENTED WITH THE BEAM WEB VERTICAL, UNLESS NOTED OTHERWISE. HSS ARE ORIENTED WITH THE WIDE FACE VERTICAL, UNLESS NOTED OTHERWISE. MINIMUM CONNECTIONS SHALL BE A TWO-BOLT CONNECTION USING HIGH-STRENGTH BOLTS IN SINGLE SHEAR.

THE CONTRACTOR SHALL BE RESPONSIBLE FOR COORDINATING THE SELECTION OF OPTIONAL DETAILS SHOWN ON THE DRAWINGS.

THE CONTRACTOR SHALL BE RESPONSIBLE FOR ALL ERECTION AIDS THAT INCLUDE, BUT ARE NOT LIMITED TO, ERECTION ANGLES, LIFT HOLES, AND OTHER AIDS.

STRUCTURAL STEEL BOLTING

ALL BOLTS 7/8-INCH DIAMETER AND SMALLER SHALL BE GROUP A HIGH-STRENGTH BOLTS AS DEFINED BY AISC, UNLESS NOTED OTHERWISE.

ALL BOLTS 1-INCH DIAMETER AND LARGER SHALL BE GROUP B HIGH-STRENGTH BOLTS AS DEFINED BY AISC, UNLESS NOTED OTHERWISE.

EITHER HEAVY HEX OR TENSION-CONTROL BOLTS MAY BE USED AT THE CONTRACTOR'S OPTION. ALL HIGH-STRENGTH BOLTS SHALL BE INSTALLED, TIGHTENED, AND INSPECTED IN ACCORDANCE WITH THE RCSC.

BOLTS DESIGNATED "A307" IN THE DRAWINGS REFER TO ASTM A307 BOLTS. INSTALL A307 BOLTS TO SNUG-TIGHT CONDITION UNLESS NOTED OTHERWISE AND PROVIDE LOCK WASHERS UNDER NUTS OR SELF-LOCKING NUTS. WHERE A307 BOLTS OCCUR IN SLOTTED HOLES, PROVIDE WASHERS TO COMPLETELY COVER THE HOLES.

BOLTS DESIGNATED "THRU" IN THE DRAWINGS REFER TO ASTM A307 BOLTS, UNLESS NOTED OTHERWISE. ALL "THRU" BOLTS ARE INSTALLED TO FINGER-TIGHT REQUIREMENTS UNLESS NOTED OTHERWISE.

HIGH-STRENGTH BOLT THREADS ARE ALLOWED IN THE SHEAR PLANE EXCEPT WHERE DESIGNATED "X" IN THE DRAWINGS.

ALL HIGH-STRENGTH BOLTED CONNECTIONS SHALL BE PRETENSIONED, UNLESS NOTED OTHERWISE. PRETENSIONED AND SLIP-CRITICAL CONNECTIONS MAY USE TURN-OF-NUT PRETENSIONING, TWIST-OFF-TYPE TENSION-CONTROL BOLT PRETENSIONING, OR DIRECT-TENSION-INDICATOR PRETENSIONING.

BOLTS IN CONNECTIONS OF BEAM-TO-BEAM/GIRDER MAY BE SNUG-TIGHT, UNLESS SPECIFICALLY CALLED OUT PRETENSIONED, SLIP-CRITICAL, OR FINGER-TIGHT. SNUG-TIGHT CONNECTIONS SHALL BE INSTALLED PER THE CRITERIA FOR SNUG-TIGHT BOLTS PER THE RCSC.

CONNECTIONS (OR JOINTS) DESIGNATED "SC" IN THE DRAWINGS SHALL HAVE BOLTS INSTALLED TO SLIP-CRITICAL REQUIREMENTS WITH CLASS A FAYING SURFACES. CONNECTIONS DESIGNATED "SCB" IN THE DRAWINGS SHALL HAVE BOLTS INSTALLED TO SLIP-CRITICAL REQUIREMENTS WITH CLASS B FAYING SURFACES. WHERE SHIMS OR FILLERS ARE REQUIRED TO MAKE A SLIP-CRITICAL CONNECTION, A SINGLE PLY FILLER SHALL BE USED BETWEEN CONNECTING PARTS, UNLESS NOTED OTHERWISE. ALL SURFACES OF FILLERS IN SLIP-CRITICAL JOINTS SHALL HAVE FAYING SURFACES MATCHING THE DESIGNATED FAYING SURFACE CLASS.

CONNECTIONS DESIGNATED "FT" IN THE DRAWINGS SHALL HAVE BOLTS INSTALLED TO FINGER-TIGHT REQUIREMENTS. WHERE FINGER-TIGHT CONNECTIONS ARE NOTED, NUTS SHALL BE HAND-TIGHTENED AND THEN BACKED OFF ONE-QUARTER TURN. DEFORM THE BOLT THREADS, PROVIDE A LOCKING NUT, OR PROVIDE A JAMB NUT TO ENSURE THE NUT IS UNABLE TO BACK OFF FURTHER.

ALL HIGH-STRENGTH BOLTS SHALL HAVE WASHERS INSTALLED PER THE RCSC, UNLESS NOTED OTHERWISE.

ALL BOLT HOLES SHALL BE STANDARD SIZE, UNLESS NOTED OTHERWISE. CENTER FINGER-TIGHT BOLTS IN SLOTS AT LONG SLOTS, UNLESS NOTED OTHERWISE.

STRUCTURAL STEEL WELDING

STRUCTURAL STEEL SHOP DRAWINGS SHALL SHOW ALL WELDING WITH AWS A2.4 SYMBOLS. ALL WELDING SHALL BE DONE BY AWS CERTIFIED WELDERS AND IN ACCORDANCE WITH AWS D1.1. WELDS SHOWN ON THE DRAWINGS ARE THE MINIMUM SIZES. INCREASE WELD SIZE TO AWS MINIMUM SIZES, BASED ON PLATE THICKNESS. THE MINIMUM WELD SIZE SHALL BE 3/16 INCH. FIELD WELDING SYMBOLS HAVE NOT NECESSARILY BEEN INDICATED ON THE DRAWINGS. WHERE SHOWN, PROPER FIELD WELDING PER AWS D1.1 SHALL BE USED. WHERE NO FIELD WELDING SYMBOLS ARE SHOWN, IT IS THE CONTRACTOR'S RESPONSIBILITY TO COORDINATE THE USE OF SHOP AND FIELD WELDS. ALL PARTIAL JOINT PENETRATION GROOVE WELD SIZES SHOWN ON THE DRAWINGS REFER TO EFFECTIVE THROAT THICKNESS. FLARE-GROOVE WELDS SHALL BE FILLED FULL TO MAXIMUM EFFECTIVE THROAT USING ANY OF THE PREQUALIFIED WELDING PROCESSES PER AISC 360-16 TABLE J2.2, UNLESS NOTED OTHERWISE. FOR BASE METALS WITH MAXIMUM YIELD STRENGTH EQUAL TO 50 KSI, ALL WELDS SHALL BE MADE USING LOW HYDROGEN ELECTRODES WITH MINIMUM TENSILE STRENGTH PER AWS D1.1 (MINIMUM 70 KSI). LOW HYDROGEN SMAW ELECTRODES SHALL BE STORED IN AN OVEN OR USED WITHIN THE ATMOSPHERIC TIME PERIODS SPECIFIED IN AWS D1.1 TABLE 5.1, OR SHALL BE REBAKED PER AWS D1.1 CLAUSE 5.3. ELECTRODES SHALL BE REBAKED NO MORE THAN ONE TIME, AND ELECTRODES THAT HAVE BEEN WET SHALL NOT BE USED.

FILLER METALS FOR ALL COMPLETE JOINT PENETRATION GROOVE WELDED T- AND CORNER JOINTS WITH BACKING LEFT IN PLACE AND COMPLETE JOINT PENETRATION GROOVE WELDED SPLICES IN HEAVY SECTIONS AS DEFINED IN AISC 360 A3.1c SHALL HAVE A MINIMUM CHARPY V-NOTCH TOUGHNESS OF 20 FOOT-POUNDS AT 40 DEGREES FAHRENHEIT.

ALL WELDING SHALL BE PERFORMED IN STRICT ADHERENCE TO A WRITTEN WELDING PROCEDURE SPECIFICATION (WPS) PER AWS D1.1. ALL WELDING PARAMETERS SHALL BE WITHIN THE ELECTRODE MANUFACTURER'S RECOMMENDATIONS. WELDING PROCEDURES SHALL BE SUBMITTED TO THE OWNER'S TESTING AGENCY FOR REVIEW PRIOR TO STARTING FABRICATION OR ERECTION. COPIES OF THE WPS SHALL BE ON SITE AND AVAILABLE TO ALL WELDERS AND THE SPECIAL INSPECTOR.

ALL COMPLETE JOINT PENETRATION WELDS SHALL BE ULTRASONICALLY TESTED UPON COMPLETION OF THE CONNECTION, EXCEPT PLATE LESS THAN OR EQUAL TO 1/4 INCH THICK, WHICH SHALL BE MAGNETIC PARTICLE TESTED. REDUCTION IN TESTING MAY BE MADE IN ACCORDANCE WITH THE BUILDING CODE WITH APPROVAL OF THE ENGINEER.

THE CONTRACTOR SHALL BE RESPONSIBLE FOR THE JOINT PREPARATIONS AND WELDING PROCEDURES THAT INCLUDE, BUT ARE NOT LIMITED TO: REQUIRED ROOT OPENINGS, ROOT FACE DIMENSIONS, GROOVE ANGLES, BACKING BARS, COPES, SURFACE ROUGHNESS VALUES, AND TAPERS AND TRANSITIONS OF UNEQUAL PARTS.

SHOP DRAWINGS

SHOP DRAWINGS FOR REINFORCING STEEL, AND STRUCTURAL STEEL SHALL BE SUBMITTED FOR REVIEW PRIOR TO FABRICATION OF THESE ITEMS.

REFER TO THE ARCHITECTURAL DRAWINGS FOR ALL CONCRETE DIMENSIONS NOT SHOWN ON THE STRUCTURAL DRAWINGS. IT SHALL BE THE CONTRACTOR'S RESPONSIBILITY TO DEVELOP DETAILED SLAB EDGE AND CONCRETE OUTLINE DRAWINGS THAT ARE BASED ON THE ARCHITECTURAL, STRUCTURAL, AND MEP DRAWINGS. THE DETAILED EDGE AND OUTLINE DRAWINGS SHALL BE SUBMITTED FOR REVIEW. SUBMITTED DRAWINGS SHALL CONTAIN ALL CONCRETE CURBS, FORM OUTLINES, AND EMBEDDED ITEMS. DIMENSIONS AND OUTLINES DEVELOPED BY THE CONTRACTOR MAY VARY FROM THOSE SHOWN BY THE ARCHITECT AND ENGINEER AS NECESSARY BASED ON THE DEPENDENCY ON ADJACENT MATERIALS THAT ARE DETERMINED BY THE CONTRACTOR AND/OR SUPPLIER (EXTERIOR CLADDING, ELEVATOR EQUIPMENT, FINAL MEP SHAFT SIZES, ETC.). CONCRETE OUTLINES SHALL BE ADJUSTED AS NECESSARY TO ACCOUNT FOR CONSTRUCTION METHODS. THE CONCRETE OUTLINE DEVELOPED BY THE CONTRACTOR SHALL NOT MATERIALLY ALTER THE DESIGN INTENT SHOWN IN THE STRUCTURAL DRAWINGS.

DIMENSIONS AND QUANTITIES ARE NOT REVIEWED BY THE ENGINEER OF RECORD; THEREFORE, THEY SHALL BE VERIFIED BY THE CONTRACTOR. THE CONTRACTOR SHALL REVIEW AND STAMP DRAWINGS PRIOR TO REVIEW BY THE ENGINEER OF RECORD. THE CONTRACTOR SHALL REVIEW DRAWINGS FOR CONFORMANCE WITH THE MEANS, METHODS, TECHNIQUES, SEQUENCES, AND OPERATIONS OF CONSTRUCTION, AND ALL SAFETY PRECAUTIONS AND PROGRAMS INCIDENTAL THERETO.

SUBMITTALS SHALL BE PROVIDED ELECTRONICALLY WHENEVER POSSIBLE AND WILL BE MARKED AND RETURNED ELECTRONICALLY. WHEN HARD COPY SUBMITTALS ARE REQUIRED, ONE ORIGINAL AND ONE COPY SHALL BE PROVIDED; THE REPRODUCIBLE COPY WILL BE MARKED AND RETURNED.

SHOP DRAWING SUBMITTALS PROCESSED BY THE ENGINEER ARE NOT CHANGE ORDERS. THE PURPOSE OF SHOP DRAWING SUBMITTALS BY THE CONTRACTOR IS TO DEMONSTRATE TO THE ENGINEER THAT THE CONTRACTOR UNDERSTANDS THE DESIGN CONCEPT, BY INDICATING WHICH MATERIAL IS INTENDED TO BE FURNISHED AND INSTALLED, AND BY DETAILING THE INTENDED FABRICATION AND INSTALLATION METHODS. IF DEVIATIONS, DISCREPANCIES, OR CONFLICTS BETWEEN SHOP DRAWINGS SUBMITTALS AND THE CONTRACT DOCUMENTS ARE DISCOVERED EITHER PRIOR TO OR AFTER SHOP DRAWING SUBMITTALS ARE PROCESSED BY THE ENGINEER, THE DESIGN DRAWINGS AND SPECIFICATIONS SHALL CONTROL AND SHALL BE FOLLOWED.

SPECIAL INSPECTION

THE FOLLOWING ITEMS REQUIRE SPECIAL INSPECTION AND TESTING PER CBC SECTION 1705. THIS WORK SHALL BE PERFORMED BY A CERTIFIED SPECIAL INSPECTOR CERTIFIED TO PERFORM THE TYPES OF INSPECTIONS AND TESTS SPECIFIED. THE FREQUENCY OF INSPECTIONS AND TESTING SHALL BE AS OUTLINED IN THE CBC TABLE ITEMS LISTED BELOW. DEFICIENCIES SHALL BE REPORTED DAILY TO THE CONTRACTOR. SUMMARY REPORTS SHALL BE DISTRIBUTED WEEKLY TO THE OWNER, ARCHITECT, CONTRACTOR, BUILDING OFFICIAL, AND STRUCTURAL ENGINEER. SEE THE SPECIFICATIONS FOR ADDITIONAL REQUIREMENTS FOR SPECIAL INSPECTION AND TESTING.

ITEM	DESCRIPTION (REFER TO CBC SECTION 1705)	CBC REFERENCE
STRUCTURAL STEEL AND WELDING	STRUCTURAL STEEL THAT IS PART OF THE STRUCTURE.	SECTION 1705.2
HIGH STRENGTH BOLTING	SEE SPECIFICATIONS FOR PROCEDURES FOR INSPECTION AND TESTING.	SECTION 1705.2

MILLER

HULL

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Kevin Carpenter

GSA

FDA

U.S. FOOD & DRUG
ADMINISTRATION

STAMP

REGISTERED PROFESSIONAL ENGINEER
GREGORY L. ROGERS
C 73602
CIVIL
STATE OF CALIFORNIA

KEY PLAN

FDA IRVINE

RM 2309
EXHAUST
DESIGN

19701 Fairchild Rd, Irvine, CA 92612

SUBMITTAL

BID SET

AUGUST 15, 2025

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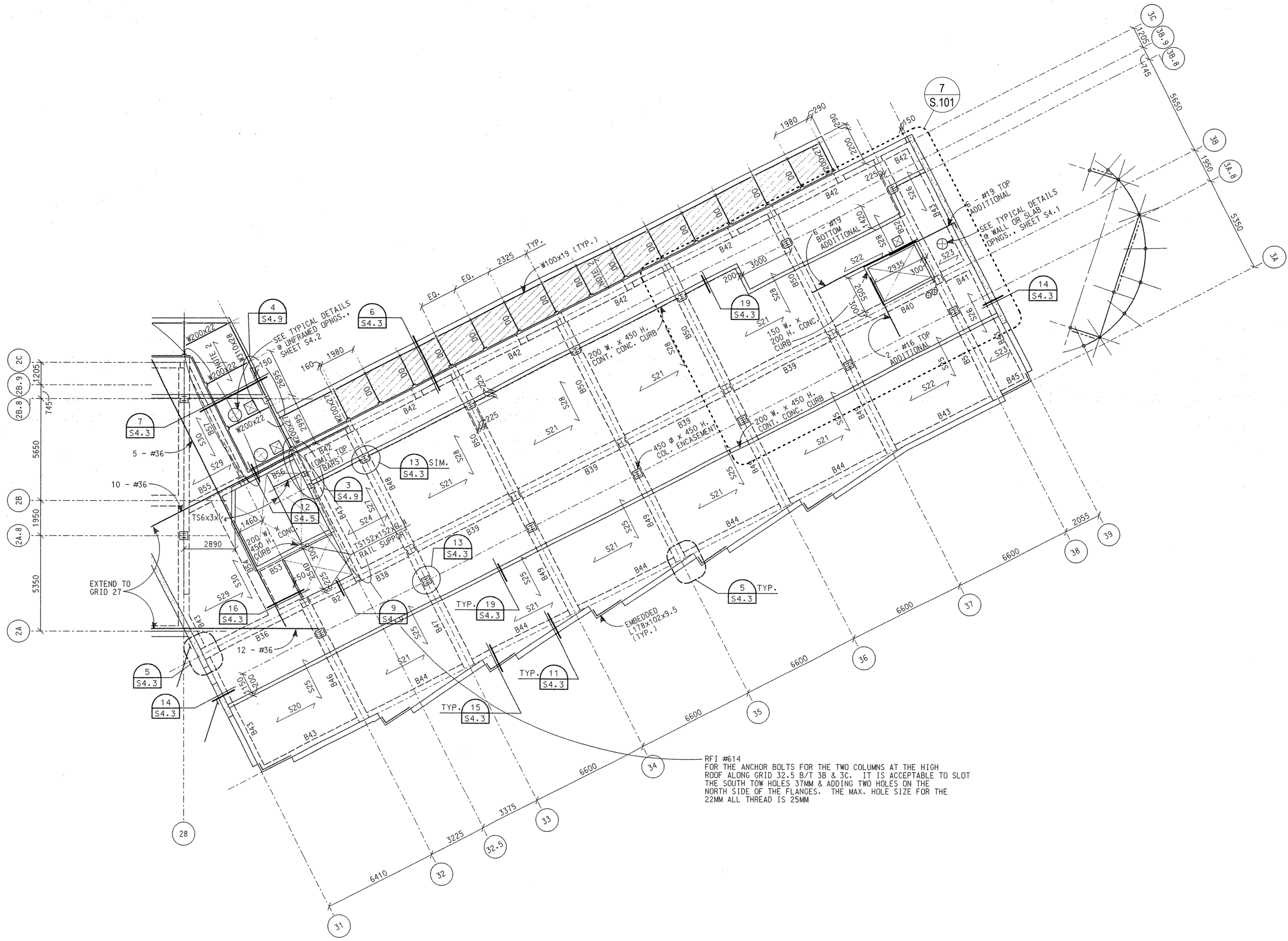
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GENERAL NOTES

S.002

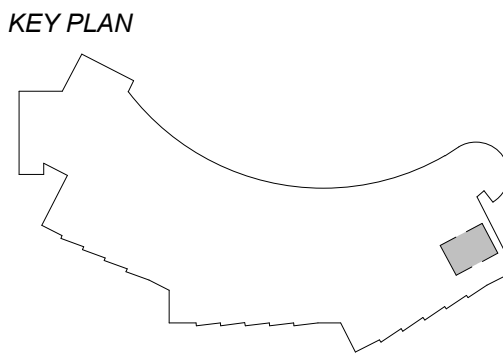
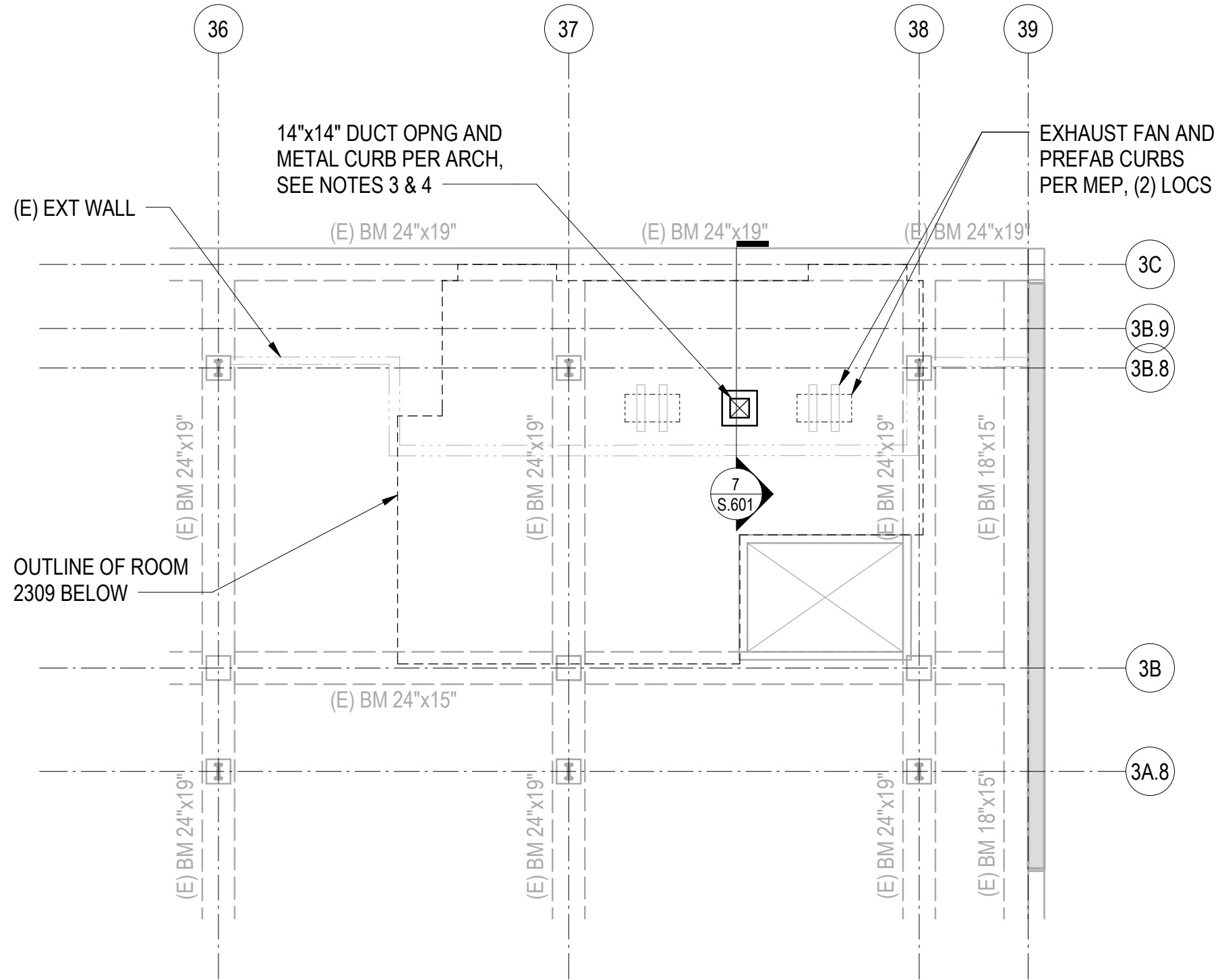


5 ROOF LEVEL - WING THREE - EXISTING FRAMING PLAN

NOTES:

- EXISTING CONSTRUCTION HAS BEEN ESTABLISHED BASED ON REVIEW OF THE AVAILABLE RECORD CONSTRUCTION DOCUMENTS FOR THE ORIGINAL BUILDING. ONLY LIMITED VERIFICATION HAS BEEN PERFORMED.
- CONTRACTOR SHALL VERIFY EXISTING CONDITIONS TO CONFIRM CONFORMANCE WITH THE DOCUMENTS. THE CONTRACTOR SHALL ALERT THE ARCHITECT WHERE EXISTING CONDITION IS DIFFERENT FROM WHAT IS SHOWN.
- CONTRACTOR TO COORDINATE LOCATION OF NEW ROOF SLAB OPENING WITH ARCHITECTURAL AND MEP DRAWINGS, AND WITH NEW EQUIPMENT.
- DO NOT OVERCUT CORNERS OF NEW CONCRETE ROOF SLAB OPENING.

7 ROOF LEVEL - WING THREE - PARTIAL FRAMING PLAN
1/8" = 1'-0"



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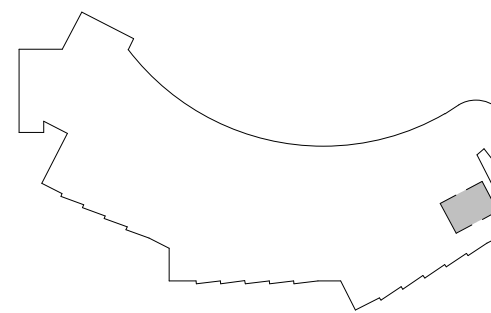
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ROOF LEVEL
DRAWINGS
S.101



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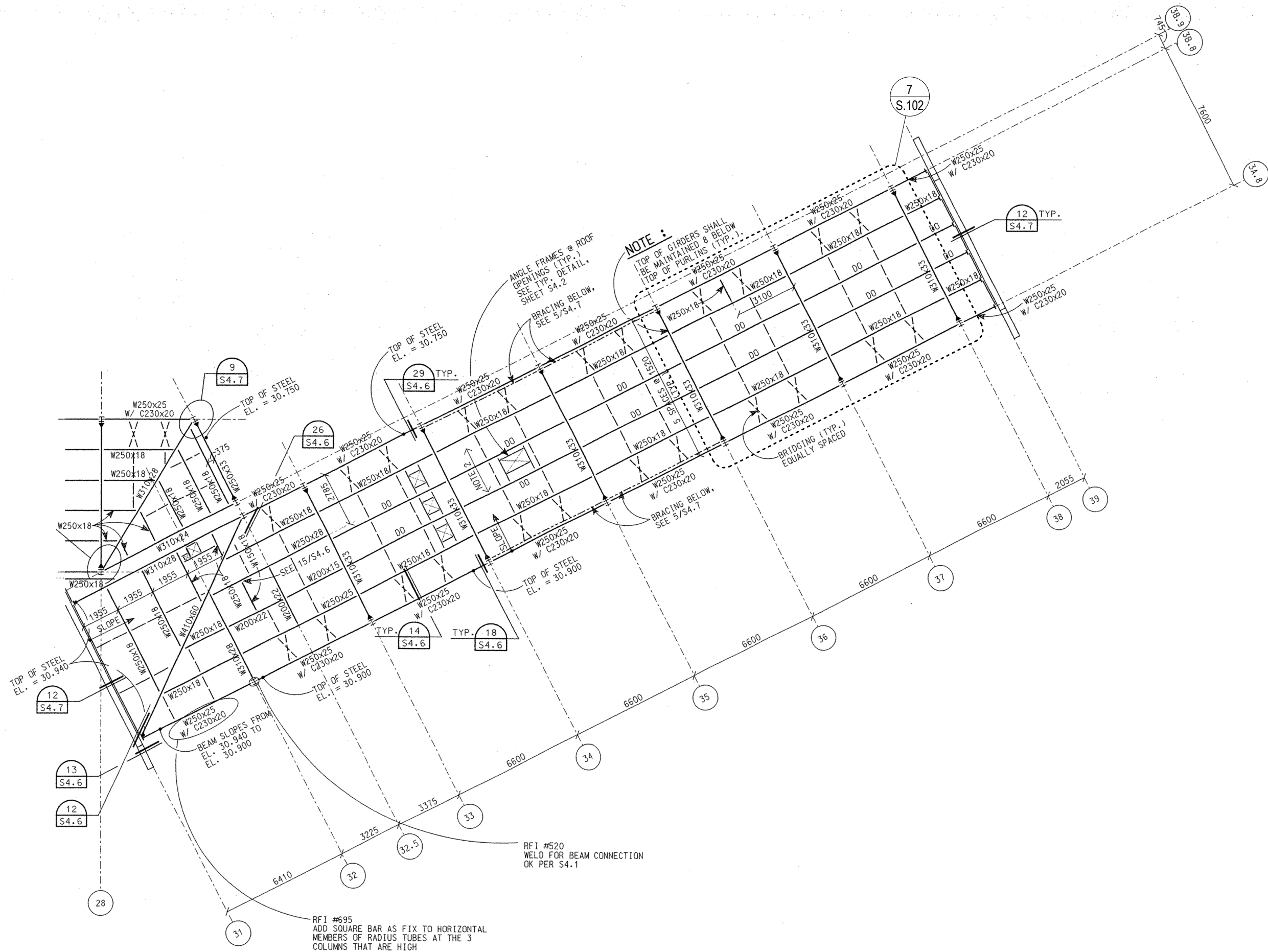
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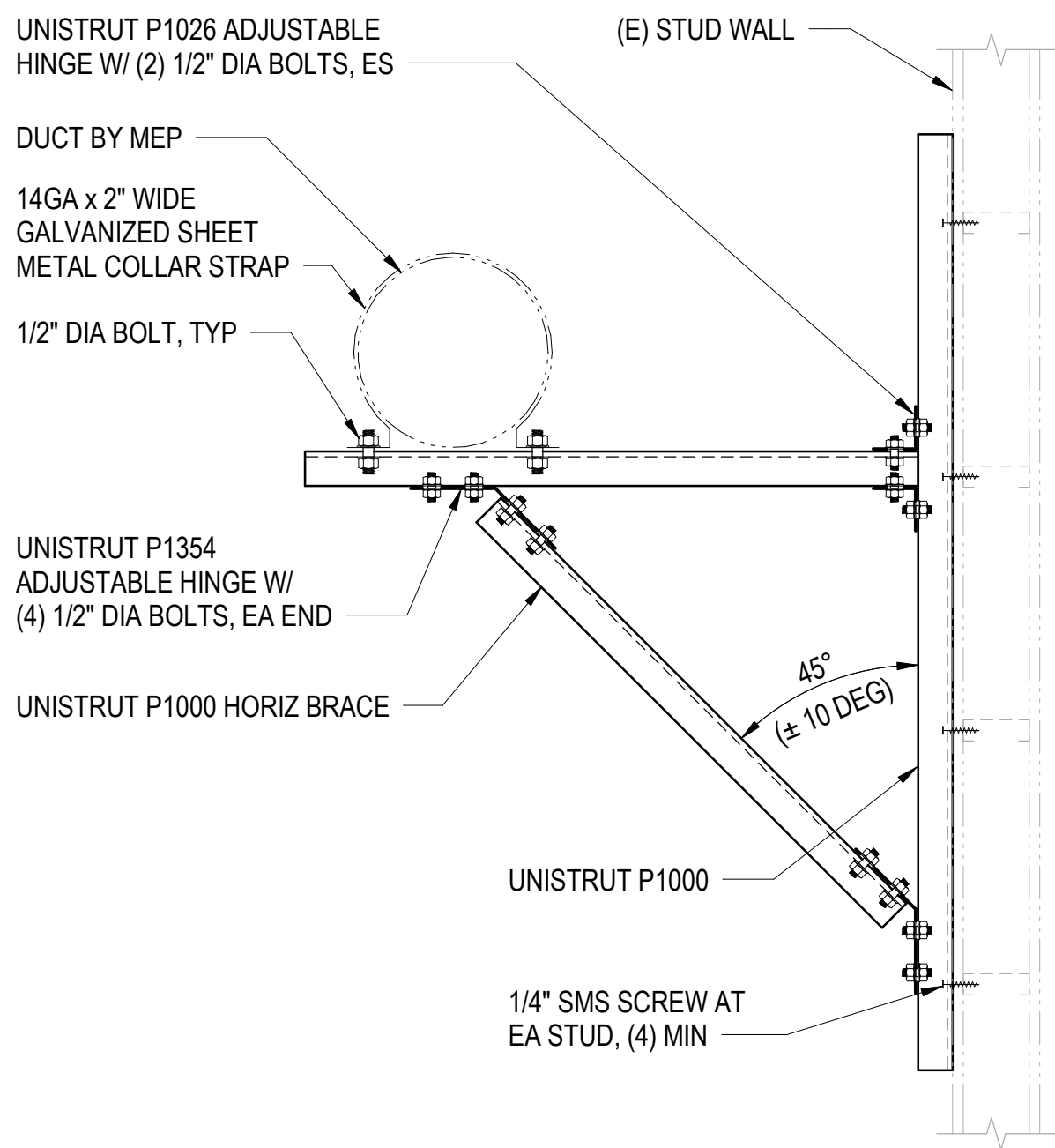
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SHED ROOF LEVEL
DRAWINGS AND
DETAILS

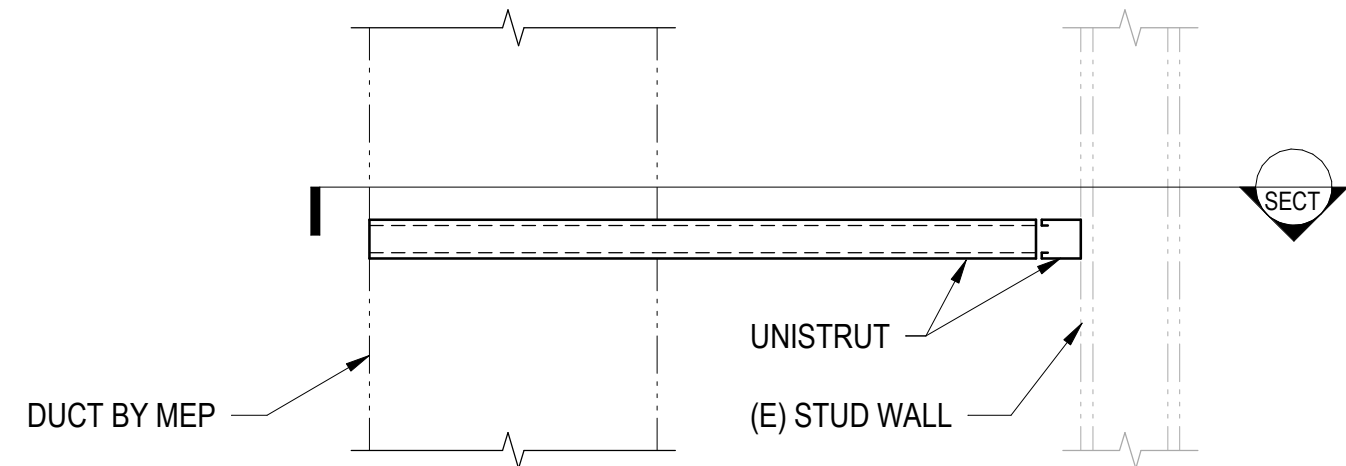
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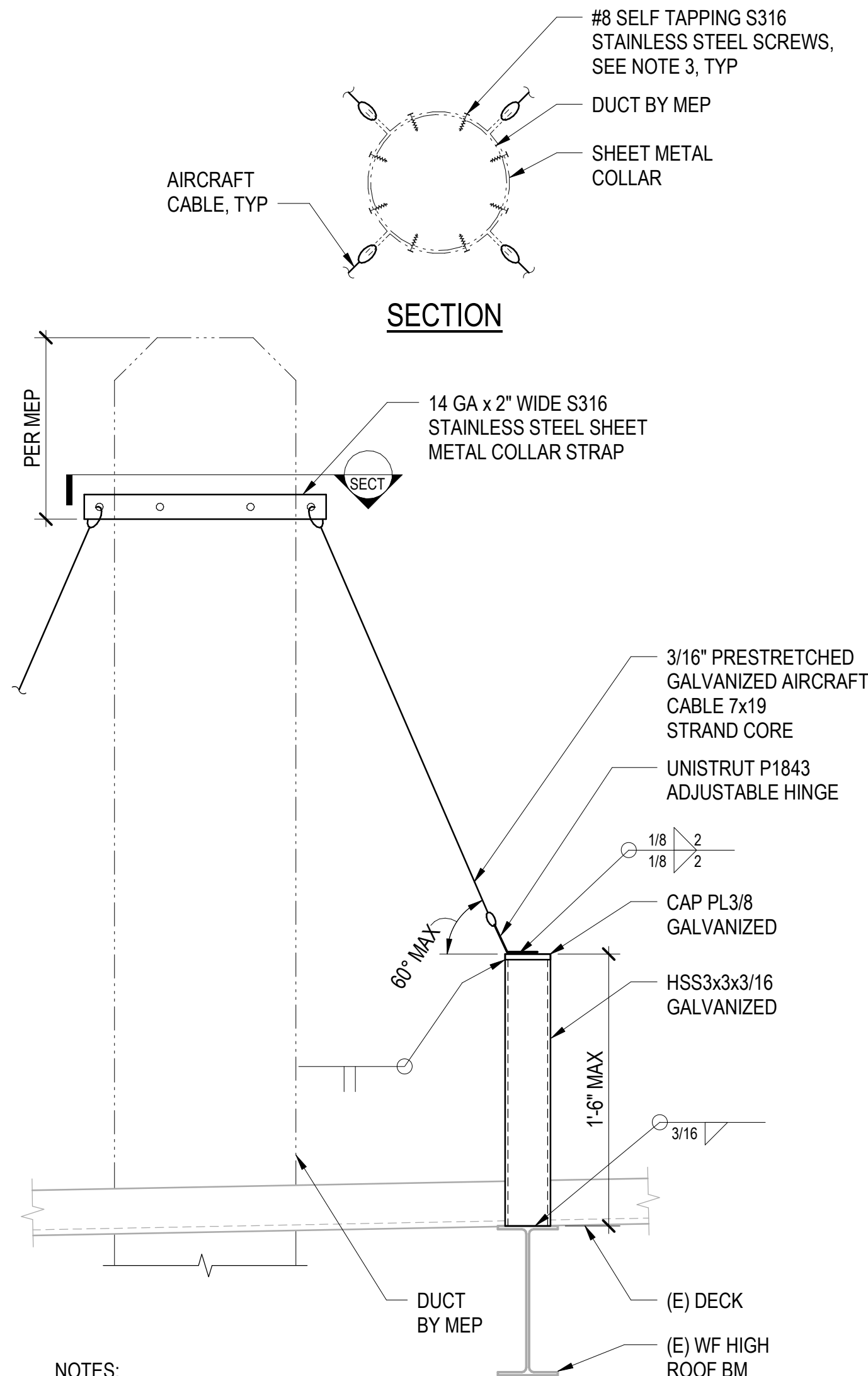
SECTION



NOTES:

1. ALL FITTINGS, FASTENERS, BOLTS AND CONNECTORS TO BE GALVANIZED OR STAINLESS STEEL.

5 DETAIL
1 1/2" = 1'-0"

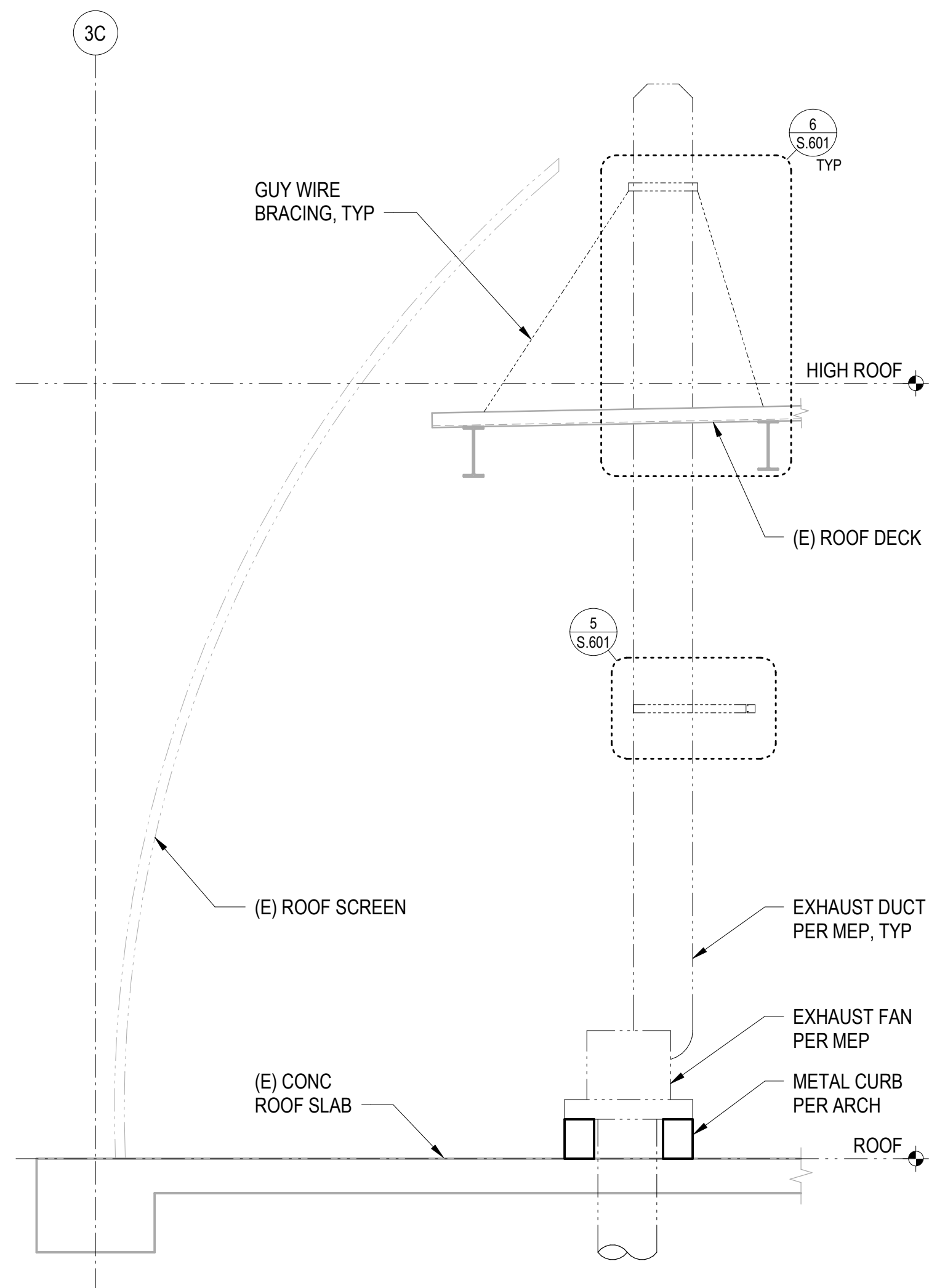


NOTES:

1. ALL FITTINGS, FASTENERS, BOLTS AND CONNECTORS TO BE GALVANIZED OR STAINLESS STEEL.
2. REMOVE GALVANIZING AT FIELD WELDS, AND REPAIR AFTER WELDING WITH GALVANIC PAINT.
3. IN LIEU OF STAINLESS STEEL SCREWS, DUCT COLLAR CAN BE WELDED TO DUCT AT CONTRACTOR'S OPTION.

6 DETAIL
1 1/2" = 1'-0"

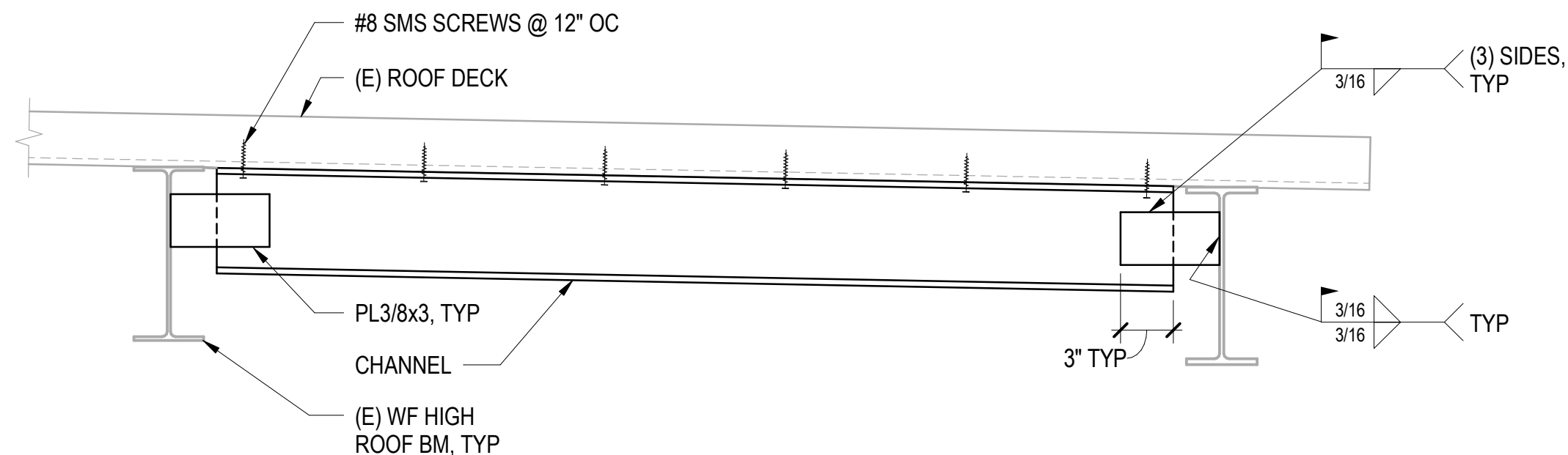
7 SECTION
1/2" = 1'-0"



NOTES:

1. INSTALL CHANNEL TIGHT TO UNDERSIDE OF EXISTING DECK. PROVIDE SHIMS AS REQUIRED.
2. ALL CHANNELS AND PLATES TO BE GALVANIZED OR TO RECEIVE PROTECTIVE HIGH PERFORMANCE COATING. REMOVE GALVANIZING OR PAINT AT FIELD WELDS, AND REPAIR AFTER WELDING WITH GALVANIC PAINT OR APPROVED HIGH PERFORMANCE COATING.
3. ALL FASTENERS TO BE GALVINZED OR STAINLESS STEEL.

11 SECTION
1 1/2" = 1'-0"



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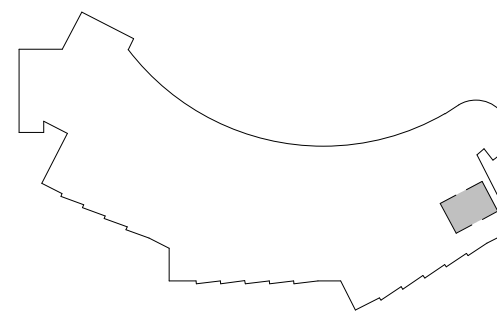
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KEY PLAN



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SECTIONS AND
DETAILS
S.601

MECHANICAL SYMBOL LIST

NOTE: This is a standard symbol list and not all items listed may be used.

Abbreviations

AHU	AIR HANDLING UNIT
BDD	BACKDRAFT DAMPER
CU	CONDENSING UNIT
CONT.	CONTINUATION
CT	COOLING TOWER
(X)	DEMOLISH
DIA	DIAMETER
EXH	EXHAUST
EF	EXHAUST FAN
(E)	EXISTING
FC	FAN COIL
FT	FEET
FD	FIRE DAMPER
GPH	GALLONS PER HOUR
GPM	GALLONS PER MINUTE
HX	HEAT EXCHANGER
HP	HORSEPOWER
HWC	HOT WATER COIL
IN	INCHES
ID	INSIDE DIAMETER
MAX	MAXIMUM
MIN	MINIMUM
MD	MOTORIZED DAMPER
(N)	NEW
N/A	NOT APPLICABLE
NIC	NOT IN CONTRACT
NTS	NOT TO SCALE
NO.	NUMBER
OC	ON CENTER
OA	OUTSIDE AIR
OD	OUTSIDE DIAMETER
PH	PHASE
PSI	POUNDS PER SQUARE INCH
QTY	QUANTITY
REF	REFRIGERANT
RS	REFRIGERANT SUCTION
RLD	RELIEF DAMPER
(R)	RELOCATE/RELOCATED LOCATION
RET	RETURN
RA	RETURN AIR
RPM	REVOLUTIONS PER MINUTE
SOV	SHUT OFF VALVE
SF	SQUARE FEET
SA	SUPPLY AIR
T, TEMP	TEMPERATURE
MBH	THOUSAND BTU'S PER HOUR
VAV	VARIABLE AIR VOLUME
VFD	VARIABLE FREQUENCY DRIVE
VEL	VELOCITY
VD	VOLUME DAMPER (HAND OPERATOR)
W	WATT
WB	WET BULB
W/	WITH

Control Symbols

	CARBON DIOXIDE SENSOR
	SPACE TEMPERATURE SENSOR/THERMOSTAT

Dampers

	BACK DRAFT DAMPER
	FIRE/SMOKE DAMPER
	MOTORIZED DAMPER
	SMOKE DAMPER
	VOLUME DAMPER

Diffusers and Grilles

	DIFFUSER OR GRILLE IDENTIFICATION
	EXHAUST AIR
	RETURN AIR
	SUPPLY AIR

Ductwork Fittings

	ACOUSTICALLY LINED DUCT (SIZES SHOWN ARE NET INSIDE)
	BELLMOUTH
	CONCENTRIC SQUARE TO ROUND
	CONCENTRIC TRANSITION, RECTANGULAR OR ROUND
	ECCENTRIC TRANSITION, RECTANGULAR OR ROUND
	FLEXIBLE CONNECTION
	NON-SYMMETRICAL WYE
	RECTANGULAR DUCT DROP
	RECTANGULAR DUCT RISER
	RECTANGULAR MAIN WITH RECTANGULAR BRANCH
	RECTANGULAR MAIN WITH ROUND BRANCH
	RECTANGULAR OFFSET LESS THAN 15% Slope
	RECTANGULAR OFFSET MORE THAN 15% Slope

ROUND DUCT DROP

ROUND DUCT RISER

ROUND DUCT WITH ROUND BRANCH

ROUND WYE

SYMMETRICAL WYE

MITERED ELBOW WITH TURNING VANES

RADIUSED ELBOW

General

LIMIT OF DEMOLITION

DEMOLISH

EXISTING WORK

NEW WORK

30X16 RECTANGULAR DUCT SIZING

30"Ø ROUND DUCT SIZING

CONTROL DIAGRAM SYMBOLS LIST

ABBREVIATIONS

AFMS	AIR FLOW MEASURING STATION
AI	ANALOG INPUT
AO	ANALOG OUTPUT
BMS/BAS	BUILDING MANAGEMENT SYSTEM
CO	CARBON MONOXIDE SENSOR
CO2	CARBON DIOXIDE SENSOR
CT	ELECTRIC CURRENT TRANSMITTER
DDC	DIRECT DIGITAL CONTROL
DI	DIGITAL INPUT
DO	DIGITAL OUTPUT
DP	PRESSURE SENSOR
EA	EXHAUST AIR
ECM	ELECTRONICALLY COMMUTATED MOTOR
EF	EXHAUST FAN
FE	FLOW ELEMENT
HS	HIGH STATIC
HWR	HEATING HOT WATER RETURN
HWS	HEATING HOT WATER SUPPLY
M	MOTORIZED
MAU	MAKE UP AIR UNIT
M/S	MOTOR STARTER
MD	MOTORIZED DAMPER
OA	OUTSIDE AIR
P	PRESSURE SENSOR
PPM	PARTS PER MILLION
RA	RETURN AIR / RELIEF AIR
RF	RETURN FAN / RELIEF FAN
SA	SUPPLY AIR
SF	SUPPLY FAN OR SQUARE FOOT
SD	SMOKE DETECTOR / ROOM THERMOSTAT
T	TEMPERATURE SENSOR
TT	TEMPERATURE TRANSMITTER
TPI	THIRD PARTY INTERFACE
VFD	VARIABLE FREQUENCY DRIVE

SYMBOLS

	ANALOG INPUT
	ANALOG OUTPUT
	ELECTRIC CURRENT TRANSMITTER
	DIRECT DIGITAL CONTROL
	DIGITAL INPUT
	DIGITAL OUTPUT
	DIFFERENTIAL PRESSURE SENSOR
	FLOW METER
	FLOW SWITCH
	ROOM / DUCT RELATIVE HUMIDITY SENSOR / TRANSMITTER
	HIGH STATIC PRESSURE SENSOR
	MOTOR STARTER
	MOTORIZED DAMPER

NETWORK

	PRESSURE SENSOR
	SMOKE DETECTOR
	SPACE PRESSURE SENSOR
	TEMPERATURE SENSOR/ROOM THERMOSTAT
	THIRD PARTY INTERFACE (BACNET)
	VARIABLE FREQUENCY DRIVE
	ELECTRONICALLY COMMUTATED MOTOR

GENERAL SHEET NOTES

- PROVIDE ALL THIRD PARTY BACNET INTERFACES WHERE NOT PROVIDED BY EQUIPMENT MANUFACTURER.
- SEE ELECTRICAL DRAWINGS FOR 120 VOLT JUNCTION BOXES FOR POWER SUPPLY. PROVIDE ADDITIONAL 120V POWER SUPPLY AS REQUIRED, AND PROVIDE 120V TO 24V TRANSFORMERS FOR LOW VOLTAGE CONTROL POWER. PROVIDE 24V POWER TO ALL CONTROL DEVICES. PROVIDE CONTROL WIRING PER DIV. 23 & 26 SPECIFICATIONS. PROVIDE PLENUM RATED CABLE ABOVE REMOVABLE TILE CEILING, AND RACEWAY ABOVE HARD LID CEILINGS.
- PROVIDE 120V POWER FOR AIRFLOW MEASURING STATIONS, LARGE VALVES, AND BMS CONTROL SYSTEMS INCLUDING ALL PANELS. FOLLOW DIV. 26 SPECIFICATIONS FOR PRODUCTS AND INSTALLATION RELATED TO 120V POWER DISTRIBUTION. PROVIDE EMERGENCY POWER FOR ASSOCIATED BMS CONTROLLERS AND TERMINAL UNITS WHERE AHU IS ON EMERGENCY POWER.
- PROVIDE COMMUNICATION WIRING AND CONNECT TO THE BUILDING AND/OR CAMPUS' IP NETWORK.
- CONTROL DIAGRAMS, SEQUENCES, AND SPECIFICATIONS ARE PERFORMANCE BASED. PROVIDE FINAL SEQUENCES, POINT LIST, TRENDING LIST, CONTROL DIAGRAMS, AND ARCHITECTURAL RISER DIAGRAMS (SEE SPECIFICATIONS).
- PROVIDE FLOOR LEVEL AND BUILDING LEVEL CONTROLLERS, AS NEEDED.
- MODIFY ADJUSTABLE SETPOINTS DURING FUNCTIONAL TESTING AND COMMISSIONING STAGE FOR PROPER SYSTEMS' OPERATION.

SHEET INDEX

M.001	SYMBOL & NOTES - MECHANICAL
M.101	FLOOR PLANS - MECHANICAL
M.201	PARTIAL FLOOR PLAN - MECHANICAL
M.301	SECTIONS - MECHANICAL
M.401	SCHEDULE & DIAGRAM - MECHANICAL
M.402	DIAGRAM & DETAILS - MECHANICAL

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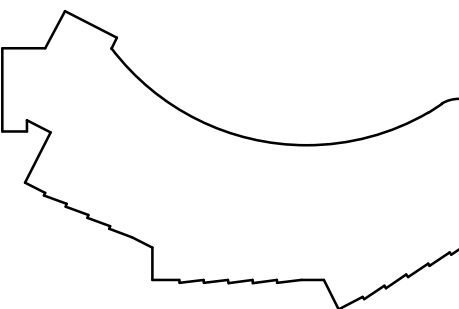


FDA U.S. FOOD & DRUG
ADMINISTRATION

STAMP



KEY PLAN



FDA IRVINE

RM 2309

EXHAUST

DESIGN

19701 Fairchild Rd, Irvine, CA 92612

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AUGUST 15, 2025

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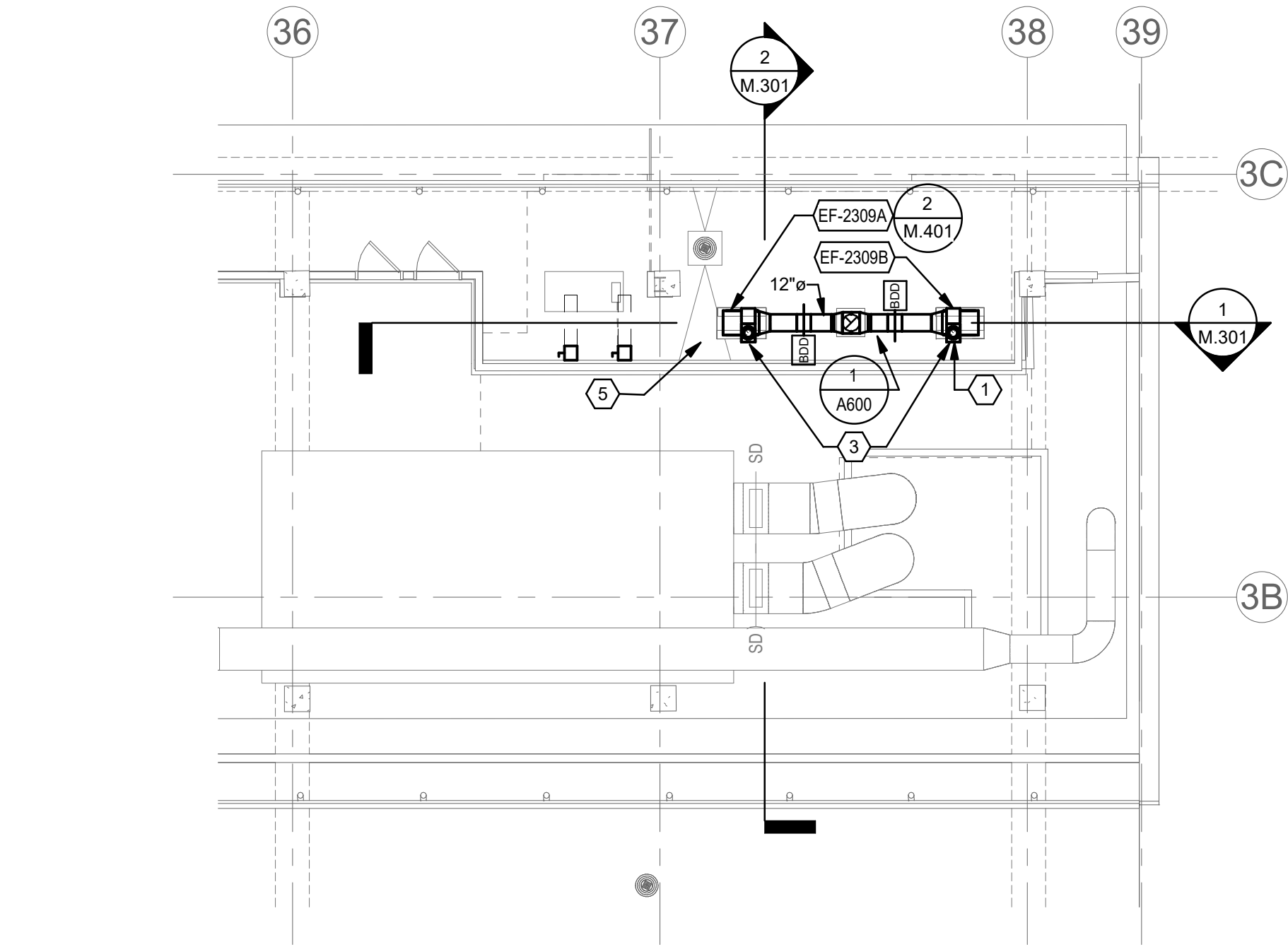
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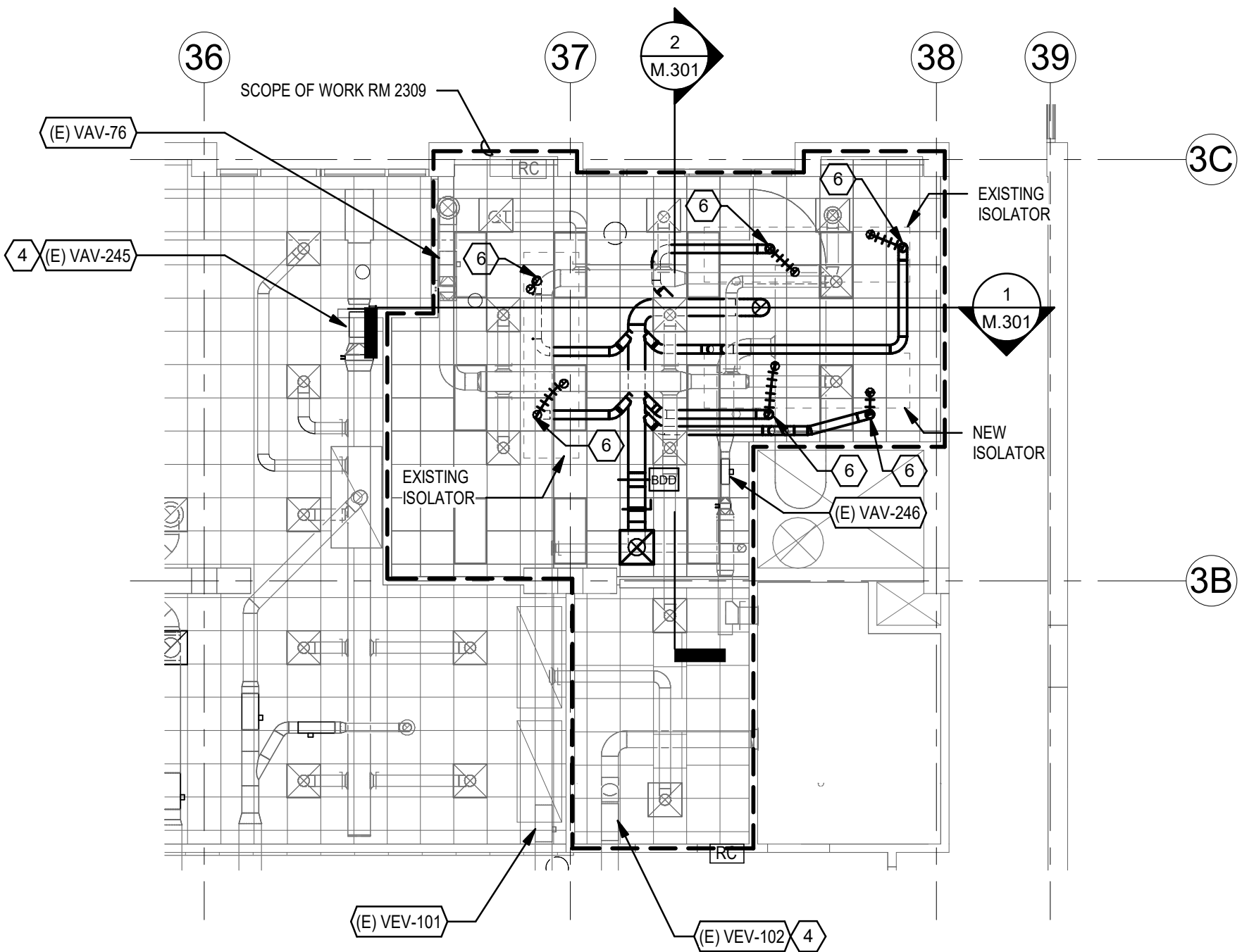
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SHEET

SYMBOL &
NOTES -
MECHANICAL
M.001



1 ROOF PLAN - ABOVE RM 2309 - MECHANICAL
M.101 1/8" = 1'-0"



2 LEVEL 2 - RM 2309 - CEILING PLAN - MECHANICAL
M.101 1/8" = 1'-0"

SHEET KEYNOTES

1. REFER TO ARCHITECTURAL DRAWINGS FOR EXACT LOCATION OF FAN AND DUCT RISER TO HIGH ROOF PENETRATION.
2. REFER TO 1/A1.00 FOR EXACT CEILING PENETRATION LOCATION, TYPICAL 5.
3. PROVIDE COUNTERCLOCKWISE AND CLOCKWISE FAN ROTATION TO MATCH FAN DISCHARGE LOCATION NEAR WALL.
4. AIR VALVE NOT IN SCOPE OF WORK. SHOWN FOR REFERENCE ONLY.
5. FIELD VERIFY LOCATION OF EXISTING CONDUITE UP THROUGH ROOF LOCATED NEAR NEW FAN EF-2309A AND RELOCATE AS REQUIRED TO AVOID CONFLICT WITH FAN CURB.
6. REFER TO ARCHITECTURAL FOR EXACT CEILING PANEL PENETRATION LOCATION AND CEILING TILE DUCT TERMINATION AT ESCUTCHEON. NEW INSTALLED LOCATION OF THE ISOLATORS AND RELOCATED CEILING EXHAUST PENETRATION AS REQUIRED TO MAINTAIN THE MINIMUM DISTANCE BETWEEN THE ISOLATOR EXHAUST CONNECTIONS AND CEILING PENETRATIONS.

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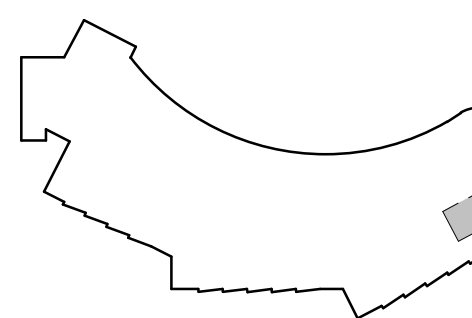


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KEY PLAN



FDA IRVINE

RM 2309

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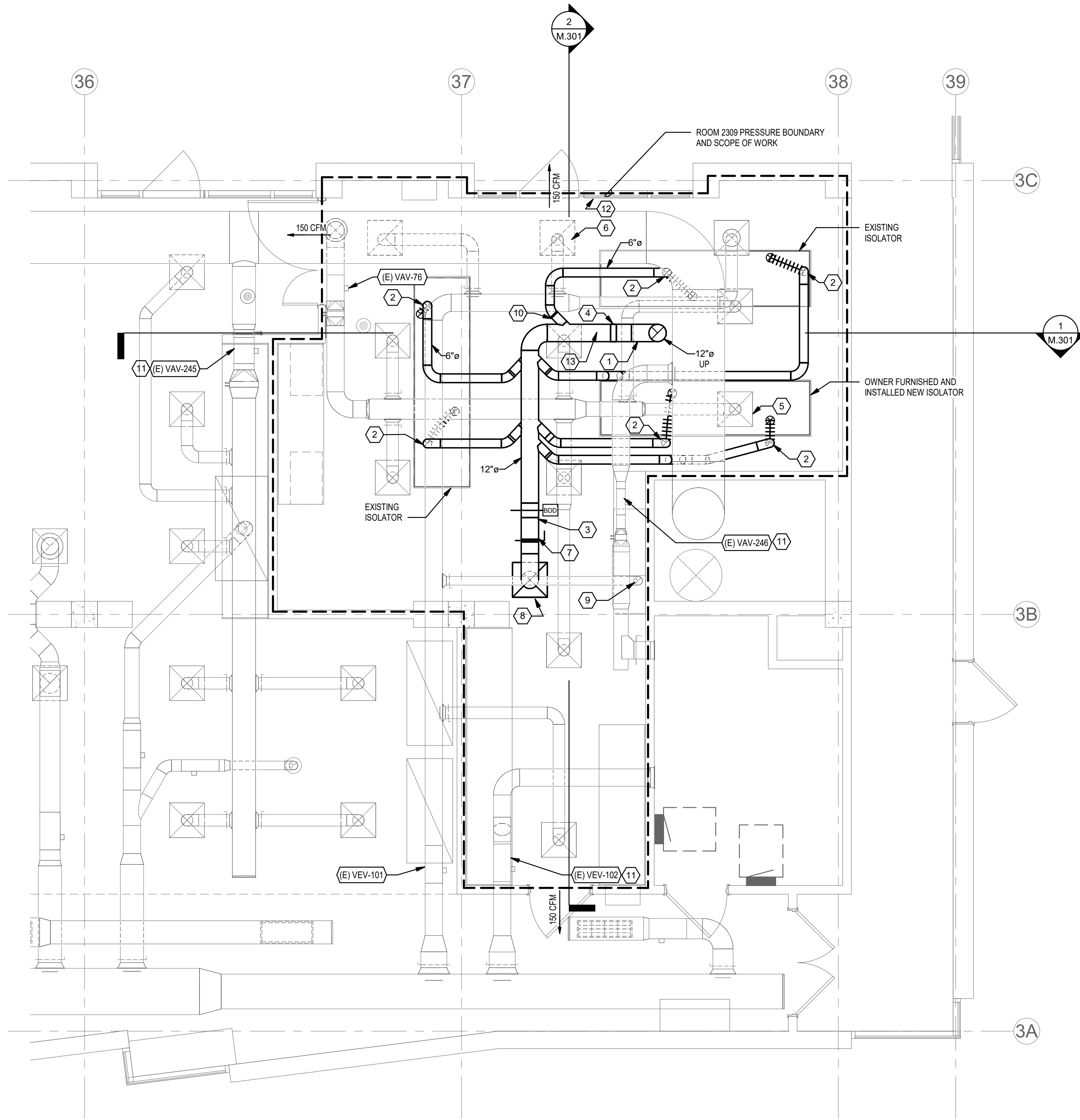
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SHEET

FLOOR PLANS
- MECHANICAL

M.101



1 PARTIAL PLAN LEVEL 2 - RM 2309 - MECHANICAL
M.201 1/4" = 1'-0"

GENERAL SHEET NOTES

- ALL NEW DUCTWORK, BALANCING DAMPERS AND ALL ASSOCIATED APPURTENANCES THAT COMES INTO CONTACT WITH ISOLATOR DISCHARGE AIR STREAM CONSTRUCTED WITH CORROSION RESISTANT 316L STAINLESS STEEL.
- REPLACE THE EXISTING SUPPLY AND EXHAUST AIR VALVE CONTROLLERS WITH NEW CONTROLLERS FOR INTEGRATION WITH THE EXISTING BUILDING AUTOMATION SYSTEM.

SHEET KEYNOTES

- EXHAUST DUCT UP THROUGH LOWER ROOF TO EXHAUST FAN LOCATED ON ROOF. REFER TO STRUCTURAL FOR EXACT ROOF PENETRATION LOCATION. REFER TO SHEET M.101 FOR CONTINUATION.
- 6" EXHAUST DUCT DROP TO ISOLATOR EXHAUST OUTLET COLLAR. USE HYDROGEN PEROXIDE CORROSION RESISTENT FLEX DUCT FROM ISOLATOR COLLAR TO STAINLESS STEEL HARD DUCT TERMINATING AT CEILING TILE ESCUTCHEON. EXTEND DUCT IMMEDIATELY UP THROUGH LAY-IN CEILING TO CONNECT TO DUCT MAIN LOCATED ABOVE CEILING. REFER TO ARCHITECTURAL FOR EXACT CEILING PANEL PENETRATION LOCATION AND CEILING TILE DUCT TERMINATION AT ESCUTCHEON.

FLEX DUCT SPECIFICATIONS:

- BASIS OF DESIGN: MASTERDUCT MASTER-CLIP SILICONE
- HYDROGEN PEROXIDE CORROSION RESISTANT COATING WITH SILICONE OR PTFE
- PRESSURE RATING: MINIMUM OF 4-INCH WG POSITIVE PRESSURE AND 4-INCH WG NEGATIVE PRESSURE.

- SET BACK DRAFT DAMPER TO OPEN AT - 0.05" W.C.
- LOCATE AIRFLOW MONITORING STATION IN DUCTWORK PER MANUFACTURER INSTALLATION REQUIREMENTS.
- BALANCE EXISTING SUPPLY DIFFUSER TO 214 CFM, TYPICAL 7.
- BALANCE EXISTING EXHAUST GRILLES TO 130 CFM, TYPICAL 4.
- SET VOLUME DAMPER TO MAINTAIN A - 0.10" W.C. EXHAUST DUCT STATIC PRESSURE AT 600 CFM EXHAUST RATE WHILE ISOLATORS ARE OFF.
- LAY-IN EXHAUST GRILLE: 24"X24" FACE/MODULE SIZE, 12" ROUND NECK, BASIS OF DESIGN: PRECISION AIR / LAMI-VENT PAT-C, WITHOUT BAFFLES.
- BALANCE BIOCAPINET TO 30 CFM.
- INSTALL 45 DEG DUCT TAP AT ALL BRANCH CONNECTIONS FROM ISOLATOR, TYPICAL 5.
- AIR VALVE NOT IN SCOPE OF WORK. SHOWN FOR REFERENCE ONLY.
- EXISTING ACCUTROL ISO-TEK ROOM PRESSURE MONITOR. REFER TO 4/M.402 FOR REQUIREMENTS.
- DUST STATIC PRESSURE SENSOR LOCATION.

RM 2309 - AIR BALANCE

	OCC		UNOCC
	MAX	MIN	MIN
SUPPLY AIR: (E) VAV-76	1500	1500	1500
TRANSFER AIR OUT	450	450	450
GENRERAL EA: (E) VEV-101	450	450	450
ISOLATOR EA: EF-2309A/B	600	600	600

NOTE: UNITS IN CFM

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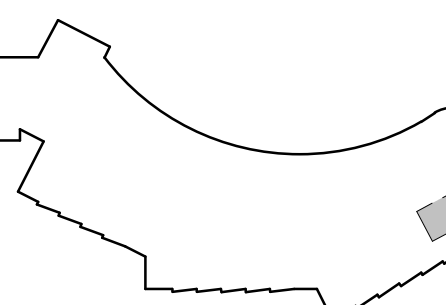
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KEY PLAN



FDA IRVINE

RM 2309 EXHAUST DESIGN

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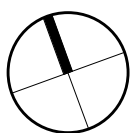
SHEET

PARTIAL FLOOR PLAN - MECHANICAL M.201

**INTERFACE
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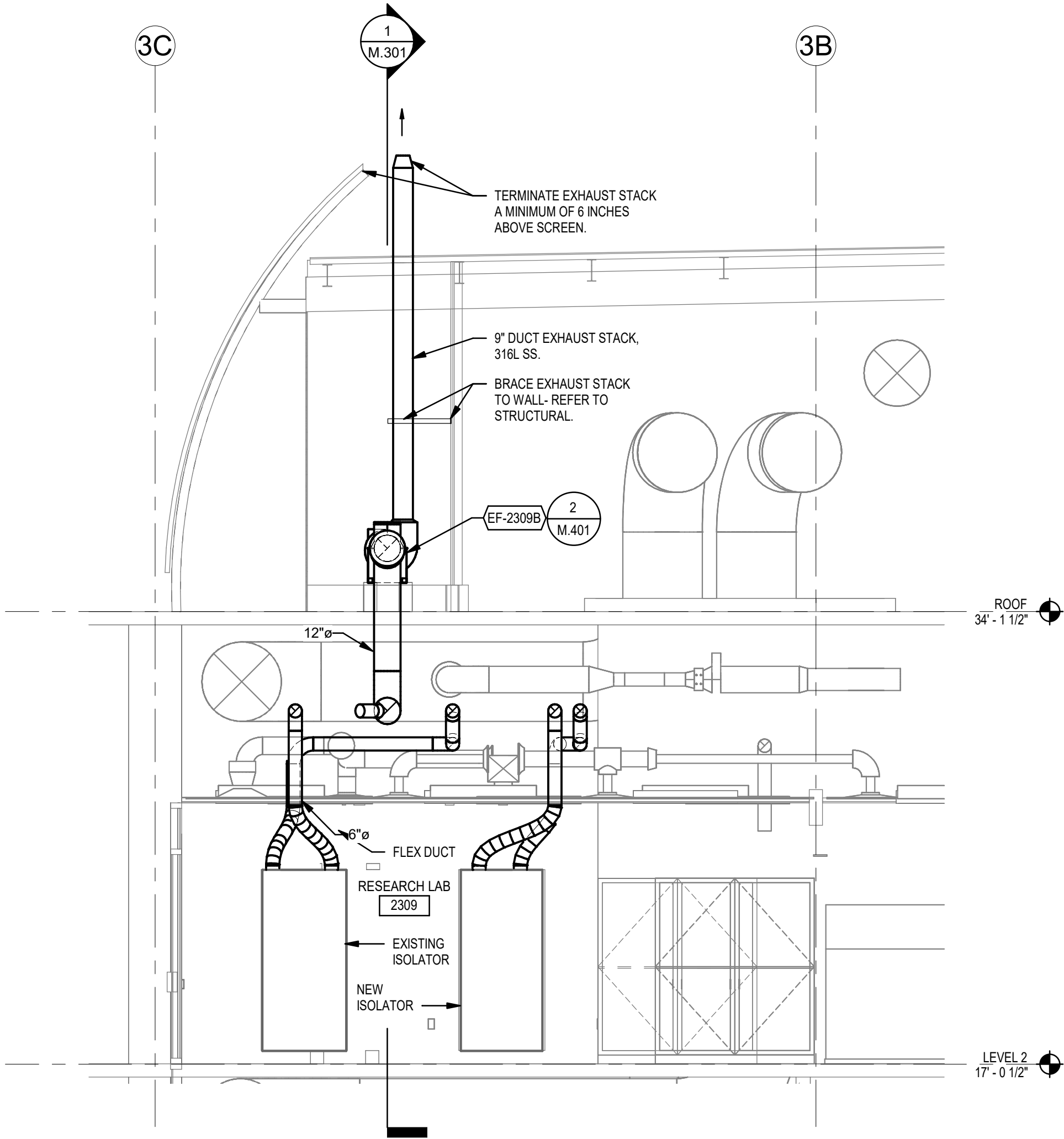
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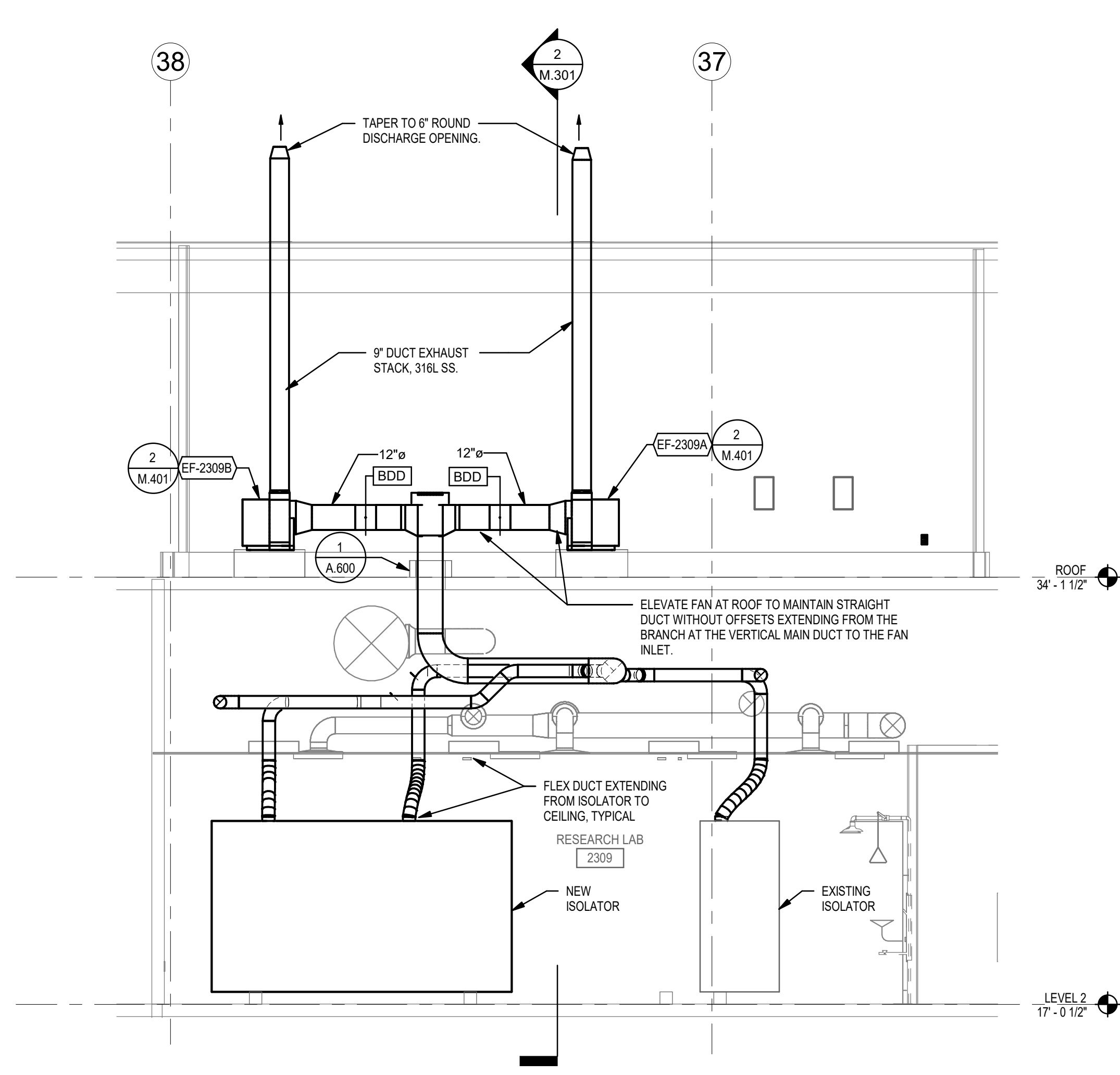
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SECTIONS -
MECHANICAL
M.301



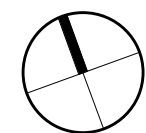
2 SECTION 2 - RM 2309
M.301 1/4" = 1'-0"



1 SECTION 1 - RM 2309
M.301 1/4" = 1'-0"

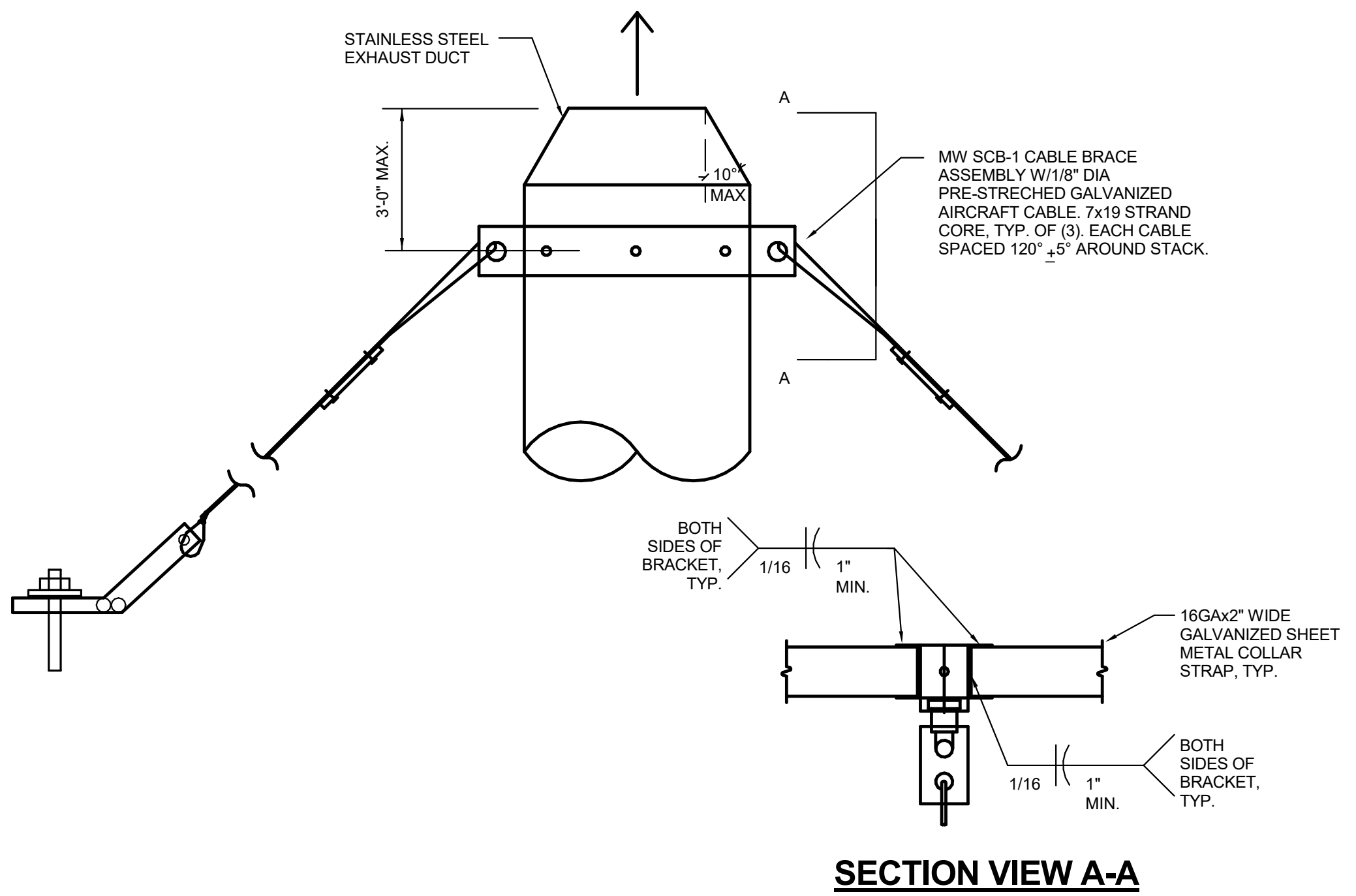


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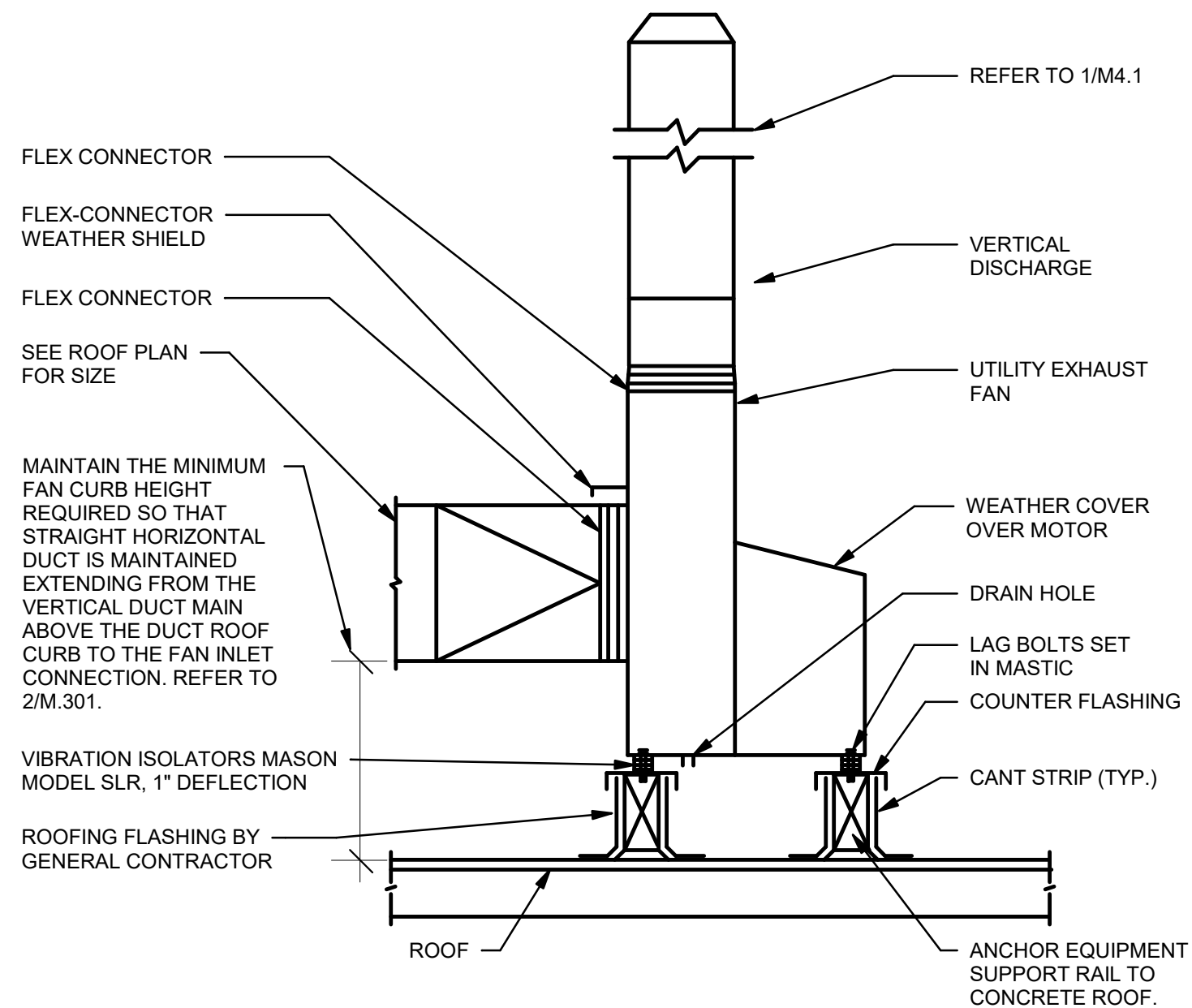
FAN SCHEDULE															
SYMBOL	AREA SERVED	BASIS OF DESIGN		TYPE	DRIVE	AIR FLOW (CFM)	ESP (IN H2O)	FAN RPM	MAX RPM	INLET SOUND (dB)	ELECTRICAL			MAX WT (LBS)	NOTES
		MFR	MODEL								VOLTS	PH	MHP		
EF-2309A	RESEARCH LAB 2309	GREENHECK	USF-4-B6	CENTRIFUGAL BLOWER	DIRECT	600	1.5	3029	4500	76	208	3	1	200	ALL
EF-2309B	RESEARCH LAB 2309	GREENHECK	USF-4-B6	CENTRIFUGAL BLOWER	DIRECT	600	1.5	3029	4500	76	208	3	1	200	ALL
NOTES:															
1	PROVIDE HI-PRO Z COATING (EXCEED 4,000-HOUR SALT SPRAY UNDER ASTM B117, EPOXY BASE COAT WITH POLYESTER TOP COAT) TO FAN AND ALL INTERNAL COMPONENTS EXPOSED TO AIR STREAM FOR CORROSION RESISTANCE TO HYDROGEN PEROXIDE.														
2	PROVIDE HI-PRO Z COATING (EXCEED 4,000-HOUR SALT SPRAY UNDER ASTM B117, EPOXY BASE COAT WITH POLYESTER TOP COAT) TO FAN HOUSING AND ACCESSORIES FOR COASTAL ENVIORNMENT CORROSION RESISTANCE.														
3	PROVIDE VFD.														
4	PROVIDE FAN ARRANGEMENT FOR VERTICAL DISCHARGE. SEE PLANS FOR ADDITIONAL INFORMATION.														
5	PROVIDE CLOCKWISE FAN ROTATION FOR ONE FAN AND COUNTERCLOCKWISE FOR THE OTHER FAN ROTATION SO THAT FAN DISCHARGE DISTANCE FROM THE EXTERIOR WALL IS THE SAME FOR BOTH FANS.														

AIR VALVE SCHEDULE										
AREA SERVED	SUPPLY AIR	SUPPLY (CFM)		INLET SIZE (INCHES)	RE-HEAT COIL		EXHAUST AIR VALVE (E) VEV-01	INLET SIZE (INCHES)	EXHAUST (CFM)	
		MAX	MIN		HEAT CAP. (MBH)	HTG FLOW RATE (GPM)			MAX	MIN
ISOLATOR 2309	(E) VAV-76	1,500	1,500	12	32.1	3.2		12	450	450
GENERAL NOTES										
1. BALANCE EXISTING AIR VALVE AIRFLOW AND HEATING WATER FLOW TO VALUES SCHEDULED.										
2. CONTRATOR TO TEST VAV-76 CONTROL VALVE FOR CORRECT OPERATION AND REPLACE AS NECESSARY.										
3. HEATING COIL: 180F EWT, 160F LWT, 3/4" EXISTING HEATING WATER PIPE EXTENDING TO COIL.										
4. MANUFACTURER/MODEL: ACCUTROL AV3000										



M.401

NO SCALE



M.401

NO SCALE

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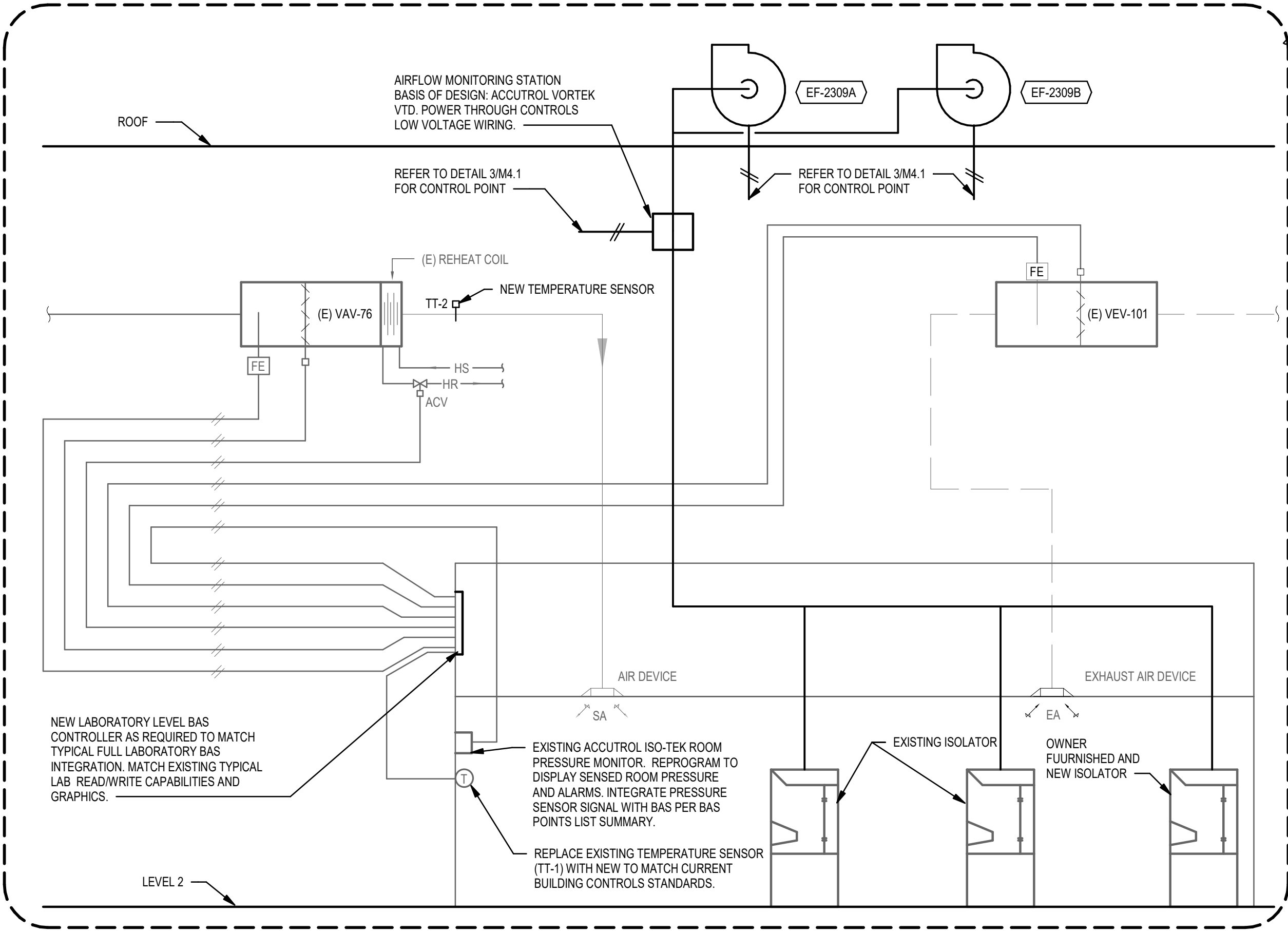
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1
M.402

ROOM 2309 CONTROLS DIAGRAM

NO SCALE

BAS POINTS LIST SUMMARY (RM 2309 LABORATORY)																														
SYSTEM APPARATUS OR AREA POINT DESCRIPTION	INPUTS										OUTPUTS		SYSTEM FEATURES				GENERAL		SUPPLEMENTARY NOTES											
	ANALOG					BINARY					DIGITAL	ANALOG	ALARM	PROGRAMS			COLOR GRAPHICS	HARDWIRED TO EA STATIC												
	MEASURED		CALCULATED			STATUS	METER	FLOW SWITCH	DIFF. PRESS. SWITCH	ON OFF	OPEN/CLOSE	DAMPER CONTROL	VALVE CONTROL	SET POINT ADJ	HI ANALOG	LOW ANALOG				PROOF	TIME SCHEDULING	TREND	ALARM INSTRUCT	NIGHT SET-BACK	DDC CONTROL					
	TEMPERATURE	PRESSURE	AIR FLOW	SASH POSITION	KWH																					ENTHALPY	RUN TIME	EFFICIENCY	POSITION	RELATIVE HUMIDITY
(N) ROOM AIR TERMINAL CONTROL (RC) FOR TRACKING PAIRS														X	X	X							X	X						
(N) SPACE THERMOSTAT (TT-1)	X																													
(N) DISCHARGE THERMOSTAT (TT-2)	X												X																	
(E) SUPPLY TERMINAL UNIT (VAV)			X										X	X																
(N) EXHAUST TERMINAL UNIT (VEV)			X										X																	
(E) ROOM PRESSURE MONITOR	X														X	X														



CONTROLS SEQUENCE

GENERAL NOTES:

PROVIDE LABORATORY ROOM LEVEL CONTROL INTEGRATION TO EXISTING AUTOMATED LOGIC BUILDING AUTOMATION SYSTEM (BAS) AS REQUIRED TO MATCH TYPICAL FULL LABORATORY BAS INTEGRATION WITH EXISTING ACCUTROL LAB CONTROLS. MATCH EXISTING TYPICAL LAB READWRITE CAPABILITIES AND BAS GRAPHICS. PROVIDE ADDITIONAL BAS INTEGRATION WITH LAB ACCUTROL PRESSURE MONITORING, NEW EXHAUST FANS AND NEW AIRFLOW MONITORING STATION POINTS.

BAS CONTROL CONTRACTOR AND ACCUTROL LAB CONTROL REPRESENTATIVE TO FIELD VERIFY CONTROL SYSTEM MODIFICATIONS REQUIRED FOR BAS INTEGRATION FOR LAB CONTROLS AND EQUIPMENT ASSOCIATED WITH ROOM 2309 INCLUDING: LAB AIR VALVES, ROOM SENSORS AND NEW EXHAUST ISOLATOR EXHAUST FAN AND AIR FLOW MONITORING.

THE INDICATED CONTROL SEQUENCE OF OPERATION AND POINTS LIST PROVIDED HERE MATCH THE EXISTING TYPICAL LAB ROOM CONTROLS BAS INTEGRATION.

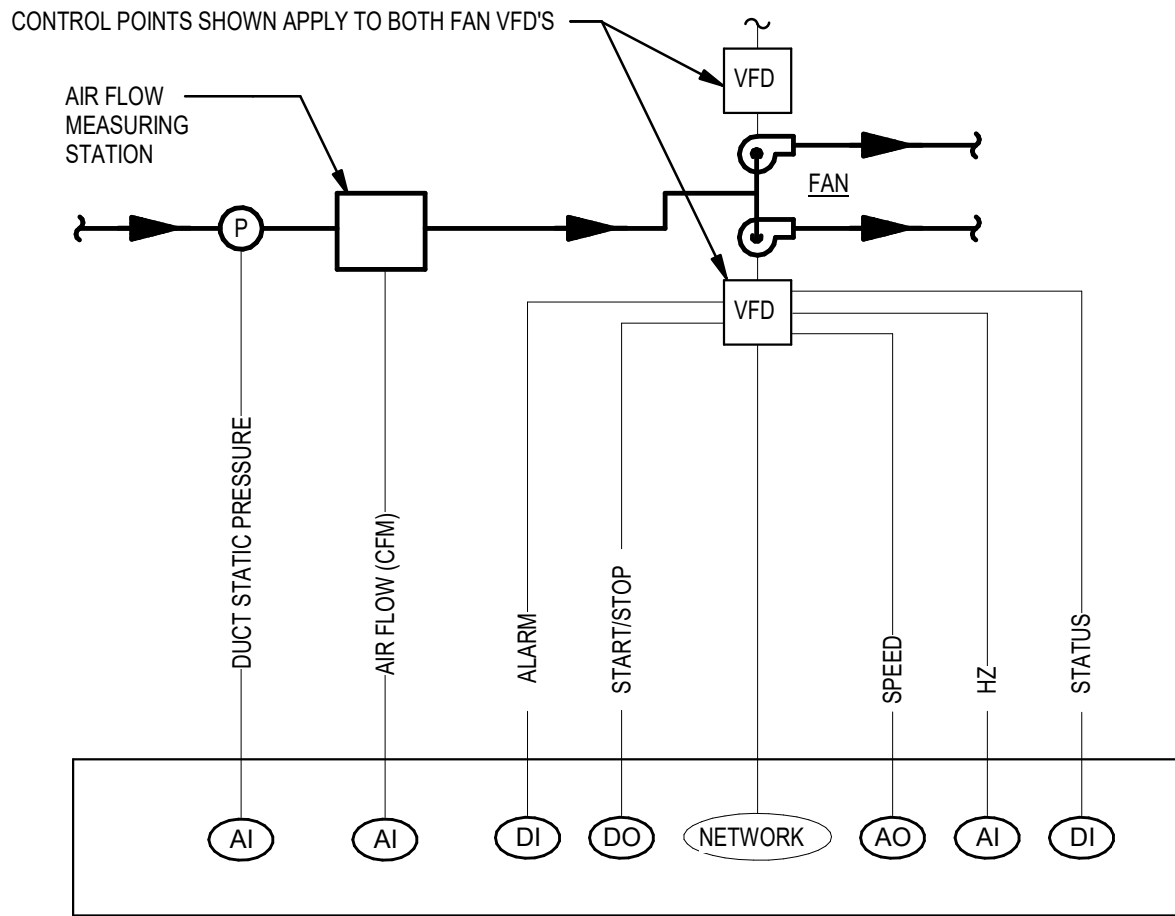
EXISTING AIR VALVES VAV-76 AND VEV-101 ARE CAPABLE OF CONSTANT VOLUME OPERATION WITH TWO AIRFLOW RATE SETTINGS. EXISTING OPERATION MAINTAINS SAME AIRFLOW RATES FOR OCCUPIED AND UNOCCUPIED MODE. THE NEW SEQUENCE OF OPERATION MAINTAINS THE SAME AIRFLOW RATES BETWEEN OCCUPIED AND UNOCCUPIED AS EXISTING WITH THE CAPABILITY OF PROVIDING UNOCCUPIED MODE AIRFLOW SET BACK IF NEEDED IN FUTURE.

ROOM 2309 LABORATORY CONTROLS

- OCCUPIED/UNOCCUPIED MODES SHALL BE AS DETERMINED BY THE OCCUPIED/UNOCCUPIED PROGRAM OF THE BAS. INDIVIDUAL LABORATORY/EQUIPMENT SPACES SHALL HAVE SEPARATE OCCUPIED/UNOCCUPIED PROGRAM SETPOINTS TO BE ESTABLISHED DURING SYSTEM COMMISSIONING. INDIVIDUAL PROGRAM SETPOINTS SHALL BE ADJUSTABLE THROUGH THE BAS.
- THE EXISTING ROOM THERMOSTAT (TT-1) SHALL SENSE AND PROVIDE LOCAL INDICATION OR ROOM TEMPERATURE. THERMOSTATS SHALL BE PROGRAMMED FOR THE INDICATED SPACE TEMPERATURE AND HAVE LOCAL "WARMER/COOLER" CONTROL BY THE SPACE OCCUPANTS. THE RANGE OF TEMPERATURE CONTROL AND PROGRAMMED DEAD BAND SHALL BE LIMITED BY THE +/- VARIABLE INDICATED. ALL THERMOSTAT SETPOINTS INCLUDING THE +/- VARIABLE SHALL BE ADJUSTABLE THROUGH THE BAS.
- A NEW SUPPLY AIR DISCHARGE TEMPERATURE SENSOR (TT-2) SHALL PROVIDE INDICATION TO THE BAS OF THE RESULTING AIR TEMPERATURE BEING DELIVERED TO THE OCCUPIED ZONE.
- OCCUPIED MODE.
 - WHEN THE TEMPERATURE SETPOINT OF 20.5°C (~69.0°F) CANNOT BE MAINTAINED, THE RC SHALL GRADUALLY MODULATE SUPPLY AIR TERMINAL REHEAT COIL CONTROL VALVE OPEN TO MAINTAIN SET POINT.
 - ON A RISE IN ROOM TEMPERATURE ABOVE THE OCCUPIED SETPOINT, THE REVERSE SHALL OCCUR.
 - THE EXISTING SUPPLY AIR VALVE (E) VAV-76, EXHAUST AIR VALVE (E) VEV-101 AND NEW EXHAUST FANS EF-2309A/EF-2309B WILL BE MODULATED TO MAINTAIN A CONSTANT AIRFLOW AND A CONSTANT PREDETERMINED AIRFLOW OFFSET AS INDICATED IN THE ROOM AIR BALANCE SCHEDULE FOR OCCUPIED MODE. THE AIRFLOW OFFSET VALUE SHALL BE ADJUSTABLE THROUGH THE BAS.
 - SUPPLY AND EXHAUST AIR VALVES SHALL PROVIDE PRESSURE INDEPENDENT AIRFLOW THROUGH INTEGRAL FLOW SENSORS AS REQUIRED BY THEIR CONTROL SIGNALS.
- UNOCCUPIED MODE.
 - THE EXISTING SUPPLY AIR VALVE (E) VAV-76, EXHAUST AIR VALVE (E) VEV-101 AND NEW EXHAUST FANS EF-2309A/EF-2309B WILL BE MODULATED TO MAINTAIN A CONSTANT AIRFLOW AND A CONSTANT PREDETERMINED AIRFLOW OFFSET AS INDICATED IN THE ROOM AIR BALANCE SCHEDULE FOR UNOCCUPIED MODE. THE AIRFLOW OFFSET VALUE SHALL BE ADJUSTABLE THROUGH THE BAS.
 - ON A FALL IN ROOM TEMPERATURE BELOW THE UNOCCUPIED MINIMUM SETPOINT OF 18°C (~65°F) AS SENSED BY THE ROOM THERMOSTAT, THE SUPPLY AIR VALVE REHEAT UNIT COIL CONTROL VALVES SHALL GRADUALLY BE MODULATED OPEN TO MAINTAIN SETPOINT.
 - TT-1 THERMOSTAT SHALL BE PROVIDED WITH AN OVERRIDE BUTTON TO RETURN INDIVIDUAL SPACES TO THE OCCUPIED MODE REGARDLESS OF THE MASTER SETPOINT SCHEDULE AT THE BAS. THE OVERRIDE BUTTON SHALL TIME OUT IN 2 HOURS AND REVERT BACK TO THE UNOCCUPIED MODE.
- FAILURE
 - UPON FAILURE OR LOSS OF NETWORK INTERFACE, ALL TERMINAL UNIT COMPONENTS SHALL REMAIN IN THEIR LAST COMMANDED STATE. UNDER NO CONDITION SHALL SUPPLY AIR TERMINAL UNIT REMAIN OPEN WITH EXHAUST TERMINAL UNITS CLOSED. THE EXHAUST VALVE AIR TERMINAL SHALL FAIL SAFE/OPEN.

ISOLATOR EXHAUST FANS - EF-2309A AND EF-2309B

- ENABLE FANS AT ALL TIMES UNLESS SHUTDOWN ON SAFETIES.
- FAN CONTROL
 - BOTH FANS OPERATE IN PARALLEL TO MAINTAIN A COMBINED CONSTANT VOLUME AIRFLOW SET POINT OF 500 CFM. WHILE THERE IS NO EXTERNAL GAS DISCHARGED INTO THE EXHAUST SYSTEM FROM THE ROOM ISOLATOR UNITS, TEST AND BALANCER AGENT TO SET FAN SPEED AND ADJUST BALANCING DAMPER IN THE EXHAUST DUCT SYSTEM SO THAT 50% AIRFLOW IS MAINTAINED BY EACH FAN DURING NORMAL OPERATION AND STATIC PRESSURE SET POINT IS MAINTAINED.
 - MODULATE BOTH EXHAUST FANS IN UNISON, AT THE SAME SPEED TO MAINTAIN CONSTANT VOLUME AIR FLOW AS SENSED BY THE AIRFLOW MONITORING STATION.
 - IF A FAN HAS FAILED OR HAS BEEN DESIGNATED "OUT OF SERVICE" PER THE SEQUENCE BELOW, THE REMAINING FAN IN OPERATION WILL INCREASE FAN SPEED TO MAINTAIN AIRFLOW SET POINT OF 500 CFM.
 - INSTALL STATIC PRESSURE SENSING PROBES IN THE MAIN EXHAUST DUCT LOCATED AT APPROXIMATELY 3/4 OF THE WAY DOWN THE MAIN EXHAUST DUCT OR AS SHOWN ON THE PLANS. STATIC PRESSURE WILL BE FOR MONITORING ONLY.
- GENERATE ALARMS AS FOLLOWS
 - EXHAUST FAN FAILURE: COMMANDED ON, BUT THE STATUS IS OFF.
 - EXHAUST FAN OPERATING IN HAND MODE: COMMANDED OFF, BUT THE STATUS IS ON.
 - EXHAUST FAN RUNTIME EXCEEDED: STATUS RUNTIME EXCEEDS A USER DEFNABLE LIMIT.
 - VFD FAULT.



EXHAUST FANS (EF-2309A & EF-2309B)
CONTROL DIAGRAM

NO SCALE

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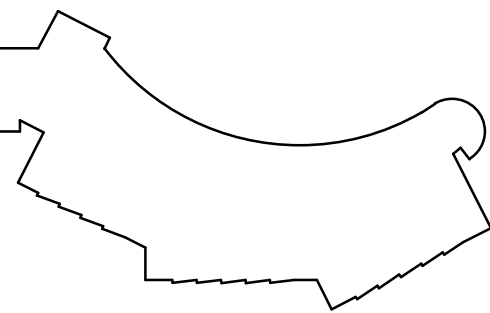
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STAMP



KEY PLAN



FDA IRVINE

RM 2309
EXHAUST
DESIGN

19701 Fairchild Rd, Irvine, CA 92612

SUBMITTAL

BID SET

AUGUST 15, 2025

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SHEET

DIAGRAM &
DETAILS -
MECHANICAL
M.402

INTERFACE
ENGINEERING

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ELECTRICAL SYMBOL LIST

NOTE: This is a standard symbol list and not all items listed may be used.

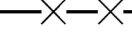
Abbreviations

AWG	AMERICAN WIRE GAUGE
BAS	BUILDING AUTOMATION SYSTEM
CLG	CEILING
C	CONDUIT, CLOSE, CONTROL
(X)	DEMOLISH
DTL	DETAIL
DIA	DIAMETER
DIM	DIMENSION
DN	DOWN
DWG	DRAWING
EMT	ELECTRICAL METALLIC TUBING
EF	EXHAUST FAN
(E)	EXISTING TO REMAIN
FF	FINISH FLOOR
FMC	FLEXIBLE METAL CONDUIT
FT	FOOT, FEET
G, GND	GROUND
GFCI	GROUND FAULT CIRCUIT INTERRUPTER
GFI	GROUND FAULT INTERRUPTER
GFP	GROUND FAULT PROTECTION
HT	HEIGHT
ID	IDENTIFICATION
IN	INCH, INCHES
IMC	INTERMEDIATE METAL CONDUIT
IG	ISOLATED GROUND
KV	KILOVOLT
KVA	KILOVOLT AMPERES
KW	KILOWATT
LV	LOW VOLTAGE
mA	MILLIAMPERES
MIN	MINIMUM
MCA	MINIMUM CIRCUIT AMPS
MISC	MISCELLANEOUS
MCC	MOTOR CONTROL CENTER
MT, MTD	MOUNT, MOUNTED
NEC	NATIONAL ELECTRIC CODE
NESC	NATIONAL ELECTRIC SAFETY CODE
NEMA	NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION
N	NEUTRAL
(N)	NEW
N/A	NOT APPLICABLE
N.I.C.	NOT IN CONTRACT
NTS	NOT TO SCALE
PNL	PANEL
PH	PHASE
PWR	POWER
QTY	QUANTITY
REF	REFERENCE
RFI	REQUEST FOR INFORMATION
RMC	RIGID METAL CONDUIT
RM	ROOM
SHT	SHEET
SIM	SIMILAR
STD	STANDARD
SWBD	SWITCHBOARD
TBD	TO BE DETERMINED
XFMR	TRANSFORMER
TVSS	TRANSIENT VOLTAGE SURGE SUPPRESSOR
TYP	TYPICAL
UG	UNDERGROUND
UPS	UNINTERRUPTIBLE POWER SUPPLY
UON	UNLESS OTHERWISE NOTED
VFD	VARIABLE FREQUENCY DRIVE
V	VOLTS, VOLTAGE
WP	WEATHERPROOF
WG	WIRE GUARD
W/	WITH
W/O	WITHOUT

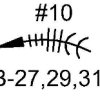
Connections / Equipment

	HEAVY DUTY FUSED DISCONNECT SWITCH
	MOTOR CONNECTION
	NON-FUSED DISCONNECT SWITCH

General

	DETAIL NUMBER AND SHEET LOCATION
	EQUIPMENT IDENTIFICATION
	KEYED NOTE
	DEMOLISH
	EXISTING WORK
	NEW WORK

Miscellaneous

	BRANCH CIRCUIT WIRING. ARROW INDICATES HOME RUN TO PANEL WITH CIRCUITS AS NOTED. WIRE SIZE IS #12 AWG MINIMUM UNLESS NOTED OTHERWISE. SHORT TICK MARKS INDICATE PHASE CONDUCTORS. LONG TICK MARKS INDICATE NEUTRAL CONDUCTORS. A SINGLE CURVED TICK MARK INDICATES INSULATED GREEN GROUND CONDUCTOR. SECOND CURVED TICK MARK INDICATES "ISOLATED GROUND" (GREEN INSULATION WITH YELLOW STRIPE) CONDUCTOR.
	BRANCH PANEL
	CIRCUIT BREAKER
	DRY TYPE TRANSFORMER
	MAIN DISTRIBUTION PANEL / SUB DISTRIBUTION PANEL

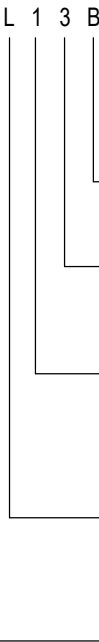
Raceways

	CONDUIT CONCEALED IN WALL OR CEILING SPACE
---	--

Switches and Receptacles

	DUPLEX RECEPTACLE (MULTIPLE LETTERS INDICATE MULTIPLE OPTIONS) A = ABOVE COUNTER B = CLOCK HANGER C = FLUSH CEILING MOUNTED E = EMERGENCY F = ARC FAULT PROTECTED BY BREAKER IN PANEL G = GROUND FAULT CIRCUIT INTERRUPTER H = HOSPITAL GRADE K = CHILD RESISTANT COVER L = ISOLATED GROUND P = PENDANT MOUNTED WITH CORD GRIPS. VERIFY PENDANT LENGTH R1 = HALF SWITCHED BY OCCUPANCY SENSOR RELAY R2 = FULLY SWITCHED BY OCCUPANCY SENSOR RELAY S = SPLIT WIRED T = TAMPER RESISTANT SHUTTERED RECEPTACLE U = USB PORT(S) W = WEATHERPROOF CONTINUOUS USE COVER, GFCI PROTECTED, WITH WEATHER-RESISTANT RECEPTACLE
	SINGLE POLE SWITCH
	2 = DOUBLE POLE SWITCH
	3 = THREE-WAY SWITCH
	4 = FOUR-WAY SWITCH
	a THRU z (LOWERCASE) = LUMINAIRE CONTROL DESIGNATION
	D = DIMMER
	F = FAN SPEED CONTROL
	K = KEY OPERATED SWITCH
	L = LIGHTED HANDLE
	M = MANUAL MOTOR STARTER WITH THERMAL OVERLOAD
	P = SWITCH WITH PILOT LIGHT
	S = SENTRY SWITCH
	T = INTERVAL TIMER
	W = WEATHERPROOF SWITCH
	V = LOW VOLTAGE SWITCH

PANELBOARD DESIGNATIONS

	L L 1 3 B A-L SEQUENCE OF PANEL IN SERIES LAB PANELS ONLY 1 WING 1 2 WING 2 3 WING 3 B BASEMENT 1 FIRST LEVEL 2 SECOND LEVEL P PENTHOUSE LEVEL H B277/480V L 120/208V C CABINET E EMERGENCY-LIFE SAFETY L LAB POWER M MOTOR CONTROL CENTER OR DISTRIBUTION PANEL N NORMAL POWER Q EMERGENCY-EQUIPMENT BRANCH R LOW VOLTAGE RELAY LIGHTING CONTROL
--	---

SHEET INDEX

E.001	SYMBOL & NOTES - ELECTRICAL
E.101	FLOOR PLANS - ELECTRICAL
E.501	SINGLE LINE DIAGRAMS - ELECTRICAL
E.601	SCHEDULES - ELECTRICAL



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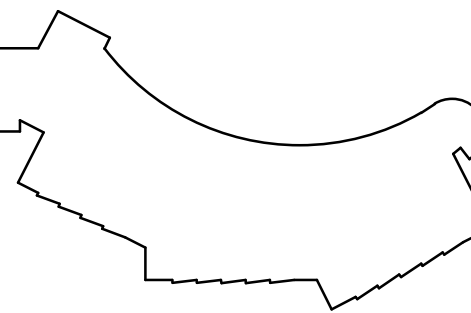
Phone: 619-220-0984
Kevin Carpenter



STAMP



KEY PLAN



FDA IRVINE

RM 2309

EXHAUST

DESIGN

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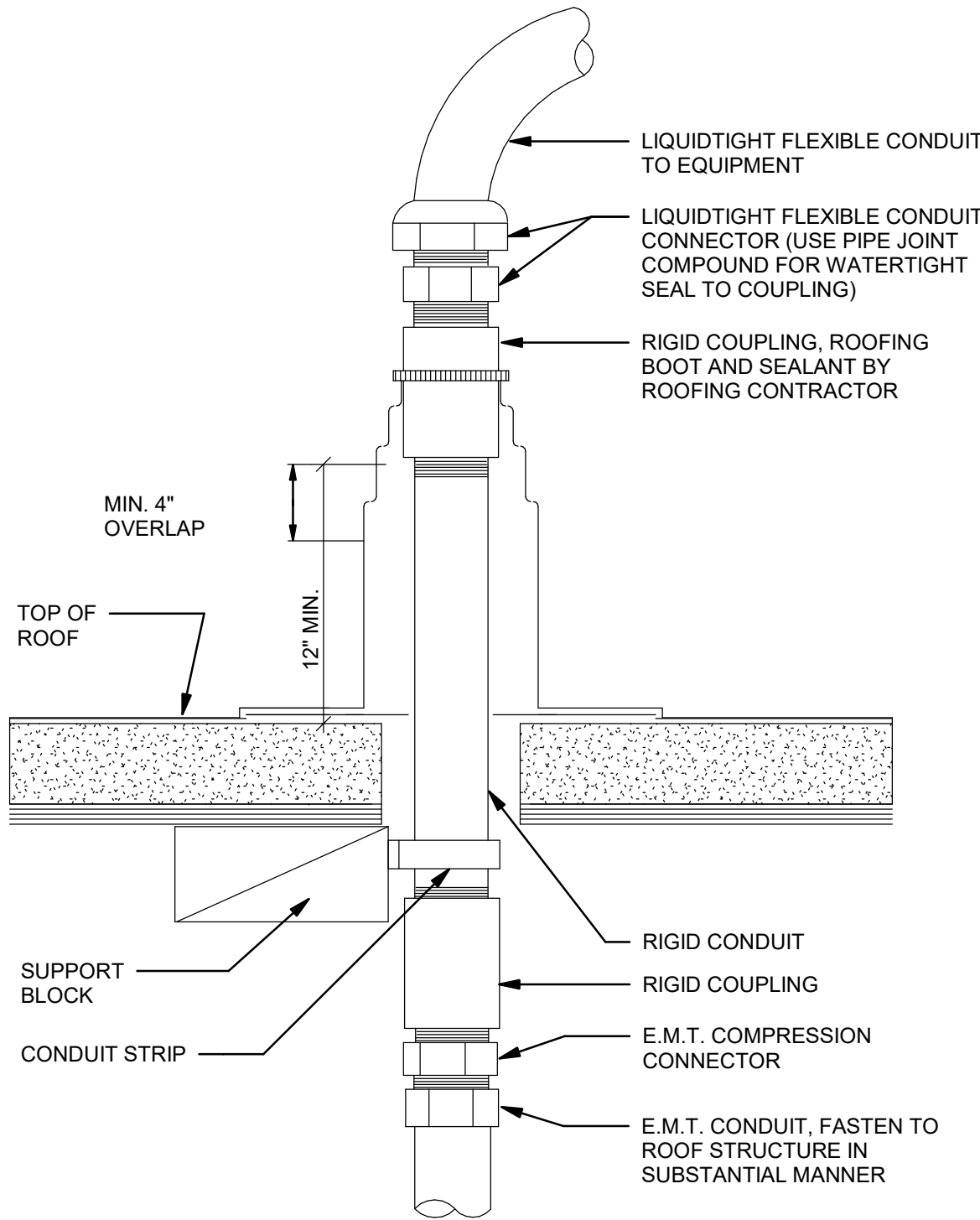
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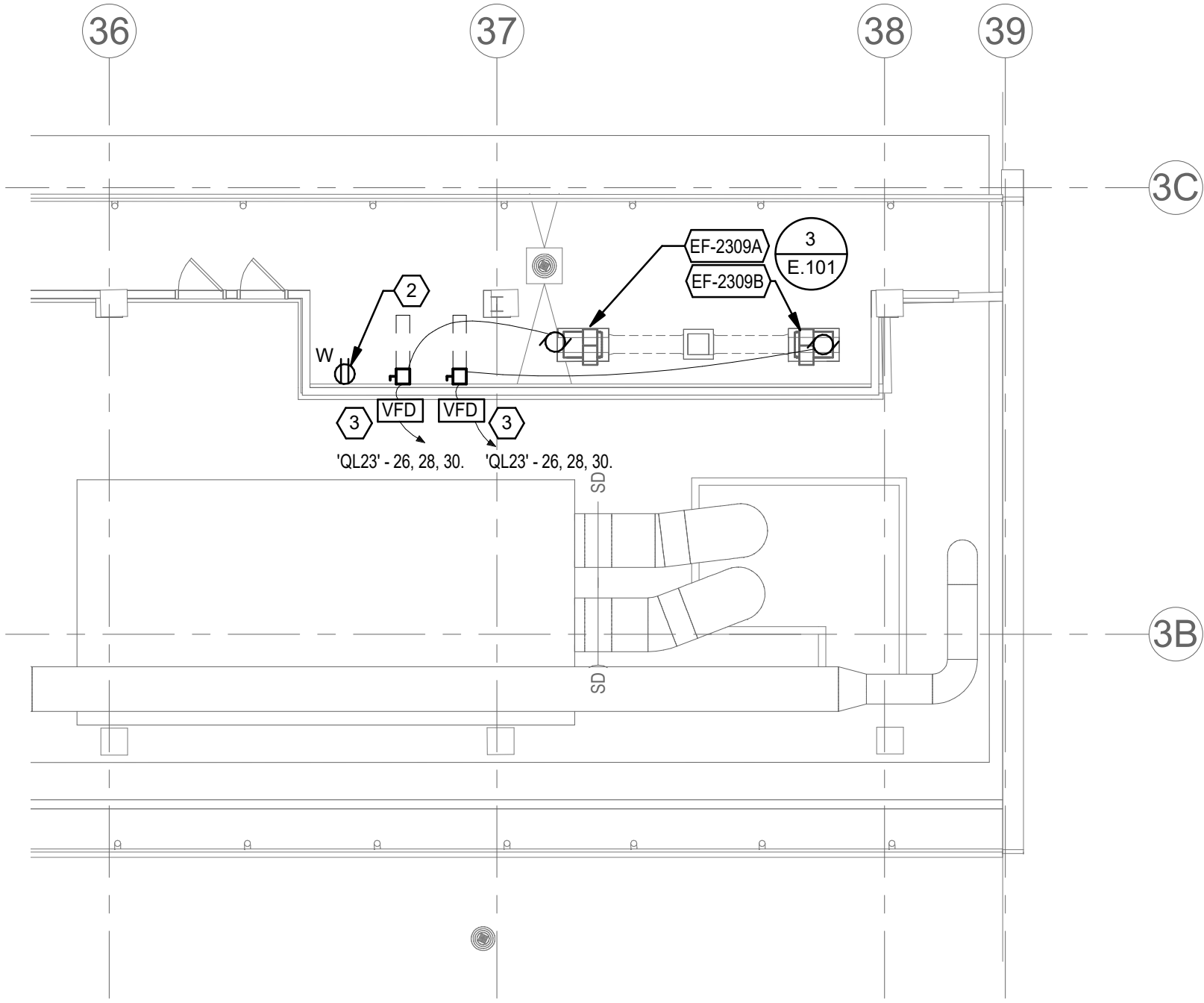
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SYMBOL & NOTES - ELECTRICAL
E.001

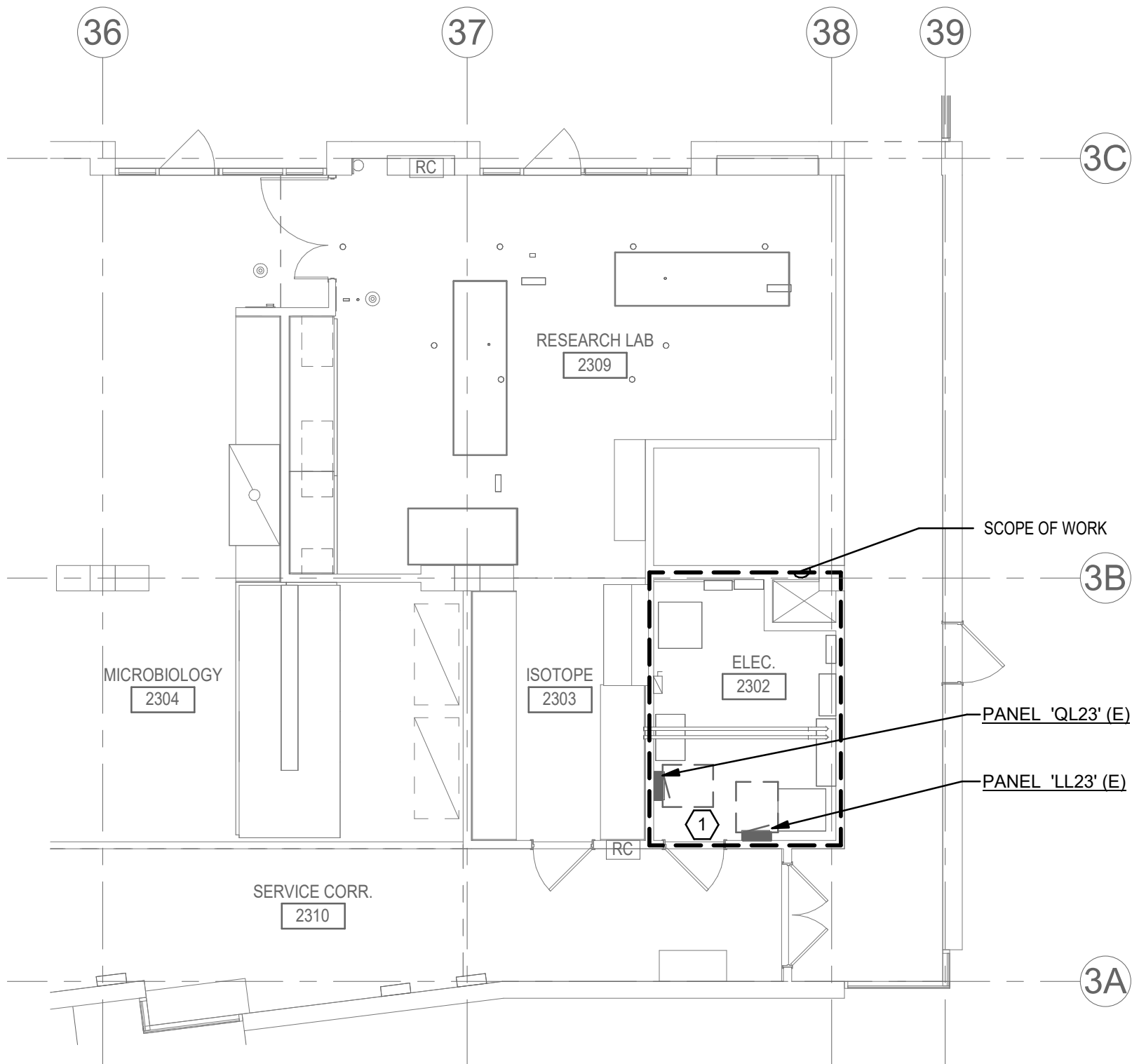
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3
E.101
CONDUIT THROUGH ROOF
NO SCALE



2
E.101
ROOF PLAN - ABOVE RM 2309 - POWER
1/8" = 1'-0"



1
E.101
LEVEL 2 PLAN - RM 2309 - POWER
1/8" = 1'-0"

GENERAL SHEET NOTES

- REFER TO MECHANICAL DRAWINGS AND SPECIFICATIONS FOR ADDITIONAL INFORMATION ON CONTROL OF THE ISOLATOR HOOD.
- PROVIDE 30 DAY METERING FOR PANELS THAT LOAD ARE BEING ADDED TO PER NEC 220.87 TO ENSURE THAT THE ELECTRICAL DISTRIBUTION SYSTEM IS NOT OVERLOADED.
- REVIEW EXISTING CONDITIONS TO IDENTIFY ANY CONFLICTS WITH THIS DESIGN SET PRIOR TO BID.

SHEET KEYNOTES

- ROUTE CONDUIT THRU LAB CEILING SPACE AND UP TO ELECTRICAL DEVICES. DO NOT PENETRATE ROOF ABOVE ELECTRICAL ROOM.
- CONNECT TO NEAREST 120V NORMAL POWER RECEPTACLE CIRCUIT THAT IS FED FROM PANELBOARD LL23. AS-BUILTS SHOW CIRCUITS 39 AND 41 ARE UTILIZED IN THAT AREA WITH 360 VA LOAD ON EACH CIRCUIT. CONNECT WITH (2)#10, #10G IN 3/4" CONDUIT.
- CONFIRM FINAL LOCATON OF VFD WITH OWNER PRIOR TO BID.

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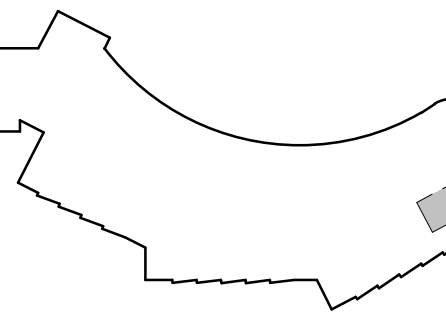
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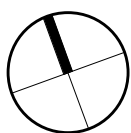
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FLOOR PLANS - ELECTRICAL E.101

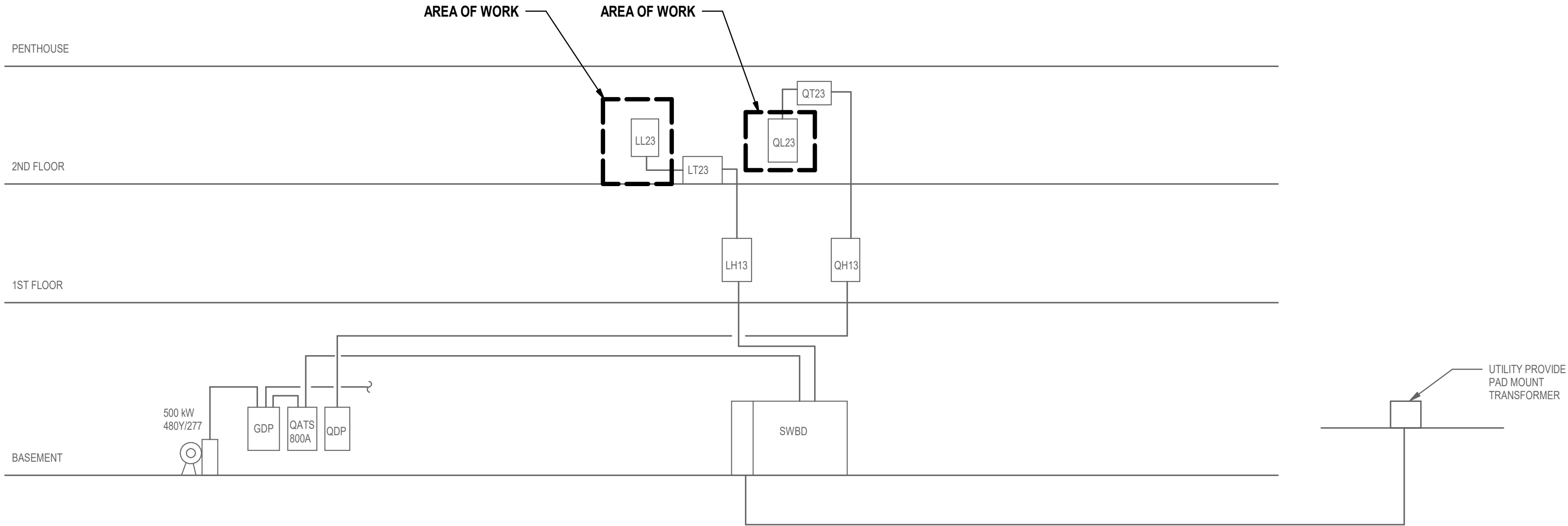
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1
E.501

ONE-LINE POWER DISTRIBUTION DIAGRAM - WING 3

NO SCALE



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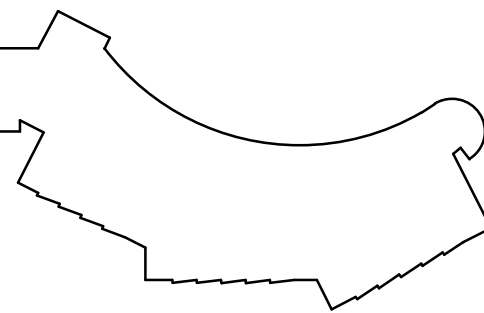
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SINGLE LINE DIAGRAMS - ELECTRICAL E.501

